From Traces to Guarded Programs: Evidence-Gated Compilation of Recurrent Agent Workflows

Anonymous authors
Paper under double-blind review

Abstract

Guarded agentic compaction (GAC) compiles the recurrent initial read-only prefix of an agent workflow into a deterministic program and leaves the rest of the agent unchanged. On three GitHub workflows the compiled agent satisfies 90/90 held-out contracts against 89/90 for the unchanged agent with 66.6% fewer model requests, and a separately frozen comparison with adapted offline Agent Workflow Memory over three workflows, two models, and 1,362 fresh records (8,172 attempted episodes) cuts estimated deployment cost by 36.9–79.1% and requests by 50–75%, with five of six cells meeting a preregistered five-point accuracy non-inferiority margin under multiplicity correction. On BIRD, a schema-first SQL opening compiles on 26/31 databases with 45–50% fewer requests and no detected accuracy loss, and a post-cutoff cohort of 132 pull requests shows no end-to-end error (upper bound 1.73% at 90% confidence).

GAC admits a program only when every tool argument is grounded in the entry state, a prior result, or an invariant literal; every tool has a declared read-only effect; induced input/output contracts hold; and an exact finite-sample test under independent-group assumptions bounds the dispatch risk. Otherwise it retires the candidate and keeps the agent, which is what happens to question-specific table selection on BIRD, to every NESTFUL, API-Bank, and BFCL family (too little calibration support), and, at dispatch time, to AppWorld's admitted program on nearly all ReAct and plan-and-execute runs. The evidence supports bounded compilation of recurrent prefixes, not universal accuracy preservation or learned-gate benefit: most certificates concern replay rather than final answers, the statistical stage adds no measured runtime benefit on admitted candidates, one second-model cell is inconclusive, and independence beyond the evaluated workloads is assumed.

1 Introduction

The standard agent loop pays for a model decision at every tool boundary (Yao et al., 2023). That flexibility is valuable while a workflow is novel. When an agent repeatedly takes the same read-only evidence path with invariant data dependencies, those decisions offer a candidate for specialization. Across 157,280 released trajectories from four agent scaffolds, between 4% and 41% of episodes open with a read-only prefix of two or more calls, and 2–13% with one whose arguments are groundable from the entry state (Appendix F.4); which end of that range an agent sits on is set by its architecture, not its workload.

Removing that decision is not ordinary caching: a tool argument may depend on an earlier observation, a repeated sequence may cross an approval or write, a read-only tool may hide an effect, and replaying the same outputs may still change the answer. The question is when a recurrent model-mediated region may be replaced by a deterministic program, and when it must be refused.

Consider held-out issue #4420. Given only `issue_number=4420`, the observed trace calls `record(4420)`, passes the *returned* `record.issue_number` to `labels` and `comments`, and invokes the latter with `limit=100`. All 132 discovery executions take that three-read route (130 with `limit=100`, two with `limit=1`), so a recurrence-only optimizer would emit all three reads. GAC instead retires the *region*: `comments.limit` is not an invariant literal and is reachable neither from the entry state nor from either earlier result, so one unwitnessed slot blocks the three-read candidate as `ungroundable_slot`. What survives is the maximal justified prefix `record` → `labels`; the agent

still chooses the limit and writes the answer. That two-read artifact passes 92/92 calibration groups at $\alpha = 0.05$ (upper bound 0.0498).

An earlier 18-record study supplies the counterexample: its three-read artifact, admitted at a looser $\alpha = 0.10$ (a 45-group floor), passed all 45 calibration replays and the tool contract on every held-out record, yet issue #6602 lost the URL from a Markdown link and only 17/18 downstream answers stayed exact; a provider-free continuation check, run after the model's answer, detects that miss. Together the two cohorts show why recurrence only proposes a candidate (Fig. 2 draws the first).

The closest precedent is trace compilation: Dynamo (Bala et al., 2000) and tracing JITs (Gal et al., 2009) compile a hot path behind *guards* and fall back when one fails, as deoptimization (Hölzle et al., 1992) restores unoptimized execution. Agent traces admit the same structure (family mining, an entry contract, fallback to the unchanged agent) but compile only *initial* read-only prefixes (Eq. (2)) and add three obligations: tool arguments must be shown *grounded* in observable state; declared *effects* must permit speculation, since the "interpreter" calls the outside world; and the residual error rate, being statistical, needs a finite-sample bound.

We study these obligations through *guarded agentic compaction* (GAC), a compile-or-retire trace-to-program system for recurrent read-only prefixes (Fig. 1). We ask: **RQ1**, which recurrent regions admit a grounded program? **RQ2**, how much work does compilation remove at a stated quality tolerance? **RQ3**, which admission safeguards add measurable protection? The contributions are:

1. A specialization pipeline combining typed provenance, effect and position barriers, bounded synthesis, verification, and compile-or-retire admission (Algs. 1 and 2). The statistical component is a standard exact binomial bound over a fixed grid, not a new inequality; our contribution is its composition with trace-derived specialization and explicit certificate scope (Proposition 1).
2. Live-provider evidence on public-record workflows and BIRD, with model/provider replications and safeguard ablations. Against adapted offline AWM, five of six workflow/model cells establish five-point accuracy noninferiority, but none establishes one-point noninferiority (Section 5.1). The ablations separate useful request-scope checks from unresolved completeness benefit and null learned-gate benefit.
3. Refusal measurements on public trace benchmarks (Section 5.3): NESTFUL, API-Bank, and BFCL v4 lack calibration support; AppWorld admits a program eligible on 2,339/2,340 full-code trajectories but 1/4,680 ReAct or plan-and-execute ones. Admission therefore does not establish deployment benefit.

2 Problem Formulation

Terminology. An *episode* is one recorded run of the agent; a *group* is the set of episodes that share an independence unit (one record, one scenario); a *region* is a contiguous span of one episode's tool calls; a *family* is the set of regions across episodes that share a canonical shape; a *candidate* is a family the compiler attempts and, once synthesized, its thresholdless tuple (P, H, V, q, M); a *program* is the deterministic code synthesized for it; an *artifact* is a program together with its guard, verifier, score, threshold, and manifest pins; and a *prefix* is a region that starts at the first model boundary. Compilation is the offline act; compaction is the runtime substitution it licenses.

Episodes and candidate regions. An episode $E = (z, M, e_{1:T}, y)$ contains an entry-state snapshot z, a content-addressed execution manifest M, ordered events e_t (model requests and responses, tool calls and results, handoffs, guardrails, approvals, errors, commit boundaries), and an observable outcome y; c_j denotes the j-th tool call and o_j its result. Each tool f carries an application-owned effect declaration $\epsilon(f)$ and capability flags such as *speculatable*; an unknown effect is not a read.

A candidate region $R = [a, b]$ begins at a model-request boundary and ends after a tool result. It is *groundable* when every call argument is either a literal invariant across the supporting traces or an expression over entry state or prior results inside R,

$$\forall c_j \in R,\ \forall u \in \mathrm{keys}(\mathrm{args}(c_j)) : \quad \mathrm{Lit}(c_j, u) \ \vee\ \exists\, s \in \mathrm{Src}(z, o_{<j}),\ \exists\, g_{j,u} \in \mathcal{L} :\ v_{j,u} = g_{j,u}(s), \quad (1)$$

where $v_{j,u}$ is the value of slot u. A witness uses the same source path and operator chain across supporting training traces; $\mathrm{Lit}(c_j, u)$ requires the same literal on those traces. A slot the schema declares

literal-only may be bound in no other way. $\mathcal{L}$ is a closed, bounded transform library, and $\mathrm{Src}(z, o_{<j})$ enumerates values at flattened paths of z and each prior result $o_i, i < j$. A region is *effect-admissible* when every tool in R is declared read-only, speculatable, replayable, and inside one principal and isolation partition. The witness is deliberately *single-source* — one chain over one observation, not a join over the history — which is the quantifier Alg. 3 (Appendix B) implements. Handoffs, approvals, writes, errors, and unknown effects are hard barriers. Because the deployed runtime resolves artifacts at the initial model boundary, the compiler also imposes a *position invariant*:

$$a = 0 \quad \text{in the normalized model-boundary index.} \tag{2}$$

A suffix program dispatched at entry can reorder or duplicate calls the agent has already made; a pilot that violated Eq. (2) produced exactly that fault (Appendix E). Everything that follows is therefore an *initial read-only prefix* compiler; the AppWorld result of Section 5.3 prices what that costs.

Selective optimization objective. The compiler emits an artifact $A = (P, H, V, q, \eta, M)$: a deterministic program, hard guard, output verifier, nonconformity score, threshold, and manifest pins. Let $x = (z, M', \omega)$ be a future entry state, runtime manifest, and external state, where $M' \simeq M$ is equality on every pinned compatibility field and ω affects outcomes and cost but is unobservable at the dispatch boundary. Dispatch is *selective*,

$$d_A(x) = \mathbf{1}\{\, H(z, M') = 1 \ \wedge\ q(z) \le \eta \ \wedge\ M' \simeq M \,\}, \tag{3}$$

and any failed precondition runs the unchanged baseline B. If execution fails before a commit and staging proves the attempt clean, B runs; a failure whose effects cannot be proved clean is an incident, not a fallback. Fix a binary loss $L(A, x)$ before calibration: replay-contract failure or incident for the primary certificate; final task-contract failure for the end-to-end study. A clean abort has replay loss zero but its baseline continuation can still fail the final task. Let $C(A, x)$ be the complete realized cost of the attempt including a clean fallback. For a future group $G = (x_1, \ldots, x_m)$, set $D_A(G) = \max_i d_A(x_i)$ and $W_A(G) = \max_i d_A(x_i) L(A, x_i)$. We seek

$$\max_A \ \mathbb{E}_{G\sim\mathcal{D}_{\mathrm{grp}}}\Big[\, \textstyle\sum_{x\in G} d_A(x)\,\{C(B,x) - C(A,x)\}\,\Big] \quad \text{s.t.} \quad \Pr_{G\sim\mathcal{D}_{\mathrm{grp}}}[W_A(G) = 1 \mid D_A(G) = 1] \le \alpha. \tag{4}$$

Both halves take the same group measure, and the constraint conditions on dispatch: pre-dispatch abstention earns no savings and enters no denominator (its conditional risk is zero by convention when population dispatch probability is zero). Returning *no* artifact is feasible, so RETIRE is a legitimate output; Eq. (4) is a design target rather than a solved program, and the guarantees of Section 3 cover only the per-candidate and fixed-m cases of the implemented ranked search. Every certificate in Section 5 except one end-to-end certificate instantiates $L(A, x) = 1$ as a *replay-contract violation* of the compiled region, so the continuation is measured, not certified; Appendix F.6 states each certificate's event and the denominator and loss conventions.

3 Guarded Agentic Compaction

Figure 1 gives the six stages; Alg. 2 (Appendix B) lists them in full. The order is part of the argument: each stage can only weaken a candidate, so a later stage may reject one an earlier stage passed, but never authorize one an earlier stage blocked.

1. Typed provenance. A capture adapter normalizes framework traces into the Episode IR; manifests pin prompt, policy, guardrail, tool, model, SDK, tracer, entry-contract, and effect-catalog identities, and episodes with different compatibility keys are partitioned before any learning. For each call-argument slot, the program-argument trace graph (PATG, Alg. 3) searches entry-state and prior-result paths for typed values reachable by a bounded transform (database-style provenance (Cheney et al., 2009) for tool arguments). Two slot marks make the region miner block rather than guess: a slot with *no* witness is unresolved in the bounded library, and one with more than $\kappa = 3$ witnesses is ambiguous (Fig. 2); either retires the whole region. A repeated tool name or a successful replay is no witness; an invariant literal is (Eq. (1); the `limit=3` slots of the PR-outcome and backlog artifacts).

2. Region mining and ranking. The miner enumerates model-boundary-to-result windows of at most $w_{\max}$ tool calls, qualifies live-ins and effects against a human-signed rather than inferred catalog,

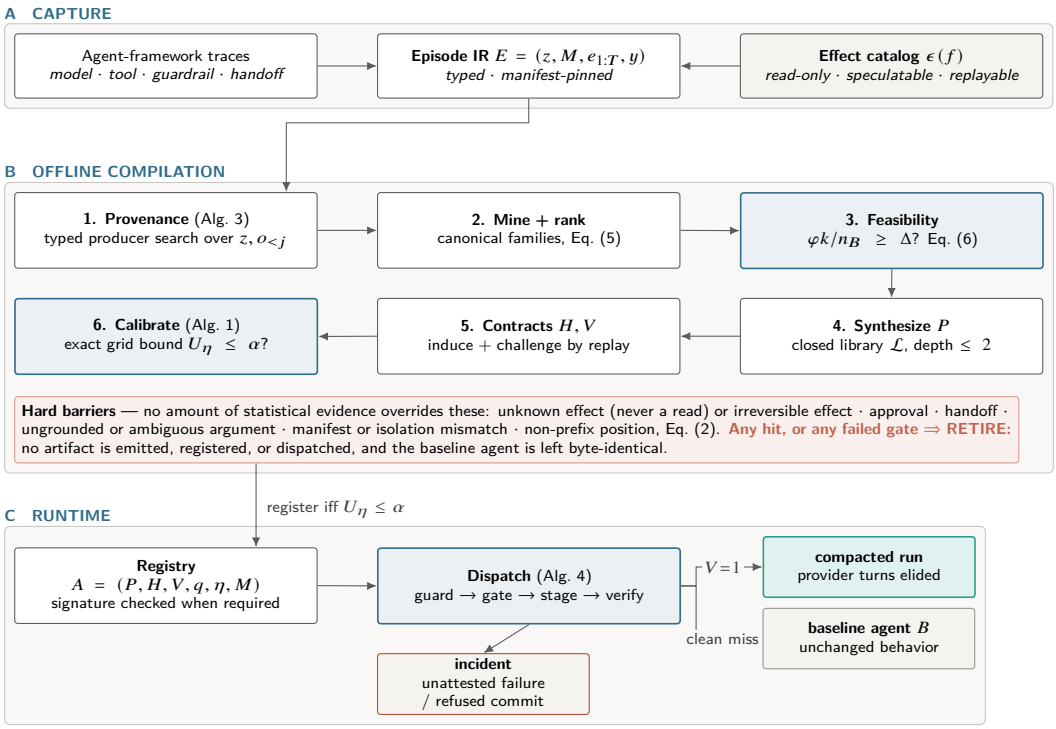


Figure 1: Architecture of GAC. **(A)** Capture normalizes traces into one typed Episode IR against a declared effect catalog. **(B)** Compilation is a cascade of independent rejection points; the shaded band lists barriers no statistical evidence can override. **(C)** At runtime one terminal edge compacts; clean rejections return the unchanged agent, and unattested failures raise an incident.

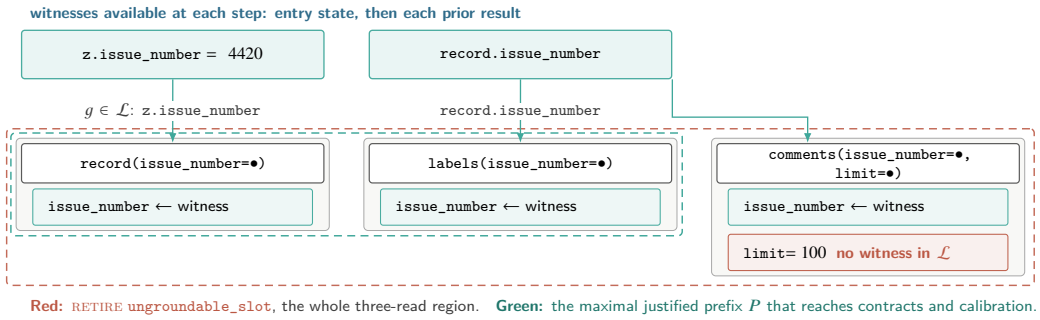


Figure 2: Equation (1) on the held-out issue-4420 record: both `issue_number` slots have an expression in $\mathcal{L}$ over the entry state or a prior result; the `limit` literal varies across the supporting traces (100 in 130 of 132, 1 in two) and has none, so the three-read region retires and only the two-read prefix is emitted.

and hashes tool signatures, dependency topology, run shape, and live-in/live-out shape while abstracting literals. Families require support across independent scenario *groups*, not raw event frequency. Survivors are ranked by an MDL-inspired trade-off that pays for expected savings and charges for heterogeneity, effect exposure, and program size:

$$\text{score}(F) = \underbrace{s_F\, \bar{k}_F\, c_m}_{\text{support-weighted saving}} - \lambda_1 \text{Ent}(F) - \lambda_2 \rho_{\text{eff}}(F) - \lambda_3 |F|, \tag{5}$$

with s_F supporting train groups, $\bar{k}_F$ the mean removable model requests, c_m their unit cost, and $\mathsf{Ent}(F)$ the Shannon entropy over canonical variants inside the family. Equation (5) affects only *which* family is tried first.

3. Feasibility ceiling. Before any synthesis runs, an estimator bounds what the compiler could achieve with a perfect verifier and a gate that never abstains:

$$\Delta_{\max} = \frac{\varphi\, k}{n_B} \;\geq\; \Delta \quad \text{(pre-synthesis filter)}, \qquad \varphi\,\rho\,k \;\geq\; \Delta\, n_B \quad \text{(realized, post-verification)}, \tag{6}$$

with n_B baseline model requests per episode, φ the fraction of episodes holding an eligible region, k the requests removed per successful dispatch, and ρ the verifier pass rate. Stage 3 evaluates only the left form ($\rho = 1$), at fixed continuation workload; downstream reasoning or fallback can change total requests. A workload below this structural target is declined.

4–5. Bounded synthesis and contracts. Synthesis searches the 23-operator library of Table 7 at depth at most two; bindings are ranked by cross-group stability, and a group-wise refit rejects bindings exploiting row-level coincidence. The hypothesis space is small enough to read, in the spirit of programming-by-example (Gulwani, 2011) and bounded sketching (Solar-Lezama et al., 2006), and never generates arbitrary Python. Hard guards then constrain entry types, required fields, categorical sets, numeric intervals, patterns, isolation keys, allowed effects, and manifest pins; verifiers constrain live-out presence, type, cardinality, empirical hulls, provenance, effect multisets, and call counts. These are candidate invariants, not proofs — which is why a statistical gate follows rather than replaces them — and under tool-layer corruption their measured protective value was null or adverse (Section 7).

6. Exact risk-gated admission. Algorithm 1 decides deployment. The score q is a regularized logistic model fitted on development windows, including synthetic perturbations, then frozen with the grid $\Lambda = \{0.02, 0.05, 0.08, 0.11, 0.14, 0.17, 0.20, 0.25, 0.30, 0.40, 0.50\}$. It predicts an unproductive replay (execution failure, verifier abstention, or verified mismatch), not a calibrated probability of task error. Training features and labels, including synthetic feature augmentation, are specified in Appendix C; calibration uses clean windows only. For each $\eta \in \Lambda$ the compiler counts admitted calibration groups (n_η) and those containing at least one post-dispatch violation (k_η), and inverts the one-sided exact binomial test (Clopper & Pearson, 1934) at a Bonferroni-corrected level, with explicit empty and all-violation cases:

$$U_\eta = \begin{cases} 1, & n_\eta = 0 \text{ or } k_\eta = n_\eta, \\ \mathrm{Beta}^{-1}\!\left(1 - \frac{\delta}{|\Lambda|}; k_\eta + 1, n_\eta - k_\eta\right), & 0 \leq k_\eta < n_\eta, \end{cases} \qquad U_\eta\big|_{k_\eta = 0} = 1 - \left(\frac{\delta}{|\Lambda|}\right)^{1/n_\eta}. \tag{7}$$

It admits the largest-coverage threshold with $U_\eta \leq \alpha$ and retires otherwise — a fixed-grid instance of learn-then-test (Angelopoulos et al., 2025). The closed form on the right is the operative constraint: at the registered $\alpha = 0.05$, $\delta = 0.1$, $|\Lambda| = 11$ (Appendix C), a zero-violation family still needs $n_\eta \geq 92$ admitted groups, and the default output is refusal.

That floor also fixes what the $|\mathcal{K}| = 92$ primary cohorts can certify: full empirical coverage or retirement, never an intermediate coverage, since abstaining on one group leaves $U_\eta = 0.0503 > \alpha$. Because the GitHub calibration labels contain no observed violation, these certificates are support-sufficiency statements that cannot tell whether q ranks dispatch risk (Appendix F.6; $\delta = 0.05$ would need 106 groups).

Proposition 1 (Per-candidate selective-risk admission)**.** *Condition on train/development data and fix one candidate (P, H, V, q), group-level violation rule, and finite grid Λ before observing calibration. If calibration groups are i.i.d. from the future-group distribution, then with probability at least $1 - \delta$, every $\eta \in \Lambda$ satisfies $r_\eta \leq U_\eta$, where r_η is $\Pr(W^\eta = 1 \mid D^\eta = 1)$ when population admission is positive and is defined as zero otherwise. Consequently the selected $\hat{\eta}$ satisfies $r_{\hat{\eta}} \leq \alpha$ whenever $U_{\hat{\eta}} \leq \alpha$.*

Appendix A gives the proof. The guarantee is per fixed candidate and conditional on i.i.d. groups, which the cohorts assume rather than establish (Section 7). A compiler-wide guarantee requires one candidate frozen before calibration (Corollary 2), or a set of m candidates fixed before calibration and per-threshold budget $\gamma = \delta/(m|\Lambda|)$. A post hoc count after calibration-dependent stopping is insufficient; Appendix C audits the retained runs and reports both bounds.

Algorithm 1 CALIBRATE — fixed-grid exact selective admission. The score q and the grid Λ are frozen *before* this procedure sees a calibration group, which is what makes the union bound legitimate.

Input: frozen candidate $A_0 = (P, H, V, q, M)$ without threshold; calibration groups $\mathcal{K}$ with fixed replay-contract labels $L(A_0, x)$; grid Λ; budgets α, δ
Output: threshold η with a simultaneous guarantee, or $\perp$ (RETIRE)

$\Pi \leftarrow \emptyset$ ▷ admissible (coverage, threshold) pairs, distinct from Alg. 2's $\mathcal{A}$
for all $\eta \in \Lambda$ **do**
  define $d^\eta(x)$ by Eq. (3) with threshold η; $D^\eta(g) \leftarrow \max_{x \in g} d^\eta(x)$; $W^\eta(g) \leftarrow \max_{x \in g} d^\eta(x) L(A_0, x)$
  $K_\eta \leftarrow \{\, g \in \mathcal{K} : D^\eta(g) = 1 \,\}$; $n_\eta \leftarrow |K_\eta|$; **if** $n_\eta = 0$ **then continue** ▷ vacuous bound $U_\eta = 1$
  $k_\eta \leftarrow |\{\, g \in K_\eta : W^\eta(g) = 1 \,\}|$ ▷ *wrong after dispatch* only, never abstention
  $U_\eta \leftarrow 1$ if $k_\eta = n_\eta$, else $\mathrm{Beta}^{-1}\big(1 - \frac{\delta}{|\Lambda|};\, k_\eta + 1,\, n_\eta - k_\eta\big)$; **if** $U_\eta \le \alpha$ **then** $\Pi \leftarrow \Pi \cup \{(n_\eta/|\mathcal{K}|, \eta)\}$ ▷ Eq. (7)
end for
return $\perp$ if $\Pi = \emptyset$, else the η component of $\mathrm{lexmax}_{(\mathrm{cov}, \eta) \in \Pi}(\mathrm{cov}, \eta)$ ▷ largest empirical group coverage; larger threshold breaks ties

4 EXPERIMENTAL SETUP

GitHub workflows. Every sample begins with one issue or pull-request number from a pinned public snapshot (helmo, 2025); the agent must make read-only calls and return exact, source-grounded fields. The three primary families use 132 discovery records each, from whose exact traces a stable hash draws 16 train, 8 dev, and 92 calibration groups ($\mathcal{K}$; the rest stay unused), and 30 balanced held-out records, with live provider calls, each pairing a read-only evidence path with an exact answer contract (a category, a title, and a verbatim excerpt; Table 15, Appendix E). The exact-trace filter retains 130–132 of 132 records but still conditions the certificate on selection; Appendix G details this limitation.

Cross-repository extension on newer records. A narrower two-read PR-outcome task spans five frozen public repositories; held-out records are newer than discovery records but inside the same pinned snapshot, selection is sealed before provider calls, retired repositories stay in the result, and a balanced rerun on three repositories controls the class mix.

Public benchmarks. On BIRD (Li et al., 2023a) a live SQL agent runs over pinned read-only databases (Appendix F.5). NESTFUL (Basu et al., 2025) and AppWorld's gold solutions (Trivedi et al., 2024) supply executable traces, plus released AppWorld agent runs; no model runs on these two, so no efficiency claim. API-Bank and BFCL v4 (Li et al., 2023b; Patil et al., 2025) behave like NESTFUL (Appendix F.3).

Provider configuration. The primary studies use OpenAI `gpt-5.6-luna` via the `openai-agents` Responses interface (OpenAI, 2026a) at reasoning effort `low`, priced at list rates (OpenAI, 2026b); the replications use `gpt-6-luna` and Anthropic `claude-sonnet-5` under the same settings (Appendix C). Conditions within a family differ in structure alone.

Conditions, metrics, and statistics. Each primary GitHub family compares the unchanged baseline B, the compiled artifact, and a hand-written pre-model program on the same records, so comparisons are paired. Quality is exact task-contract satisfaction; resources are requests, interfaces, tokens, latency, and estimated cost; tests are paired exact McNemar (McNemar, 1947), bootstrap intervals (Efron & Tibshirani, 1993), and Wilcoxon signed-rank (Wilcoxon, 1945), with gate multiplicity controlled by the Bonferroni term of Eq. (7). A separately frozen comparison adds an adapted offline Agent Workflow Memory arm on fresh records of the same snapshot (Appendix K).

5 RESULTS

Three results carry the paper. First, on recurrent GitHub workflows the compiled agent matches the unchanged agent on held-out contracts while removing about two-thirds of model requests, and against an automated reuse baseline, adapted Agent Workflow Memory, it lowers estimated deployment cost by 36.9–79.1% within a preregistered accuracy tolerance in five of six workflow/model cells (Section 5.1). Second, on the public BIRD benchmark a live SQL agent's schema-first opening compiles on 26 of 31 databases with 45–50% fewer requests and no detected accuracy loss (Section 5.2). Third, the compiler refuses what it cannot certify: NESTFUL, API-Bank, and BFCL v4

Table 1: Primary live-provider results ($n = 30$ held-out records per family). "Interfaces" counts provider-visible tool interfaces. All conditions share records, model, and grader; *Manual* is hand-written with full workflow knowledge (Table 13, Appendix D).

	Exact held-out contract			Reduction of compiled vs. baseline (%)				
Family	Base	Compiled	Manual	Requests	Interfaces	Tokens	Latency	Cost
Issue-type routing	30/30	**30/30**	30/30	50.0	0.0	39.5	51.7	32.0
PR-outcome audit	30/30	**30/30**	30/30	75.0	66.7	80.8	73.0	75.3
Backlog-attention routing	29/30	**30/30**	30/30	74.8	66.3	81.4	68.9	75.1
Weighted total ($n = 90$)	89/90	**90/90**	90/90	66.6	44.2	63.1	64.2	58.7

Table 2: Adapted offline Agent Workflow Memory (AWM) comparison on 1,344 fresh primary records and two models. $100U$ is the simultaneous one-sided bound on GAC's accuracy loss against AWM in points (six comparisons, Bonferroni); † exceeds the registered five-point margin. USD/1k is estimated deployment cost per 1,000 attempted episodes, excluding induction (Appendix K).

			Correct of n				Per episode, GAC/ AWM		
Family	Model	n	GAC	AWM	Base	$100U$	Requests	USD/1k	Saving
Backlog	`gpt-5.6-luna`	144	143	142	144	3.27	1 / 4	0.204 / 0.806	74.7%
Backlog	`gpt-6-luna`	144	142	143	144	$5.86^{\dagger}$	1 / 4	0.088 / 0.380	76.8%
Issue triage	`gpt-5.6-luna`	600	584	583	590	3.61	2 / 4	0.588 / 0.931	36.9%
Issue triage	`gpt-6-luna`	600	583	583	582	4.04	2 / 4	0.210 / 0.411	48.9%
PR outcome	`gpt-5.6-luna`	600	597	598	597	1.43	1 / 4	0.164 / 0.739	77.8%
PR outcome	`gpt-6-luna`	600	598	600	599	1.43	1 / 4	0.074 / 0.353	79.1%

fall below the calibration floor, and AppWorld's admitted program is rarely dispatchable (Section 5.3). Certificate levels, replications, and the safeguard studies are summarized in Section 5.4 and detailed in the appendix.

5.1 Primary workflows against the unchanged agent, a hand-written program, and workflow memory

Across the 90 held-out records, GAC satisfies 90/90 exact contracts and the unchanged agent 89/90 (Table 1), reducing provider requests by 66.6%, tokens by 63.1%, latency by 64.2%, and cost by 58.7% (per-record values and bootstrap intervals in Tables 9 and 12, Appendix D); issue-type routing, a two-read prefix, is the low end of every range. These are marginal per-episode savings within one snapshot: every artifact pins its snapshot digest and abstains on any other (a new snapshot is recompiled), and discovery cost alone breaks even after 411, 183, and 181 episodes per family (Appendix D). The discordant pair is one baseline-only miss (backlog record 5189, an ungrounded excerpt) against zero compiled-only misses (McNemar $p = 1$; one-sided 95% bound on compiled-only discordance 3.3%); a rerun of the 30 records scored 30/30 in every arm (Appendix G), so no accuracy difference is detected; equivalence and noninferiority were not established. Programs hand-written with full workflow knowledge also reach 90/90 and tie on requests. Pooled across families, they use fewer interfaces and tokens and cost marginally less, while compilation has lower latency (Table 13): we claim discovery and admission, not runtime dominance.

Against an automated reuse baseline, a separately frozen three-workflow, two-model comparison with an adapted offline Agent Workflow Memory (Wang et al., 2025) attempts all 8,172 episodes on 1,362 fresh records, retaining one baseline provider failure (Table 2; Appendices J and K). GAC uses 50–75% fewer requests and 36.9–79.1% less estimated deployment cost than adapted AWM, whose induced workflow text leaves every step a model decision. Five of six cells pass the registered five-point accuracy noninferiority margin with six-comparison correction; second-model backlog is inconclusive (142/144 versus 143/144; loss bound 5.86%). Only the two PR cells pass a two-point margin; none passes one point. AWM's native browser evaluation is not reproduced, and no compiler-placed comparator was run (Appendix G).

Table 3: The two trace benchmarks with no measured GAC benefit (all four in Table 18). Stage numbers follow Fig. 1; "slots" is the share of dependency slots with a recovered producer; "at floor" counts families meeting the stage-2 support minimum ($\geq$ 5 groups on NESTFUL, $\geq$ 3 elsewhere); $n_{\max}$ is the largest family support (Eq. (7) requires 92); "p/a/w" is pass/abstain/wrong on the *family-level* replay split, not the 11/11 admission test split (Appendix F.3). AppWorld shows effect arm A.

Substrate	Traces	Slots (1)	Windows (2)	At floor (2)	Synth. (4)	Replay p/a/**w** (5)	$n_{\max}$ (6)	Adm.
NESTFUL	1,415	96.3%	1,207	32	12	24 / 12 / **0**	26	**0**
AppWorld	136	63.2%	136	33	1	34 / 0 / **0**	**136**	**1**

On `gpt-6-luna` and Anthropic `claude-sonnet-5`, re-discovery from each model's own traces on the same cohorts re-derives the identical programs on two of the three families and refuses backlog routing (on Sonnet at 91 and 90 of 92 groups, the floor binding), with four compiled-only excerpt misses on Sonnet; transferring the retained artifacts to `gpt-6-luna` keeps the structure with one compiled-only miss in 90 (Appendix G).

5.2 A public benchmark where GAC helps: BIRD

A live SQL agent on BIRD (Li et al., 2023a), a public text-to-SQL benchmark, shows where GAC helps (Appendix F.5): on the five dev databases with at least 146 questions, GAC retires all five when the agent picks its tables per question, and admits the schema-first opening on all five. On 150 sealed questions the compiled agent dispatched every time, answered 97 correctly against 94 (McNemar $p = 0.45$), and cut requests 44.9% and latency 29.4% against a warm rerun of the unchanged agent. Retrospective all-scheduled counts are 93 against 91 of 150 on a second model, and 402 against 403 of 630 on 21 of 26 training databases. Requests fall 49.6% and 45.8% on 150 and 629 completed pairs, respectively. A conservative paired 95% interval for the training difference is $[-3.27, +2.96]$ points under i.i.d. pairs; Table 25 reports database sensitivity. A second provider's agent never ran a fixed opening and was refused (Table 23).

5.3 Where GAC yields no measured benefit: NESTFUL and AppWorld

Thin evidence. NESTFUL, the largest trace corpus the compiler ingests here (1,415 traces), passes provenance, synthesis, and replay with *no* wrong execution, yet its best family reaches $n_{\max} = 26 < 92$, so every family retires at stage 6 (Table 3); API-Bank and BFCL v4 retire the same way at 8 and 15 (Table 18).

Admissible, rarely dispatchable. AppWorld's gold solutions reach $n_{\max} = 136$, and GAC admits one argument-free two-call artifact at 92/92 zero-violation groups ($U_{\hat{\eta}} = 0.0498$, compiler-wide under one effect arm, per-candidate under the other; Appendix F.3). Yet across 8,190 released trajectories it is entry-eligible in 37.9%, almost always for full-code agents and almost never for ReAct and plan-and-execute, which open by reading API documentation (Table 19). Allowing that read as a prologue recovers only 2 more of 4,680 (Appendix G).

5.4 Certificates, replications, and further studies

What is certified (Appendix F.6). Each primary artifact was admitted at 92/92 zero-violation calibration groups ($U_{\hat{\eta}} = 0.0498$, conditional on i.i.d. groups), a support-sufficiency certificate that is compiler-wide for issue-type routing (one candidate at calibration) and per-candidate for PR-outcome and backlog routing (two; corrected $U = 0.0569$, so compiler-wide only at $\alpha \geq 0.057$; Corollary 2); Table 26 lists every admitted artifact, including the one end-to-end certificate, from a post-cutoff PR-outcome cohort on which the re-derived artifact dispatched on 132 fresh groups and missed 0 task contracts (one-sided bound 0.0173 under the single pre-registered rule, 0.035 under the primary grid). What the statistical stage adds at runtime is measurable, and null on these candidates: full statistical admission, support-only admission, and deterministic guards give identical routing and evidence on all 2,724 record–model cases of the AWM comparison, and the gate's contribution is its sample floor, which is what refuses the thin benchmarks of Section 5.3 and, on a second provider, backlog routing.

Newer records within a snapshot (Appendix F). On a two-read PR-outcome task over five frozen repositories, GAC completes 120/120 exact contracts across four and retires `pytorch/pytorch` at admission (no calibration group accepted at any threshold); open-class records abstain on the induced `pr.state` hull, so requests fall 44.4% (66.7% where dispatched), and a class-balanced rerun with `pytorch/pytorch` reaches 180/180 at 66.7% fewer requests.

Live recovery (Appendix I). A four-arm study of a three-comment prefetch with equal recovery tools removes request-scope admission (the entry hull on the requested comment limit) or completeness checking (the verifier clause that the prefetched page covers that request) one at a time over 2,304 scheduled episodes. On 72 confirmation issue groups with constructed requests the scope guard raises success from 186/216 to 205/216 (+8.80 points, descriptive 95% interval $[3.70, 13.89]$); removing the completeness check yields 204/216 versus 205/216 under capped responses (the agent repairs 22/27 incomplete prefetches), so its benefit is unresolved; and full does not beat baseline: mechanism evidence under hand-specified contracts, not a live-answer certificate.

6 Related Work

Trace compilation and specialization. Beyond the trace-compilation precedent of the introduction, GAC takes meta-tracing's unit of specialization, the trace (Bolz et al., 2009). Partial evaluation is the older frame: a specialized program is correct only where its static assumptions hold (Futamura, 1971; Jones et al., 1993); here grounded arguments, declared effects (Lucassen & Gifford, 1988), and manifest pins are the guard, and each can veto. Family mining is a narrow form of process mining (van der Aalst, 2016); synthesis is bounded programming by example and sketching (Gulwani, 2011; Solar-Lezama et al., 2006) with explicit provenance (Cheney et al., 2009); and likely invariants inferred from executions (Ernst et al., 2001) are the ancestor of the induced verifier, with the same caveat (Appendix G): an inferred hull is evidence, not proof.

Reuse, caching, and optimization of agents. AWM induces reusable workflows that guide subsequent model generations (Wang et al., 2025); GAC executes grounded prefixes without those decisions. Plan caching reuses plans (Zhang et al., 2025); EvoC2F compiles an annotated orchestration IR and tests skills before admission (Wei et al., 2026); Agent JIT generates and schedules executable plans with tool pre/postconditions (Winston et al., 2026). These are distinct optimization units, not interchangeable implementations. Other systems optimize workflows, compile plans, or prune topology (Abuzakuk et al., 2026; Li et al., 2026; Chen et al., 2026; Wang et al., 2026; Chundru, 2026); skill libraries grow reusable procedures from experience (Wang et al., 2024a; Zheng et al., 2025); semantic caches reuse responses (Bang, 2023), and prompt caching and cache-aware compression reuse shared prefixes (Lumer et al., 2026; Song, 2026); DSPy, GEPA, TextGrad, and Trace optimize prompts and programs against task feedback (Khattab et al., 2024; Opsahl-Ong et al., 2024; Agrawal et al., 2026; Yuksekgonul et al., 2025; Cheng et al., 2024); RouteLLM changes models (Ong et al., 2025), LLMCompiler parallelizes calls (Kim et al., 2024), and Yue et al. (2026) survey the space. Our contribution combines trace reuse with a finite-sample admission bound and exact fallback on clean rejection; the reviewed systems motivate its components rather than establishing a universal priority claim. Counterfactual trace auditing (Zhou et al., 2026), which pairs each trace with a counterfactual run of the same task, is the closest analogue of GAC's continuation check.

Selective prediction and finite-sample risk control. Dispatch-or-fall-back is selective prediction with a reject option (Chow, 1970; Geifman & El-Yaniv, 2017). The admission bound is a fixed-grid instance of learn-then-test (Angelopoulos et al., 2025) using the exact binomial interval (Clopper & Pearson, 1934); the risk-controlling prediction-set framework gives related high-probability risk guarantees (Bates et al., 2021); conformal prediction and conformal risk control give marginal-coverage and expected-loss guarantees, which differ (Vovk et al., 2005; Angelopoulos et al., 2024).

Tool-use safety and evaluation. ToolEmu and AgentDojo evaluate agents against hazardous or injected tool environments (Ruan et al., 2024; Debenedetti et al., 2024), the regime our perturbation studies probe (Appendix G). BFCL, ToolLLM, NESTFUL, API-Bank, AppWorld, ToolSandbox, and τ^2-bench score tool use broadly (Patil et al., 2025; Qin et al., 2024; Basu et al., 2025; Li et al., 2023b; Trivedi et al., 2024; Lu et al., 2025; Barres et al., 2025); GAC runs a live agent on BIRD (Li et al., 2023a) and uses four of these as compiler substrates, reporting what each refuses (Appendix F.3).

7 Discussion and Limitations

Across 810 held-out GitHub episodes run under a compiled artifact on the calibrated model in six studies, one produced a compiled-only contract failure: 1 of 270 on the excerpt-bearing contracts (one-sided bound 1.7%) and 0 of 540 on the outcome-only cross-repository contract (that subset covers 354 distinct records) (Appendix D); Appendix G pairs limitations with protocols and Appendix L maps claims to evidence.

What the certificates cover. The primary registered certificates (Table 26, Appendix F.6) are support-sufficiency statements about a replay-contract event with no observed failures here: they certify neither graded selectivity of q — the cross-repository learned and support-only arms were indistinguishable, and an amended 184-group issue study retired (Appendix H) — nor the continuation, so the compiled-only misses lie outside them; the end-to-end certificate is per-candidate and covers one family. Every bound assumes i.i.d. groups: counting a creation day or an author as one unit retires every primary family under the registered grid (Table 28; single-rule sensitivity in Appendix C), and AppWorld's bound retires at scenario or supervisor level (Appendix F.3).

External validity, guards, and comparison. Evidence spans pinned GitHub snapshots, a post-cutoff cohort and BIRD. The six-study compiled-only misses are excerpts departing from identical evidence, without a counterfactual separating model and context effects. Earlier drift studies show no protection: provider-free perturbations are null by construction, and live continuation grading is adverse (39/60 records wrong in both arms under the original grader, 40 under its correction; Appendix G). Constructed requests support scope admission, not completeness checking or contract discovery (Section 5.4). Fresh adapted AWM comparisons span three workflows and two models, but one cell does not establish five-point noninferiority; no owned backlog records and only one open PR limit branch coverage (Appendix K). Native browser and compiler-only causal comparisons remain missing. Manual authoring is unpriced; the statistical gate adds no observed runtime value on these candidates. Writes, handoffs, hosted tools, undeclared effects and removed-turn guardrails lie outside Proposition 1.

8 Conclusion

GAC saves model work without detected accuracy loss in the primary comparisons, certifies one fresh cohort's end-to-end contract, and refuses thin evidence. Request-scope checks help on constructed tasks, completeness benefit remains unresolved, and earlier drift tests show no protection. This supports bounded compilation, not universal quality preservation or learned-gate superiority.

AI Use Statement

The authors used generative AI tools to restructure and copy-edit the manuscript, suggest section organization, draft and revise paper text, critique methodological exposition and internal consistency, convert repository-grounded results into LaTeX tables, and assist in designing and implementing the experiment and analysis scripts used to produce the reported results. The reported results distinguish live-provider execution, replay, provider-free audits, and simulation, and are linked to retained artifacts. The authors are responsible for the final content, experimental design, and reported results.

Ethics Statement

The main risk in removing model-mediated decisions is over-trusting a compiled prefix, and thereby accelerating unsafe automation or deleting a guardrail evaluation that happened to run at the removed boundary. The method mitigates this by compiling only read-only prefixes, treating unknown effects, approvals, handoffs, and writes as hard barriers, and defaulting to retirement or fallback when evidence is insufficient. Empirical inputs are public GitHub records and public benchmarks; separate simulation studies are labeled. Public records can include usernames and user-authored text; no private repository or private loan records were acquired. The artifact is not a license to bypass human review, legal or compliance controls, or safety guardrails; unsupported regimes are part of the result, not exceptions to hide.

Reproducibility Statement

Algorithm 2 gives the compiler and Alg. 1 the admission protocol, Algs. 3 and 4 give provenance construction and runtime dispatch, Appendix A proves Proposition 1, and Appendix C records the registered statistical and synthesis settings. The accompanying artifact contains source manifests, raw result files, deterministic table and figure generation, and scripts for the live GitHub studies, the cross-repository extension, the BIRD SQL-agent study with its extensions, the fresh-record and live-recovery safeguard studies (Appendix I), the adapted workflow-memory and fresh-generalization comparisons (Appendices J and K), and all four compiler substrates: NESTFUL, API-Bank, executed BFCL, and executed AppWorld. Appendix F.3 states how the latter two obtain observed results, and Appendix C gives their exact provider-free entry points. Before upload, these materials are packaged by an anonymizing builder with a separate checksum manifest for shipped bytes; original repository hashes remain provenance records. The archive and upload stage require author review. The artifact pins the provider model string, the `openai-agents` and `openai` SDK versions, the effective generation settings, cache policy, the 120 s timeout and no-retry policy, the per-record condition order, concurrency, the token fields used for cost, the rate-card date, the tool and effect-catalog digests, and every split and candidate manifest (Table 4); the second-model and second-provider replications pin their model strings, the Anthropic adapter's request mapping, and their protocols the same way. The added safeguard studies retain their configurations and execution schedules in separate prospective seals (Appendix I).

References

Sami Abuzakuk, Anne-Marie Kermarrec, Rishi Sharma, Rasmus Moorits Veski, and Martijn de Vos. Optimizing agentic workflows using meta-tools. *arXiv preprint arXiv:2601.22037*, 2026. URL `https://arxiv.org/abs/2601.22037`.

Lakshya A. Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziems, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J. Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alexandros G. Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, and Omar Khattab. GEPA: Reflective prompt evolution can outperform reinforcement learning. In *International Conference on Learning Representations*, 2026. URL `https://arxiv.org/abs/2507.19457`.

Anastasios N. Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. Conformal risk control. In *International Conference on Learning Representations*, 2024. arXiv:2208.02814.

Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, Michael I. Jordan, and Lihua Lei. Learn then test: Calibrating predictive algorithms to achieve risk control. *The Annals of Applied Statistics*, 19(2):1641–1662, 2025. doi: 10.1214/24-AOAS1998. URL `https://arxiv.org/abs/2110.01052`.

Anthropic. Claude API pricing, 2026. URL `https://platform.claude.com/docs/en/about-claude/pricing`. Official documentation; accessed October 8, 2026.

Vasanth Bala, Evelyn Duesterwald, and Sanjeev Banerjia. Dynamo: A transparent dynamic optimization system. In *Proceedings of the ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, pp. 1–12, 2000. doi: 10.1145/349299.349303.

Fu Bang. GPTCache: An open-source semantic cache for LLM applications enabling faster answers and cost savings. In *Proceedings of the 3rd Workshop for Natural Language Processing Open Source Software (NLP-OSS)*, pp. 212–218, 2023. URL `https://aclanthology.org/2023.nlposs-1.24/`.

Victor Barres, Honghua Dong, Soham Ray, Xujie Si, and Karthik Narasimhan. τ^2-Bench: Evaluating conversational agents in a dual-control environment. *arXiv preprint arXiv:2506.07982*, 2025. URL `https://arxiv.org/abs/2506.07982`.

Kinjal Basu, Ibrahim Abdelaziz, Kiran Kate, Mayank Agarwal, Maxwell Crouse, Yara Rizk, Kelsey Bradford, Asim Munawar, Sadhana Kumaravel, Saurabh Goyal, Xin Wang, Luis A. Lastras, and Pavan Kapanipathi. NESTFUL: A benchmark for evaluating LLMs on nested sequences of API calls. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 33538–33547, 2025. doi: 10.18653/v1/2025.emnlp-main.1702. URL https://aclanthology.org/2025.emnlp-main.1702/.

Stephen Bates, Anastasios N. Angelopoulos, Lihua Lei, Jitendra Malik, and Michael I. Jordan. Distribution-free, risk-controlling prediction sets. *Journal of the ACM*, 68(6), 2021. doi: 10.1145/3478535. URL https://arxiv.org/abs/2101.02703. Article 43.

Carl Friedrich Bolz, Antonio Cuni, Maciej Fijałkowski, and Armin Rigo. Tracing the meta-level: PyPy's tracing JIT compiler. In *Proceedings of the 4th Workshop on the Implementation, Compilation, Optimization of Object-Oriented Languages and Programming Systems (ICOOOLPS)*, pp. 18–25, 2009. doi: 10.1145/1565824.1565827.

Yulang Chen, Haoxuan Peng, Jinyan Liu, Zichen Wen, Dongrui Liu, and Linfeng Zhang. AgentSlimming: Towards efficient and cost-aware multi-agent systems. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, pp. 30064–30086, 2026. doi: 10.18653/v1/2026.acl-long.1387. URL https://aclanthology.org/2026.acl-long.1387/.

James Cheney, Laura Chiticariu, and Wang-Chiew Tan. Provenance in databases: Why, how, and where. *Foundations and Trends in Databases*, 1(4):379–474, 2009. doi: 10.1561/1900000006.

Ching-An Cheng, Allen Nie, and Adith Swaminathan. Trace is the next AutoDiff: Generative optimization with rich feedback, execution traces, and LLMs. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2406.16218.

C. K. Chow. On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory*, 16(1):41–46, 1970. doi: 10.1109/TIT.1970.1054406.

Jagadeesh Chundru. Agentic compilation: Mitigating the LLM rerun crisis for minimized-inference-cost web automation. *arXiv preprint arXiv:2604.09718*, 2026. URL https://arxiv.org/abs/2604.09718.

C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404.

Edoardo Debenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2024. arXiv:2406.13352.

Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall/CRC, 1993. doi: 10.1201/9780429246593.

Michael D. Ernst, Jake Cockrell, William G. Griswold, and David Notkin. Dynamically discovering likely program invariants to support program evolution. *IEEE Transactions on Software Engineering*, 27(2):99–123, 2001. doi: 10.1109/32.908957.

Yoshihiko Futamura. Partial evaluation of computation process—an approach to a compiler-compiler. *Systems, Computers, Controls*, 2(5):45–50, 1971. URL https://fi.ftmr.info/PE-Museum/PE-Original-English1971.pdf.

Andreas Gal, Brendan Eich, Mike Shaver, David Anderson, David Mandelin, Mohammad R. Haghighat, Blake Kaplan, Graydon Hoare, Boris Zbarsky, Jason Orendorff, Jesse Ruderman, Edwin W. Smith, Rick Reitmaier, Michael Bebenita, Mason Chang, and Michael Franz. Trace-based just-in-time type specialization for dynamic languages. In *Proceedings of the ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, pp. 465–478, 2009. doi: 10.1145/1542476.1542528.

Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL https://proceedings.neurips.cc/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html.

Sumit Gulwani. Automating string processing in spreadsheets using input-output examples. In *Proceedings of the 38th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages*, pp. 317–330, 2011. doi: 10.1145/1926385.1926423.

Headroom Labs. Headroom: Compressing tool outputs, logs, and retrieved context before they reach the LLM. Software repository, `https://github.com/headroomlabs-ai/headroom`, 2026. Apache-2.0; self-reported 60–95% token reduction on JSON payloads and 73% on a GitHub triage workload. Accessed 7 August 2026.

helmo. GitHub issues dataset card. Hugging Face Datasets, 2025. URL `https://huggingface.co/datasets/helmo/github-issues`. Revision e344be7b84d199661a9956036991e1fc25715a47.

Urs Hölzle, Craig Chambers, and David Ungar. Debugging optimized code with dynamic deoptimization. In *Proceedings of the ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, pp. 32–43, 1992. doi: 10.1145/143095.143114. URL `https://bibliography.selflanguage.org/dynamic-deoptimization.html`.

Huiqiang Jiang, Qianhui Wu, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. LLMLingua: Compressing prompts for accelerated inference of large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023. URL `https://aclanthology.org/2023.emnlp-main.825/`.

Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik R. Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? In *The Twelfth International Conference on Learning Representations*, 2024. URL `https://proceedings.iclr.cc/paper_files/paper/2024/hash/edac78c3e300629acfe6cbe9ca88fb84-Abstract-Conference.html`.

Neil D. Jones, Carsten K. Gomard, and Peter Sestoft. *Partial Evaluation and Automatic Program Generation*. Prentice Hall, 1993. URL `https://studwww.itu.dk/people/sestoft/pebook/`.

Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan, Saiful Haq, Ashutosh Sharma, Thomas T. Joshi, Hanna Moazam, Heather Miller, Matei Zaharia, and Christopher Potts. DSPy: Compiling declarative language model calls into self-improving pipelines. In *International Conference on Learning Representations*, 2024. URL `https://proceedings.iclr.cc/paper_files/paper/2024/file/f1cf02ce09757f57c3b93c0db83181e0-Paper-Conference.pdf`.

Sehoon Kim, Suhong Moon, Ryan Tabrizi, Nicholas Lee, Michael W. Mahoney, Kurt Keutzer, and Amir Gholami. An LLM compiler for parallel function calling. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 24370–24391, 2024. URL `https://proceedings.mlr.press/v235/kim24y.html`.

LangChain. Build a SQL agent, 2026. URL `https://docs.langchain.com/oss/python/langchain/sql-agent`. Official documentation; accessed October 8, 2026.

Jinyang Li, Binyuan Hui, Ge Qu, Jiaxi Yang, Binhua Li, Bowen Li, Bailin Wang, Bowen Qin, Ruiying Geng, Nan Huo, Xuanhe Zhou, Chenhao Ma, Guoliang Li, Kevin C. C. Chang, Fei Huang, Reynold Cheng, and Yongbin Li. Can LLM already serve as a database interface? A BIg Bench for large-scale database grounded text-to-SQLs. In *Advances in Neural Information Processing Systems 36, Datasets and Benchmarks Track*, 2023a. URL `https://proceedings.neurips.cc/paper_files/paper/2023/hash/83fc8fab1710363050bbd1d4b8cc0021-Abstract-Datasets_and_Benchmarks.html`.

Junyan Li, Zhang-Wei Hong, Maohao Shen, Yang Zhang, and Chuang Gan. FlowCompile: An optimizing compiler for structured LLM workflows. *arXiv preprint arXiv:2605.13647*, 2026. URL `https://arxiv.org/abs/2605.13647`.

Minghao Li, Yingxiu Zhao, Bowen Yu, Feifan Song, Hangyu Li, Haiyang Yu, Zhoujun Li, Fei Huang, and Yongbin Li. API-Bank: A comprehensive benchmark for tool-augmented LLMs. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 3102–3116, 2023b. doi: 10.18653/v1/2023.emnlp-main.187. URL https://aclanthology.org/2023.emnlp-main.187/.

Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, Minlie Huang, Yuxiao Dong, and Jie Tang. AgentBench: Evaluating LLMs as agents. *arXiv preprint arXiv:2308.03688*, 2023. URL https://arxiv.org/abs/2308.03688.

Jiarui Lu, Thomas Holleis, Yizhe Zhang, Bernhard Aumayer, Feng Nan, Felix Bai, Shuang Ma, Shen Ma, Mengyu Li, Guoli Yin, Zirui Wang, and Ruoming Pang. ToolSandbox: A stateful, conversational, interactive evaluation benchmark for LLM tool use capabilities. In *Findings of the Association for Computational Linguistics: NAACL 2025*, 2025. doi: 10.18653/v1/2025.findings-naacl.65. URL https://aclanthology.org/2025.findings-naacl.65/.

John M. Lucassen and David K. Gifford. Polymorphic effect systems. In *Proceedings of the 15th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (POPL)*, pp. 47–57, 1988. doi: 10.1145/73560.73564.

Elias Lumer, Faheem Nizar, Akshaya Jangiti, Kevin Frank, Anmol Gulati, Mandar Phadate, and Vamse Kumar Subbiah. Don't break the cache: An evaluation of prompt caching for long-horizon agentic tasks. *arXiv preprint arXiv:2601.06007*, 2026. URL https://arxiv.org/abs/2601.06007.

Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12:153–157, 1947. doi: 10.1007/BF02295996.

Grégoire Mialon, Clémentine Fourrier, Craig Swift, Thomas Wolf, Yann LeCun, and Thomas Scialom. GAIA: A benchmark for general AI assistants. *arXiv preprint arXiv:2311.12983*, 2023. URL https://arxiv.org/abs/2311.12983.

Nebius. SWE-agent trajectories: 80,036 issue-resolution trajectories from open-weight action generators. Hugging Face dataset https://huggingface.co/datasets/nebius/SWE-agent-trajectories, 2024.

Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed Kadous, and Ion Stoica. RouteLLM: Learning to route LLMs from preference data. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/5503a7c69d48a2f86fc00b3dc09de686-Abstract-Conference.html.

OpenAI. Agents SDK guide. OpenAI Developer Documentation, 2026a. URL https://developers.openai.com/api/docs/guides/agents. Accessed 2026-08-03.

OpenAI. API pricing. OpenAI Developer Documentation, 2026b. URL https://developers.openai.com/api/docs/pricing. Accessed 2026-08-03.

Krista Opsahl-Ong, Michael J. Ryan, Josh Purtell, David Broman, Christopher Potts, Matei Zaharia, and Omar Khattab. Optimizing instructions and demonstrations for multi-stage language model programs. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 9340–9366, 2024. doi: 10.18653/v1/2024.emnlp-main.525. URL https://aclanthology.org/2024.emnlp-main.525/.

Shishir G. Patil, Huanzhi Mao, Fanjia Yan, Charlie Cheng-Jie Ji, Vishnu Suresh, Ion Stoica, and Joseph E. Gonzalez. The Berkeley function calling leaderboard (BFCL): From tool use to agentic evaluation of large language models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 48371–48392, 2025. URL https://proceedings.mlr.press/v267/patil25a.html.

Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, Sihan Zhao, Lauren Hong, Runchu Tian, Ruobing Xie, Jie Zhou, Mark Gerstein, Dahai Li, Zhiyuan Liu, and Maosong Sun. ToolLLM: Facilitating large language models to master 16000+ real-world APIs. In *The Twelfth International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2307.16789.

Yangjun Ruan, Honghua Dong, Andrew Wang, Silviu Pitis, Yongchao Zhou, Jimmy Ba, Yann Dubois, Chris J. Maddison, and Tatsunori Hashimoto. Identifying the risks of LM agents with an LM-emulated sandbox. In *International Conference on Learning Representations*, 2024. arXiv:2309.15817.

Armando Solar-Lezama, Liviu Tancau, Rastislav Bodik, Sanjit Seshia, and Vijay Saraswat. Combinatorial sketching for finite programs. In *Proceedings of the 12th International Conference on Architectural Support for Programming Languages and Operating Systems*, pp. 404–415, 2006. doi: 10.1145/1168857.1168907.

Yan Song. Cache-aware prompt compression: A two-tier cost model for LLM API caching. *arXiv preprint arXiv:2607.15516*, 2026. URL https://arxiv.org/abs/2607.15516.

Harsh Trivedi, Tushar Khot, Mareike Hartmann, Ruskin Manku, Vinty Dong, Edward Li, Shashank Gupta, Ashish Sabharwal, and Niranjan Balasubramanian. AppWorld: A controllable world of apps and people for benchmarking interactive coding agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 16022–16076, 2024. doi: 10.18653/v1/2024.acl-long.850. URL https://aclanthology.org/2024.acl-long.850/.

Maria Trofimova, Anton Shevtsov, Ibragim Badertdinov, Konstantin Pyaev, Simon Karasik, and Alexander Golubev. OpenHands trajectories with Qwen3-Coder-480B-A35B-Instruct. Nebius blog and Hugging Face dataset https://huggingface.co/datasets/nebius/SWE-rebench-openhands-trajectories, 2025.

Wil M. P. van der Aalst. *Process Mining: Data Science in Action*. Springer, second edition, 2016. doi: 10.1007/978-3-662-49851-4.

Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005. URL https://link.springer.com/book/10.1007/b106715.

Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. *Transactions on Machine Learning Research*, 2024a. URL https://arxiv.org/abs/2305.16291.

Jincheng Wang, Min Zheng, and Tao Wei. COVENANT: Natural-language workflow compilation for aligned agent execution. *arXiv preprint arXiv:2607.25400*, 2026. URL https://arxiv.org/abs/2607.25400.

Xingyao Wang, Boxuan Li, Yufan Song, Frank F. Xu, Xiangru Tang, Mingchen Zhuge, Jiayi Pan, Yueqi Song, Bowen Li, Jaskirat Singh, Hoang H. Tran, Fuqiang Li, Ren Ma, Mingzhang Zheng, Bill Qian, Yanjun Shao, Niklas Muennighoff, Yizhe Zhang, Binyuan Hui, Junyang Lin, Robert Brennan, Hao Peng, Heng Ji, and Graham Neubig. OpenHands: An open platform for AI software developers as generalist agents. *arXiv preprint arXiv:2407.16741*, 2024b. URL https://arxiv.org/abs/2407.16741.

Zora Zhiruo Wang, Jiayuan Mao, Daniel Fried, and Graham Neubig. Agent workflow memory. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 63897–63911, 2025. URL https://proceedings.mlr.press/v267/wang25bx.html.

Jason Wei, Zhiqing Sun, Spencer Papay, Scott McKinney, Jeffrey Han, Isa Fulford, Hyung Won Chung, Alex Tachard Passos, William Fedus, and Amelia Glaese. BrowseComp: A simple yet challenging benchmark for browsing agents. *arXiv preprint arXiv:2504.12516*, 2025. URL https://arxiv.org/abs/2504.12516.

Lei Wei, Qi Liu, Ruiyang Huang, Xiao Peng, Sicong Xie, Lanbo Lin, Chenhao Jiang, Yuanwu Xu, Tianyuan Yang, Jiayao Liu, Li Cai, Zhaolu Kang, and Bin Wang. EvoC2F: Compiling tool orchestration for efficient and evolvable LLM agents. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pp. 133855–133884, 2026. URL https://proceedings.mlr.press/v306/wei26q.html.

Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945. doi: 10.2307/3001968.

Caleb Winston, Ron Yifeng Wang, Azalia Mirhoseini, and Christos Kozyrakis. Agent JIT compilation for latency-optimizing web agent planning and scheduling. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pp. 135161–135186, 2026. URL https://proceedings.mlr.press/v306/winston26b.html.

John Yang, Carlos E. Jimenez, Alexander Wettig, Kilian Lieret, Shunyu Yao, Karthik Narasimhan, and Ofir Press. SWE-agent: Agent-computer interfaces enable automated software engineering. In *Advances in Neural Information Processing Systems 37*, 2024. URL https://arxiv.org/abs/2405.15793.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=WE_vluYUL-X.

Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ-bench: A benchmark for tool-agent-user interaction in real-world domains. *arXiv preprint arXiv:2406.12045*, 2024. URL https://arxiv.org/abs/2406.12045.

Ling Yue, Kushal Raj Bhandari, Ching-Yun Ko, Dhaval Patel, Shuxin Lin, Nianjun Zhou, Jianxi Gao, Pin-Yu Chen, and Shaowu Pan. From static templates to dynamic runtime graphs: A survey of workflow optimization for LLM agents. *arXiv preprint arXiv:2603.22386*, 2026. URL https://arxiv.org/abs/2603.22386.

Mert Yuksekgonul, Federico Bianchi, Joseph Boen, Sheng Liu, Pan Lu, Zhi Huang, Carlos Guestrin, and James Zou. Optimizing generative AI by backpropagating language model feedback. *Nature*, 639:609–616, 2025. doi: 10.1038/s41586-025-08661-4. URL https://www.nature.com/articles/s41586-025-08661-4.

Qizheng Zhang, Michael Wornow, and Kunle Olukotun. Agentic plan caching: Test-time memory for fast and cost-efficient LLM agents. In *Advances in Neural Information Processing Systems*, volume 38, 2025. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/9549f7d06700f0966d5f938f1d11022a-Abstract-Conference.html.

Boyuan Zheng, Michael Y. Fatemi, Xiaolong Jin, Zheng Zhuoyang Wang, Apurva Gandhi, Yueqi Song, Yu Gu, Jayanth Srinivasa, Gaowen Liu, Graham Neubig, and Yu Su. SkillWeaver: Web agents can self-improve by discovering and honing skills. *arXiv preprint arXiv:2504.07079*, 2025.

Xiaolin Zhou, Jinbo Liu, Li Li, Ryan A. Rossi, and Xiyang Hu. Counterfactual trace auditing of LLM agent skills. *arXiv preprint arXiv:2605.11946*, 2026. URL https://arxiv.org/abs/2605.11946.

A Proof of Proposition 1

Condition on the train/development data, fix $\eta \in \Lambda$, and write $\gamma = \delta/|\Lambda|$. Let D_i^η and W_i^η be the group admission and violation indicators defined in Section 2. Because the entire candidate and grid were frozen before calibration and groups are i.i.d., the pairs (D_i^η, W_i^η) are i.i.d. Conditional on $n_\eta > 0$ admitted groups, their violation count therefore satisfies $k_\eta \mid n_\eta \sim \text{Binomial}(n_\eta, r_\eta)$, where $r_\eta = \Pr(W^\eta = 1 \mid D^\eta = 1)$. Inverting the one-sided exact binomial test (Clopper & Pearson, 1934) gives, for $0 \le k_\eta < n_\eta$, the upper bound $U_\eta = \text{Beta}^{-1}(1-\gamma;\, k_\eta+1,\, n_\eta-k_\eta)$ of Eq. (7), with $U_\eta = 1$ when $k_\eta = n_\eta$; either way $\Pr\{r_\eta > U_\eta \mid n_\eta\} \le \gamma$. When $n_\eta = 0$ the event is impossible under the convention that empty-admission thresholds receive the vacuous bound $U_\eta = 1$. Averaging over the random value of n_η therefore yields $\Pr\{r_\eta > U_\eta\} \le \gamma$ for each fixed threshold. Applying the union bound over the finite grid gives $\Pr\{\exists \eta \in \Lambda : r_\eta > U_\eta\} \le |\Lambda|\gamma = \delta$, i.e. simultaneous coverage with probability at least $1 - \delta$. On that event, every threshold whose upper bound is at most α satisfies the group-level target in Eq. (4), including the threshold returned by Alg. 1. ■

Two scope conditions are worth restating. First, the statement is *per fixed candidate*: Alg. 2 may calibrate several families and retain survivors, and that outer search is not multiplicity-corrected — except when the search is frozen, per the corollary below. Second, i.i.d. source groups is an assumption about the calibration cohort, not a property the compiler verifies; the grouped split in Alg. 2 prevents within-group leakage but does not establish independence across groups or under temporal drift.

Corollary 2 (Compiler-wide guarantee under frozen selection)**.** *If Alg. 2 runs with candidate freezing enabled — ranking on train data, synthesizing and challenging on train and development windows only, and halting the search the instant any candidate reaches calibration, whatever its outcome — Proposition 1's guarantee holds for that partition's actual reported output with no adjustment to $\gamma = \delta/|\Lambda|$: no candidate multiplicity remains to correct for.*

Proof. Ranking, synthesis, the challenge test, and the frozen score model q are all functions of train and development windows alone, fixed before a single calibration window is inspected. The search reaches calibration for the top-ranked surviving candidate exactly once and halts immediately after that candidate's outcome, so no second candidate's program, guard, verifier, or score model is ever compared against calibration data. Proposition 1's precondition — one candidate fixed before observing calibration — is met by the run's entire search, not by a hypothetical single-candidate sub-procedure, so its conclusion transfers unchanged. ■ □

This is a corollary, not a new inequality: it is a restatement of Proposition 1 for a search that never lets a second candidate see calibration data. Without freezing, stopping after a target number of admissions can make the number actually tested calibration-dependent. A union bound must then budget for the candidate set fixed before calibration, not just the number eventually tested. In the retained primary runs, the train/development audit accounts for every candidate reaching calibration. The cross-repository extension runs with freezing enabled and every sealed repository report shows one candidate reaching calibration, so Corollary 2 applies to those admissions directly. Issue-type routing did not freeze, but its second candidate retired at synthesis (an ungroundable slot) and never saw calibration, so $m = 1$ and the same argument applies. PR-outcome and backlog-attention routing each calibrated $m = 2$ candidates on the same 92 groups and kept the higher-removal survivor: their certificates are per-candidate at $\alpha = 0.05$, and under $\gamma = \delta/(m|\Lambda|)$ the retained zero-violation tables give $U = 0.0569 > \alpha$, a compiler-wide certificate only at $\alpha \ge 0.057$ or after 106 groups (Table 5). The pre-registered repair, a fresh class-balanced 106-group calibration per family (`paper/supplementary/multiplicity-repair-protocol.md`), failed its own go/no-go check: after excluding every record used in a retained cohort, the pinned snapshot holds 33 unused `open` pull requests and 8 unused `owned` backlog issues against the 36 per class the design requires, so it was not run and the per-candidate wording stays.

Multiplicity accounting. For one fixed candidate and one fixed threshold η, $\Pr(r_\eta > U_\eta) \le \gamma = \delta/|\Lambda|$. For one fixed candidate over the whole grid, the union bound gives $\Pr(\exists \eta \in \Lambda : r_\eta > U_\eta) \le |\Lambda|\gamma = \delta$. For a set of m candidates fixed before calibration and each evaluated over Λ, choosing $\gamma = \delta/(m|\Lambda|)$ gives $\Pr(\exists\, a, \eta : r_{a,\eta} > U_{a,\eta}) \le \delta$; at $\alpha = 0.05$, $\delta = 0.1$, $|\Lambda| = 11$ the zero-violation requirement is 92 groups for $m = 1$ and 106 for $m = 2$ ($U = 0.0498$ versus 0.0569

Algorithm 2 COMPILE — the compile-or-retire cascade. Every candidate is either registered or assigned an attributed RETIRE, so refusal accounting remains explicit. Stages marked (CALLER) are separate library entry points run in this order rather than steps inside one function. Freezing applies per compatible partition; certifying several partitions requires an additional allocation of confidence.

Input: episodes $\mathcal{E}$; effect catalog C; entry schema Θ; transform library $\mathcal{L}$; ambiguity cap κ; mining window $w_{\max}$; budgets α, δ; grid Λ; feasibility target Δ; freeze flag f
Output: artifact set $\mathcal{A}$, or RETIRE with attributed reasons

$\mathcal{A} \leftarrow \emptyset$
$\mathcal{E} \leftarrow \textsc{Qualify}(\mathcal{E}, C)$; $\{\mathcal{E}_M\} \leftarrow \textsc{PartitionBy}(\mathcal{E}, \text{manifest, principal, isolation})$ ▷ drop partial runs, unpinned manifests
for all partitions $\mathcal{E}_M$ **do**
  $(\mathcal{T}, \mathcal{D}, \mathcal{K}, \mathcal{S}) \leftarrow \textsc{GroupedSplit}(\mathcal{E}_M)$ (CALLER) ▷ train / dev / calibration / sealed test, split by *group*
  $(G, S) \leftarrow \textsc{BuildPatg}(\mathcal{T}, \Theta, C, \mathcal{L}, \kappa)$ ▷ Alg. 3: annotate every slot
  $\mathcal{F} \leftarrow \textsc{Canonicalize}(\textsc{MineRegions}(G, S, C, w_{\max}))$, keeping only cross-group support ▷ rejects any window with an UNGROUNDED or AMBIGUOUS slot
  if $\textsc{Ceiling}(\mathcal{T}, \mathcal{F}) < \Delta$ **then** (CALLER)
    record RETIRE **(infeasible ceiling, Eq. (6))**; **continue** ▷ try the next compatible partition
  end if
  for all families $F \in \mathcal{F}$ ranked by Eq. (5) **do**
    $P \leftarrow \textsc{Synthesize}(F, \mathcal{L})$ at depth ≤ 2; **if** $P = \perp$ **then record** RETIRE **(synthesis); continue**
    $(H, V) \leftarrow \textsc{InduceContract}(F)$; **if** $\neg\textsc{ReplayAndChallenge}(P, V, \mathcal{D})$ **then record** RETIRE **(counterexample); continue**
    $q \leftarrow \textsc{FitScore}(\mathcal{D})$; **freeze** q ▷ dev groups only; never calibration
    $\eta \leftarrow \textsc{Calibrate}((P, H, V, q, M), \mathcal{K}, \Lambda, \alpha, \delta)$ ▷ Alg. 1; no candidate switch after calibration if f
    if $\eta = \perp$ **then**
      record RETIRE **(risk)**
    else
      $\mathcal{A} \leftarrow \mathcal{A} \cup \{\textsc{Package}(P, H, V, q, \eta, M)\}$
    end if
    if f **then**
      break ▷ stop this partition whether admitted or retired
    end if
  end for
end for
return $\textsc{SelectUndominated}(\mathcal{A})$ if nonempty, else RETIRE

at $n = 92$). `paper/scripts/frozen_candidate_coverage_simulation.py` checks the *single-threshold* quantity exactly, without Monte Carlo: for one candidate at one threshold and $n = 92$, the population rate that maximizes miscoverage is $r^* \approx 0.637$, where miscoverage is $\approx 0.00909 \approx \gamma$. Pooling m such candidates at the uncorrected per-candidate budget raises that single-threshold union to 1.8% at $m = 2$ and 13.6% at $m = 16$, while the corrected budget keeps it below 0.81% for every $m \ge 2$ tested ($m = 1$ sits at the 0.91% per-threshold budget by construction). These are illustrative single-threshold figures; the grid-wide and candidate-wide certificates are the δ statements above, not these numbers. A companion seeded Monte Carlo (seed `20260822`, 200,000 replicates per cell) draws the 92 groups in correlated blocks instead of i.i.d.: realized miscoverage matches the exact value at zero correlation and reaches 56% against the 0.9% nominal single-threshold budget at the most clustered configuration tested (block size 46, one quarter of blocks shifted by 0.3), which is why the live studies' `min_days=min_principals=1` configuration is this guarantee's weakest link.

B Additional Algorithms

Algorithm 3 constructs the typed provenance graph that decides Eq. (1). PATG annotates; MINEREGIONS blocks. The two blocking marks are conservative: a slot with no witness is unresolved in the bounded search and a slot with more than κ witnesses is ambiguous, so the analysis never resolves an argument by picking the cheapest available explanation. PATG keeps up to κ single-source witnesses per grounded slot and SYNTHESIZE commits to the one source stable across every supporting group.

Algorithm 4 is the runtime counterpart of Eq. (3). Cheap deterministic checks reject first and the calibrated gate runs only on what survives them, so the common case costs a registry lookup and a guard evaluation. Once an artifact resolves, the position check runs before any guard: any tool already observed in the episode, whether or not the compiled program uses it, returns the unchanged agent (Eq. (2)). Exactly one exit path compacts; every clean rejection returns the unmodified baseline agent, and any failure whose reversibility cannot be attested raises an incident rather than feigning a rollback. "Baseline" is byte-exact only where a staging owner holds the commit boundary: the outer runner can restore byte-identical model input, whereas a model-boundary adapter cannot un-emit a response the host has already committed to conversation history.

Algorithm 3 BUILDPATG — typed argument provenance. PATG annotates every argument slot and blocks nothing itself; MINEREGIONS (Alg. 2) rejects any window containing an UNGROUNDED or AMBIGUOUS slot, so a slot with no witness (unresolved in the bounded library) and a slot with too many witnesses both retire the region rather than defaulting to the cheapest explanation.

Input: episodes $\mathcal{T}$; entry schema Θ; effect catalog C; transform library $\mathcal{L}$; ambiguity cap κ
Output: provenance graph G (witness and order edges) and slot-status map S

$G \leftarrow \emptyset$; $S \leftarrow \emptyset$
for all episodes $E = (z, M, e_{1:T}, y) \in \mathcal{T}$ **do**
  $\Sigma \leftarrow \{(\text{“}z\text{”}, \pi, v) : (\pi, v) \in \text{FLATTEN}(z),\ \pi \in \Theta\}$ ▷ only allowlisted entry paths are eligible
  for all call–result pairs (c_j, o_j) in $e_{1:T}$, in temporal order **do**
    for all slot names $s \in \text{keys}(\text{args}(c_j))$ **do**
      $v_{j,s} \leftarrow \text{args}(c_j)[s]$ ▷ the slot's value, distinct from its name
      if (c_j, s) is declared literal-only in Θ **then**
        $S[j, s] \leftarrow$ LITERAL; **continue** ▷ preserve the declared constant; infer no provenance
      end if
      $\mathcal{H} \leftarrow \text{BESTPERSOURCE}\{(\sigma, g) \in \Sigma \times \mathcal{L} : g(\text{value}(\sigma)) = v_{j,s},\ \text{depth}(g) \le 2\}$
      if $\mathcal{H} = \emptyset$ **then**
        $S[j, s] \leftarrow$ UNGROUNDED ▷ unresolved in bounded search
      else if $|\mathcal{H}| > \kappa$ **then**
        $S[j, s] \leftarrow$ AMBIGUOUS
      else
        $S[j, s] \leftarrow$ GROUNDED; $G \leftarrow G \cup \{\sigma \xrightarrow{g} (j, s) : (\sigma, g) \in \mathcal{H}\}$ ▷ $1..\kappa$ single-source witnesses; synthesis commits to one
      end if
    end for
    $\Sigma \leftarrow \Sigma \cup \{(\text{var}(c_j), \pi, v) : (\pi, v) \in \text{FLATTEN}(o_j),\ \text{ELIGIBLE}(\pi, v)\}$ ▷ o_j witnesses only *later* slots
  end for
  for all event pairs (e_s, e_t) with $s < t$ sharing a resource where at least one writes it **do**
    $G \leftarrow G \cup \{e_s \prec e_t\}$ ▷ order edge in observed temporal order
  end for
end for
return (G, S)

C CONFIGURATION

Live-provider configuration. Every primary live study calls one provider and one model: OpenAI `gpt-5.6-luna` through the `openai-agents` SDK's Responses interface. Model settings are fixed across all conditions at reasoning effort `low` and verbosity `low`, with parallel tool calls disabled so provider-visible interface counts are comparable between the baseline and a compiled prefix, and server-side response storage disabled. Sampling parameters are not exposed for this model; the SDK sends none, and the effective request manifest is Table 4. The replications in Appendix G run the same harness on `gpt-6-luna` (same settings) and on Anthropic `claude-sonnet-5` through an adapter over the official SDK (adaptive thinking at effort `low`, parallel tool use disabled, structured answers through the same schemas). Reported dollars are an *estimate*, computed by applying the standard short-context list rates below (OpenAI, 2026b; Anthropic, 2026) — retrieved 2026-08-02 for `gpt-5.6-luna` and 2026-09-24 for the other two — to the token counts retained per record; they are not an account invoice and carry no negotiated or cached-tier discount beyond the cached-input line.

Rate (USD per million tokens)	`gpt-5.6-luna`	`gpt-6-luna`	`claude-sonnet-5`
Input	0.20	0.10	2.00
Cached input	0.02	0.01	0.20
Cache write	0.25	0.125	2.50
Output	1.20	0.50	10.00

Because every condition within a family shares this model, these settings, the same records, and the same grader, the conditions differ in execution structure alone.

The manifests retain requested model aliases and SDK/request settings, not provider-attested immutable weight checkpoints. Consequently these records support auditing the requests and results,

Algorithm 4 DISPATCH — boundary-time admission. Cheap deterministic checks reject first and the calibrated gate runs only on what survives them. Exactly one exit path compacts; clean rejections return the unmodified agent, while an unattested failure or a refused commit raises an incident.

Input: entry state z; runtime context $X = (M', \mathcal{O}, \text{SNAP}, \text{STAGE})$ — runtime manifest, tools already observed in this episode, optional snapshot provider, staging owner; registry $\mathcal{R}$; effect catalog C; mode $m \in \{\text{off}, \text{shadow}, \text{live}\}$
Output: observations for the region, FALLBACK to baseline agent B, or INCIDENT

if $m = \text{off}$ **then**
  return FALLBACK ▷ conformance: model-visible input must be byte-identical
end if
$A \leftarrow \mathcal{R}.\text{RESOLVE}(X.M'.\text{compatibility}, X.M'.\text{partition})$
if $A = \bot \lor A.\text{lifecycle} \neq \text{ACTIVE} \lor (\text{SIGNATUREREQUIRED} \land \neg \text{VERIFYSIGNATURE}(A))$ **then**
  return FALLBACK
end if
if $X.\mathcal{O} \neq \emptyset$ **then**
  return FALLBACK ▷ Eq. (2): any earlier tool call, by any tool, leaves position 0
end if
if $H_A(z, X.M') \neq 1$ **then**
  return FALLBACK ▷ manifest pins, isolation keys, typed hulls, effect set
end if
if $m = \text{live} \land \exists f \in \text{tools}(A.P) : C(f).\textit{quota_attested} \land X.\text{SNAP} = \bot$ **then**
  return FALLBACK ▷ a quota-bearing read cannot be discarded cleanly without a snapshot
end if
if $q_A(z) > \eta_A$ **then**
  return FALLBACK ▷ calibrated abstention on an uncertain input
end if
if $m = \text{shadow}$ **then**
  log would-dispatch; **return** FALLBACK ▷ observe coverage without changing behavior
end if
$S \leftarrow X.\text{STAGE}.\text{BEGIN}(X.\text{SNAP})$; $r \leftarrow \text{INTERPRET}(A.P, z, \text{FACADE}(C))$ ▷ the facade re-checks effects at execution time
if r raised a pre-commit error **then**
  return FALLBACK if $S.\text{ABORTANDATTEST}()$, else INCIDENT
end if
if $V_A(r) \neq 1$ **then**
  return FALLBACK if $S.\text{ABORTANDATTEST}()$, else INCIDENT ▷ live-outs, effect multiset, and call count are checked before use
end if
return r.observations if $S.\text{COMMIT}(r.\text{effects})$ succeeds, else INCIDENT ▷ commit refuses any effect outside the read classes; a refused commit is never a fallback

but cannot guarantee that a later request to the same alias reproduces the same model. Omitted decoding fields mean provider defaults were used; they do not establish which settings the provider exposes. Historical list prices estimate cost, not billed spend.

Admission budgets. The primary registered-grid admission runs use risk budget $\alpha = 0.05$, confidence budget $\delta = 0.1$, and a fixed grid of $|\Lambda| = 11$ thresholds, which fixes the zero-violation sample requirement at $n_\eta \geq \ln(\delta/|\Lambda|)/\ln(1-\alpha) = 91.6$, i.e. 92 admitted groups; at exactly $n_\eta = 92$ and $k_\eta = 0$ the bound is $U_\eta = 1 - (0.1/11)^{1/92} = 0.0498 \leq \alpha$. Tightening the confidence budget to $\delta = 0.05$ would raise the requirement to 106 groups, beyond these 92-group pools. The later 132-group end-to-end study uses a prospectively frozen single rule; exploratory runs with other budgets are labeled separately. A Bonferroni repair over $m = 2$ candidate families uses $\gamma = \delta/(m|\Lambda|)$ and lands on the same 106-group requirement, which is why PR-outcome and backlog routing carry per-candidate certificates while issue-type routing and the frozen cross-repository families are compiler-wide (Table 5). The grid is $\Lambda = \{0.02, 0.05, 0.08, 0.11, 0.14, 0.17, 0.20, 0.25, 0.30, 0.40, 0.50\}$ and the ambiguity cap is $\kappa = 3$. Synthesis uses the 23-operator closed library of Table 7 at depth ≤ 2; region mining uses $w_{\max}$ tool calls per window and requires support across independent scenario groups. Every bound in this paper is conditional on i.i.d. calibration groups.

Table 4: Effective live-provider request manifest. The companion JSON traces each row to a retained `run` block or source constant. Cache and cost fields are the ones the price formula consumes.

Key	Value
Provider / model	OpenAI `gpt-5.6-luna` via the `openai-agents` Responses interface
SDK versions	`openai-agents 0.19.2`, `openai 2.52.0`, Python 3.14.4, macOS-26.5.2 arm64
Model settings	reasoning effort `low`, verbosity `low`, `parallel_tool_calls=False`, `store=False`
Sampling	temperature, top-p, and seed are omitted in the retained requests
Turn and time limits	`max_turns` 8 (families) / 6 (cross-repository); 120 s per episode; no retries; failures kept in `failures[]`
Concurrency	discovery 6 (PR, backlog) / 8 (issue-type, cross-repository); held-out arms one at a time per record (PR, backlog, cross-repository); blocked Latin design, 8-way (issue-type)
Condition order	per-record permutation cycle over all six arm orders, five records per order
Cache accounting	fields `input_tokens`, `cached_tokens`, `cache_write_tokens`, `output_tokens`; no shared server state; cached tokens 0 in PR/backlog/cross-repository, non-zero in issue-type
Pricing	USD per 1M tokens: input 0.20, cached 0.02, cache write 0.25, output 1.20 (list rates, 2026-08-02); cost in USD $= 10^{-6}\big[(\text{input} - \text{cached} - \text{write}) \cdot 0.20 + \text{cached} \cdot 0.02 + \text{write} \cdot 0.25 + \text{output} \cdot 1.20\big]$
Data pins	`helmo/github-issues@e344be7b` parquet sha256 `09453eef…` (tracked); four cross-repository parquets pinned by sha256 in `source_manifest.json`
Tool and effect pins	per-family `effect_catalog_version` digest, `tools_hash`, `prompt_hash`, `policy_hash`, `entry_contract_version`
Run timestamps	one per study: 2026-08-04, 2026-08-05, 2026-08-05 (families); 2026-08-07 (core); 2026-08-22 (gate frontier); per-record timestamps not retained

Admission versus completed compaction. The indicator d_A counts an admitted attempt. Historical runtime tables use "dispatched" for successful compiled execution after verification; clean post-admission failures are reported as fallbacks. These completion counts must not replace n_η, which also includes clean aborted attempts.

Calibration labels as computed. For the replay-calibrated artifacts, Alg. 1 receives labels produced by clean replay of the frozen program over each calibration group's recorded tool results (`grc/compile.py`, `_calibration_samples`). A group whose entry state fails the frozen hard guard is ineligible and never enters n_η. An eligible group enters n_η at every η its score permits, whatever happens next. If the interpreter fails or the verifier rejects, the group is *unproductive*: a clean abstention that stays in n_η with $L = 0$ and would run the unchanged agent at runtime. Only when the program completes and the verifier accepts is each live-out compared with the recorded result under the tool's declared equivalence; a missing or non-equivalent output is the violation counted in k_η. The downstream continuation is never executed in this path, so an answer-level miss after a clean region cannot enter k_η; it is measured on held-out records instead (Section 5.1, Appendix D). Across the retained pools every admitted artifact has $k_\eta = 0$; verifier abstentions do occur in some pools and are reported as unproductive, not as violations. The incident branch of L (a dirty failure after a commit) cannot arise in replay calibration of a read-only region and is enforced at runtime by staging (Alg. 4).

Retrospective single-rule sensitivity; registered certificates unchanged. Because the admitted artifacts in this sensitivity analysis have coverage 1, the operative rule on these cohorts is "dispatch every hard-guard-eligible group", and the 11-point grid buys nothing but its Bonferroni factor. Table 6 recomputes every retained bound as if a single acceptance rule had been pre-registered instead, so the confidence budget is split only across the m candidates that reached calibration ($\gamma = \delta/m$). Under that rule the zero-violation floor is 45 groups for $m = 1$ and 59 for $m = 2$ (77 and 93 with one violation), the $n = 92$ bounds are 0.0247 and 0.0320, and every primary and cross-repository artifact would be compiler-wide at $\alpha = 0.05$; the cluster-level bounds at distinct days admit everywhere and at distinct authors admit everywhere except `streamlit/streamlit` (25 authors, 0.0880). This rule was *not* registered for these runs, so the certificates in Table 5 stand as printed; the table is a sensitivity analysis, not a repair, and the author- and day-level columns inherit every caveat of Table 28: distinct counts do not establish independent clusters, and a cluster bound targets a different population from the record-level one. A single frozen rule is adopted only prospectively, for studies whose candidates,

Table 5: Certificate level of every primary artifact. m is the number of candidates that reached calibration, fixed conditional on train and development data; U is the zero-violation upper bound at $n = 92$ under the per-candidate budget $\gamma = \delta/|\Lambda|$ and under the candidate-wide budget $\gamma = \delta/(m|\Lambda|)$. Rows are recomputed from the retained per-threshold tables (`compiler.artifact.gate.notes`) of the admitted artifacts. A provider-free recompilation from the sealed discovery checkpoints reproduces both admitted artifacts (identical ids, split digests, and bounds) and recovers the dominated two-read candidates' tables, which are also 92/0 at every $\eta \geq 0.14$ with $U = 0.0498$ (`paper/results/iclr_revision/recompiled_candidates_*.json`).

Family	m	n_η/k_η	$U_{\delta/\|\Lambda\|}$ / $U_{\delta/(m\|\Lambda\|)}$	Certificate at $\alpha = 0.05$	Groups needed
Issue-type routing	1	92/0	0.0498 / 0.0498	compiler-wide	92
PR-outcome audit	2	92/0	0.0498 / 0.0569	per-candidate; compiler-wide at $\alpha \geq 0.057$	106
Backlog-attention routing	2	92/0	0.0498 / 0.0569	per-candidate; compiler-wide at $\alpha \geq 0.057$	106
Cross-repository (7 admitted)	1	92/0	0.0498 / 0.0498	compiler-wide (frozen search)	92

Table 6: Retrospective single-rule sensitivity. Each row recomputes the retained all-eligible calibration table (n/k from the widest row of `compiler.artifact.gate.notes`) under one pre-registered acceptance rule with $\gamma = \delta/m$; "registered" is the grid certificate as printed. Day and author columns apply the same rule to the distinct-unit counts of Table 28 (count in parentheses). Hypothetical bounds for the already selected artifacts; not new certificates.

Cohort	m	n/k	U registered	U single, δ/m	single, days	single, authors
Issue-type routing	1	92/0	0.0498	0.0247	0.0253 (90)	0.0277 (82)
PR-outcome audit	2	92/0	0.0498	0.0320	0.0342 (86)	0.0487 (60)
Backlog-attention routing	2	92/0	0.0498	0.0320	0.0338 (87)	0.0402 (73)
Core: `huggingface/datasets`	1	92/0	0.0498	0.0247	0.0284 (80)	0.0488 (46)
Core: `pandas-dev/pandas`	1	92/0	0.0498	0.0247	0.0247 (92)	0.0425 (53)
Core: `psf/requests`	1	92/0	0.0498	0.0247	0.0253 (90)	0.0287 (79)
Core: `streamlit/streamlit`	1	92/0	0.0498	0.0247	0.0261 (87)	0.0880 (25)

acceptance policy, grouping and endpoint are fixed before fresh calibration; the post-cutoff study of Appendix G did so after this analysis and reports its bound under both rules. The symmetric pessimistic alternatives are not in the table: $\delta = 0.05$ raises the floor to 106 groups, and month-level units retire every cohort (`paper/results/iclr_revision/effective_units.json`).

Runtime capability vocabulary. `quota_attested` is a boolean field of a tool's signed effect declaration, not a capability flag: it marks a read whose invocation advances a counter (a rate quota or an audit log) that the staging snapshot must attest over before an attempt can be discarded as clean, so in live mode such a tool is not dispatched without a snapshot provider (Alg. 4). Position 0 of Eq. (2) is identified at runtime with "no tool observation has been committed in this episode": any earlier tool call, whether or not the compiled program uses that tool, returns the unchanged agent, the runtime form of the compile-time `prefix_only` constraint. In the shipped facade every effect outside {PURE, READ-LOCAL, READ-EXTERNAL} is refused at commit and raises an incident; no post-commit path is reachable in the reported runs, and Alg. 4's commit line is that contract.

Bindings have grammar $b ::= \text{Const}(v) \mid s \mid f_1(s) \mid f_2(f_1(s))$, with one source path s and type-compatible composition. An invariant constant must agree across supporting training windows; expressions are checked across windows and by group-wise refitting. Parameters come from bounded defaults, observed literals/paths, and templates inferred from the target; they are not unrestricted constants. Identity is represented by an empty chain; `bool` is executable but is not proposed by the default candidate generator. The default search caps depth at two, visited nodes at 20,000, and the frontier at 96. Consequently, an absent witness means unresolved in this search, not impossibility

Table 7: Closed transform library $\mathcal{L}$, version `T-v1`. These 23 operator implementations bound expressiveness; the value-directed search proposes a subset. Invalid types, missing paths, or invalid parameters reject a chain. S denotes scalar, N numeric (excluding Boolean), T string, L list/tuple, and A any supported value.

Operators	Input → output and restrictions
`id`; `str`; `bool`	$A \to A$; $S \to T$; $S \to$ Bool. Containers cannot be coerced to string or Boolean.
`int`; `float`	Convertible T or $N \to N$; Boolean input rejected.
`lower`, `upper`, `strip`	$T \to T$.
`split(sep)[i]`; `join(sep)`	$T \to T$ at a valid index; $L[T] \to T$.
`fmt(template)`	$S \to T$; exactly one placeholder.
`add(c)`, `mul(c)`, `round(k)`	$N \to N$; bounded numeric parameters.
`sum`; `len`	$L[N] \to N$ (empty sum is zero); $T, L,$ Dict $\to$ Int.
`project(path)`	Dict or $L \to A$; select path from each element; a singleton list projection returns its value.
`filter(path,op,c)`	$L \to L$; retain elements satisfying the comparison.
`first`, `last`	Nonempty $L \to A$.
`sort(path)`; `topk(path,n)`	$L \to L$; comparable keys, ascending sort or descending top-n.
`date_fmt(pattern)`	Parseable date string $T \to T$.

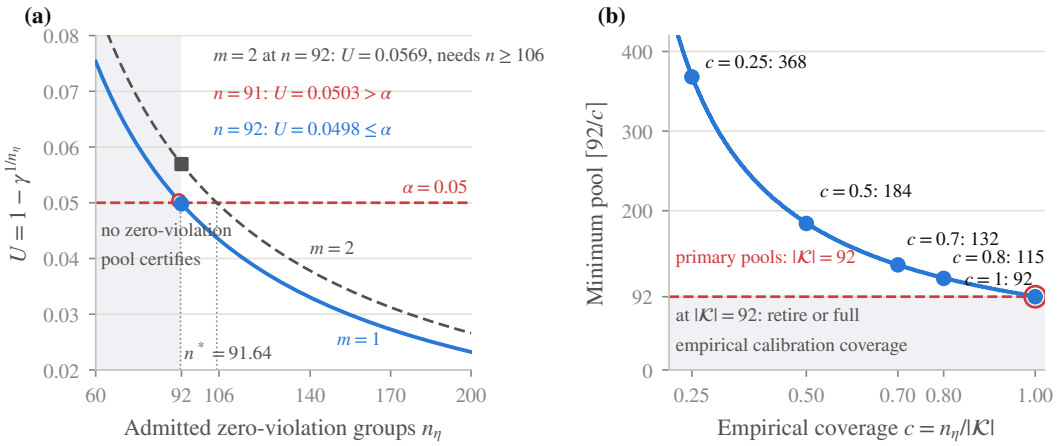


Figure 3: Why a 92-group pool certifies only full coverage. (a): the zero-violation bound $U(n) = 1 - (\delta/|\Lambda|)^{1/n}$ at $\delta = 0.1$, $|\Lambda| = 11$ crosses $\alpha = 0.05$ between $n = 91$ (0.0503) and $n = 92$ (0.0498). (b): the smallest pool $\lceil 92/c \rceil$ that can certify selective empirical coverage c. The primary pools contain 92 groups, so only retirement or full calibration coverage is possible. Later studies use 184 or 192 groups; the post-cutoff certificate uses 132 under a single rule.

in the full library. The library version is pinned: adding operators changes both coverage and the chance of coincidental matches. Implementation: `grc/dsl.py` under the package source tree.

The nonconformity score q: implemented training procedure. The compiler fits feature hulls on training windows and a logistic model on *development rows*, not group-aggregated labels. With standardized feature vector $\tilde{f}_i$, label y_i, and $p_i = \sigma(w^\top \tilde{f}_i + b)$, it minimizes

$$\tfrac{1}{2}\|w\|_2^2 + C \sum_i a_i [-y_i \log p_i - (1 - y_i) \log(1 - p_i)], \qquad C = 1,$$

where $a_i = N/(2N_{y_i})$ balances the two classes. It uses scikit-learn `LogisticRegression`, default L2 penalty and `lbfgs` solver, `max_iter=500`, and the registered seed. Development means and standard deviations standardize features; scales below 10^{-9} become one. The helper also computes up to five group-separated out-of-fold predictions, but the compiler discards them: the final development

fit scores clean calibration rows. For one observed class, weights are zero and the intercept is −6 or +6; for mixed labels in only one group, the score is clipped empirical prevalence.

Table 8: The seven implemented score features. Names are retained to make the source auditable; `knn_distance` does not compute nearest neighbors.

Feature	Definition
`unseen_category`	Fraction of enum-hull paths with an unseen nonmissing value.
`knn_distance`	Duplicate of `unseen_category`.
`missing_optional`	Fraction of guard paths resolving to missing/null, despite the historical name.
`hull_margin`	Maximum range-normalized numeric excess or regex/length violation, clipped to one.
`drift_days`	Indicator that the supplied day exceeds the maximum training day; zero if unavailable.
`provenance_ambiguity`	Candidate constant $\min(1, n_{\text{alternative bindings}}/50)$.
`branch_entropy`	Candidate constant: training-family branch entropy.

A clean development row is positive when execution fails, the verifier rejects, or verified output is missing/different from the recorded result. Guard eligibility is stored separately and does not set this label. Each window also attempts one hostile development variant; a returned variant is labeled by execution/verifier failure or an invariant perturbation's verified semantic mismatch. Its `hull_margin` and `provenance_ambiguity` are each increased by 0.5 and clipped to one. These synthetic increments are unavailable at dispatch, so the augmented training distribution need not represent runtime risk. Earlier challenge rejection does not imply that every later training label is zero. Calibration excludes these perturbations and counts only verified replay mismatches among admitted groups. Thus score training and the certified endpoint differ; the exact bound requires a frozen score, not a well-calibrated logistic probability. This implementation, including the redundant feature, is retained for reproducibility rather than repaired after seeing results (`grc/compile.py`, `grc/calibrate.py`).

What cohort size permits the gate to certify. The zero-violation floor $n_\eta \geq 92$ constrains coverage, not just admission. Certifying selective coverage $c = n_\eta/|\mathcal{K}|$ requires $|\mathcal{K}| \geq \lceil 92/c \rceil$: 92 groups for $c = 1$, 115 for $c = 0.8$, 132 for $c = 0.7$, 184 for $c = 0.5$. The primary and earlier cross-repository cohorts fix $|\mathcal{K}| = 92$, the smallest satisfiable pool. Later studies use 184 (issue recovery) or 192 (mechanism ablation) groups, without demonstrating a useful graded gate. At 92, retirement or full empirical calibration coverage are the only outcomes: abstaining on one group leaves $n_\eta = 91$ and $U_\eta = 1-(0.1/11)^{1/91} = 0.0503 > \alpha$. The step-like gate profiles reported in Table 27 and Appendix H are therefore consistent with two different worlds — a score that cannot rank, and a cohort too small to certify a ranking — and these experiments do not separate them. Two intermediate coverages were *observed* and neither is certifiable: `huggingface/datasets` accepts 10/92 groups at $\eta \leq 0.08$ in the gate-frontier study ($U = 0.375$), and `pandas-dev/pandas` accepts 90/92 at $0.08 \leq \eta \leq 0.20$ in the balanced rerun ($U = 0.0509$). A frontier study must first buy calibration groups (Fig. 3). The implementation also supports a minimum-distinct-days requirement, which the single-snapshot live study does not exercise (`min_days` = 1). Two terms of Eq. (5) are coarser in the implementation than the notation suggests: ρ_{eff} is approximated by the fraction of tool names containing a namespace separator rather than read from the effect catalog, and $|F|$ is substituted by the argument-slot count of the first observed window because ranking precedes synthesis. Both affect candidate ranking and thus which candidate is tested, but neither enters the risk bound. The effect catalog is checked during qualification and in the runtime facade.

The two executable-gold external substrates are provider-free after their pinned sources are prepared. Their exact compiler entry points are:

```
B=paper/scripts/bfcl_compiler_benchmark.py
python "$B" --source-root "$BENCHMARK_SOURCE_ROOT"
A=paper/scripts/appworld_compiler_benchmark.py
python "$A" --source-root "$BENCHMARK_SOURCE_ROOT" \
  --appworld-python "$APPWORLD_VENV/bin/python"
D=paper/scripts/appworld_dispatch_preflight.py
```

Table 9: Absolute per-record resources behind the percentages of Table 1: mean / median over the 30 held-out records of each family and over all 90. Estimated dollars are list-price estimates (Appendix C); the entire 90-record compiled evaluation cost an estimated $0.028 against $0.068 for the baseline. Cells are mean / median.

Family	Arm	Requests	Interfaces	Tokens	Latency (ms)	Cost (USD×10^{-4})
Issue-type routing	Baseline	4.00	3.00	4,259.4 / 4,320	6,160 / 5,402	8.72 / 8.74
	Compiled	2.00	3.00	2,576.5 / 2,602	2,973 / 2,881	5.93 / 5.74
	Manual	2.00	1.00	1,780.8 / 1,734	3,251 / 2,856	5.45 / 5.49
PR-outcome audit	Baseline	4.00	3.00	2,718.2 / 2,676	5,329 / 4,839	6.79 / 6.66
	Compiled	1.00	1.00	521.5 / 462	1,440 / 1,297	1.68 / 1.51
	Manual	1.00	1.00	522.2 / 458	1,827 / 1,364	1.69 / 1.51
Backlog-attention routing	Baseline	3.97	2.97	2,844.9 / 2,816	6,428 / 5,163	7.29 / 7.10
	Compiled	1.00	1.00	529.9 / 468	2,000 / 1,557	1.82 / 1.62
	Manual	1.00	1.00	535.5 / 479	1,695 / 1,501	1.88 / 1.70
All (n = 90)	Baseline	3.99	2.99	3,274.2 / 2,907	5,972 / 5,163	7.60 / 7.42
	Compiled	1.33	1.67	1,209.3 / 615	2,137 / 1,957	3.14 / 2.05
	Manual	1.33	1.00	946.2 / 616	2,258 / 1,913	3.01 / 2.18

Table 10: Per-record paired differences (compiled minus baseline) for all three families, with 10,000-sample bootstrap intervals (seed 20260802) and two-sided Wilcoxon signed-rank tests over $n = 30$ paired held-out records (exact where differences are tie-free, tie-corrected normal approximation otherwise). Latency is reported as the mean and as the median paired difference, each with its interval; cost intervals are in USD. Provider-request differences are deterministic by construction (every pair removes the same number of turns), so no rank test is reported for them: the statistic would measure the degeneracy rather than the effect. The backlog baseline's one three-request record (5189) is the only exception and appears in Table 11.

Family	Endpoint	Paired difference (95% CI)	p
Issue-type	Total tokens	$-1,682.9$ $[-1,772.5, -1,596.7]$	1.9×10^{-9}
	Wall latency (mean)	$-3,187$ ms $[-3,758, -2,660]$	1.9×10^{-9}
	Wall latency (median)	$-2,716$ ms $[-3,532, -2,309]$	—
	Estimated cost	$-\$2.79 \times 10^{-4}$ $[-3.13, -2.41] \times 10^{-4}$	3.7×10^{-9}
PR-outcome	Total tokens	$-2,196.7$ $[-2,211.1, -2,182.5]$	1.7×10^{-6}
	Wall latency (mean)	$-3,889$ ms $[-4,499, -3,346]$	1.9×10^{-9}
	Wall latency (median)	$-3,476$ ms $[-3,736, -3,245]$	—
	Estimated cost	$-\$5.11 \times 10^{-4}$ $[-5.22, -5.02] \times 10^{-4}$	1.7×10^{-6}
Backlog-attention	Total tokens	$-2,315.0$ $[-2,366.0, -2,224.1]$	1.7×10^{-6}
	Wall latency (mean)	$-4,428$ ms $[-5,400, -3,558]$	1.9×10^{-9}
	Wall latency (median)	$-3,619$ ms $[-4,201, -3,091]$	—
	Estimated cost	$-\$5.47 \times 10^{-4}$ $[-5.66, -5.23] \times 10^{-4}$	1.7×10^{-6}

```
python "$D" --source-root "$BENCHMARK_SOURCE_ROOT"
```

The AppWorld environment uses Python 3.12 with package version `0.1.3.post1` and data version `0.1.0`; the signed source manifest and requirements file pin the remaining acquisition inputs. The BFCL source is pinned to revision `6ea57973c7a6`. These commands regenerate only aggregate, non-sensitive evidence; no model or provider call is made.

D Paired Statistics for the Primary Families

Latency validity. Table 10 gives the paired differences behind Table 1. Latency is wall-clock provider evidence and is secondary to requests and tokens. In the PR-outcome and backlog families each held-out record ran its three arms one at a time on one host, cycling through all six arm orders (five records per order), with no warm-up, a 120 s per-episode timeout, and no retries; the issue-type family used a blocked six-permutation Latin design with eight-way concurrency within each block. No server-side state is shared between arms (`store=False`); cached-input tokens are zero on every PR-outcome and backlog record and are priced separately where they occur

Table 11: Exact-contract discordance per family and pooled. "Baseline-only" counts records only the unchanged agent missed; "compiled-only" records only the compiled arm missed. The upper bound is the one-sided 95% Clopper–Pearson bound on the compiled-only count (0/90 pooled).

Family	n	Both pass	Baseline-only	Compiled-only	McNemar p	Upper bound
Issue-type	30	30	0	0	1.00	9.5%
PR-outcome	30	30	0	0	1.00	9.5%
Backlog-attention	30	29	1	0	1.00	9.5%
Pooled	90	89	1	0	1.00	3.3%

Table 12: Percentage reductions of Table 1 with 95% bootstrap intervals on the ratio of paired sums (10,000 record resamples, seed 20260804). Per-family request reductions are structural (backlog's one three-request baseline record aside); the intervals on tokens and cost are within five points of the estimate, latency within ten. Discovery cost alone breaks even after 411, 183, and 181 compiled episodes per family and per snapshot (`paper/results/cache_accounting.json`); the issue-type baseline's higher cached-input share is why its cost reduction trails its token reduction.

Family	Requests	Tokens	Latency	Cost
Issue-type routing	50.0 [50.0, 50.0]	39.5 [38.7, 40.3]	51.7 [46.7, 56.4]	32.0 [28.3, 35.3]
PR-outcome audit	75.0 [75.0, 75.0]	80.8 [79.1, 82.5]	73.0 [68.2, 77.1]	75.3 [73.3, 77.2]
Backlog-attention routing	74.8 [74.4, 75.0]	81.4 [79.5, 82.9]	68.9 [61.4, 75.3]	75.1 [73.0, 76.9]
Pooled (n = 90)	66.6 [64.1, 68.9]	63.1 [58.7, 67.8]	64.2 [60.4, 67.9]	58.7 [54.1, 63.3]

(`paper/results/cache_accounting.json`). Recomputing the compiled-versus-baseline wall-latency reduction within each of the six arm orders gives reductions between 43.2% and 78.6% across the eighteen permutation–family cells (every cell positive), and provider-response latency reproduces the wall-latency reductions to within 0.3 percentage points, showing consistency across order blocks while not excluding service variance.

Table 13: The hand-written comparator priced on the same resource axes as Table 1, as percentage reduction against the same baseline and the same held-out records. *Manual* is written with full knowledge of the workflow. Pooled over the three families, it ties on provider requests, uses fewer interfaces and tokens, and costs slightly less; compilation has lower pooled wall latency (bold). Per-family token and cost comparisons vary. The interface gap is concentrated in issue-type routing, where the compiled prefix covers two reads but the agent still issues the third. We therefore claim automated discovery and guarded admission, not runtime dominance over code written by someone who already knows the workflow.

Condition	Requests	Interfaces	Tokens	Latency	Cost
Issue-type routing					
Compiled	50.0	0.0	39.5	**51.7**	32.0
Manual	50.0	66.7	58.2	47.2	37.5
PR-outcome audit					
Compiled	75.0	66.7	80.8	**73.0**	75.3
Manual	75.0	66.7	80.8	65.7	75.2
Backlog-attention routing					
Compiled	74.8	66.3	81.4	68.9	75.1
Manual	74.8	66.3	81.2	**73.6**	74.2
Weighted total (n = 90)					
Compiled	66.6	44.2	63.1	**64.2**	58.7
Manual	66.6	66.5	71.1	62.2	60.4

An extended issue-type held-out cohort. A pre-registered extension (`paper/supplementary/issue-type-extended-heldout-protocol.md`) ran the retained issue-type artifact on 120 further unused records of the pinned snapshot on the calibrated model, balanced 40/40/40 over bug,

Table 14: Extended issue-type held-out cohort on the calibrated model: per-record means over 120 unused records with a disclosed class mix (40 `bug` / 40 `enhancement` / 40 `other`). The compiled-only miss (record 2737) and the pooled bound are the pre-registered primary endpoints.

Arm	Exact	Requests	Tokens	Latency (s)	Cost (¢)
Unchanged agent	118/120	4.00	4,381	6.70	0.090
Compiled two-read prefix (GAC)	118/120	2.00	2,649	3.85	0.061
Hand-written bundle	117/120	2.00	1,813	3.68	0.055
Compiled-only failures: 1/120 here (one-sided 95% bound 3.9%); 1/210 pooled with the primary cohorts (bound 2.2%)					

Table 15: Primary GitHub benchmark at a glance. Each sample supplies one record number from a pinned public snapshot; every listed answer field is checked against the source record.

Task	Read-only evidence path	Exact answer contract
Issue type	`issue_number` → record, official labels, comments	Title, state, label-derived category, verbatim excerpt
PR outcome	`pr_number` → PR record, merge status, discussion	`open`, `merged`, or `closed_unmerged`, plus title and comment excerpt
Backlog attention	`issue_number` → open-issue record, assignees, discussion	`owned`, `discussed_unowned`, or `awaiting_first_response`, plus title, owner, and excerpt

`enhancement`, and `other` (the snapshot holds 16 exclusive `question` issues, 10 already used, so the retained three-way mix could not be reproduced), with the three retained arms in a balanced Latin order and the artifact recompiled provider-free under the current tracer pin (identical id, program, splits, and 92/0 gate). By the decision rule fixed in advance the compiled-only miss comes first: record 2737, where the compiled arm quoted an error line with backticks the source does not contain; the baseline missed record 3040 (a doubled space) and all three arms missed record 3968. Exact contracts were 118/120 baseline, 118/120 compiled, and 117/120 hand-written (Table 14); the artifact dispatched on 119/120 records and abstained cleanly on record 5102 on the induced verifier's cardinality clause, running the unchanged agent. The one-sided 95% bound on compiled-only failures is 3.9% on these 120 records and 2.2% pooled with the 90 primary records (1/210); the pre-registered targets of zero misses, 120/120 dispatch, and a pooled bound at or below 1.5% were all missed and are reported as such. Five of the 120 records had appeared in the gate-frontier pilot's cohort because the exclusion scan omitted that directory; none of the missed or abstained records is among them, and without them the counts are 113/113/112 of 115 with bounds of 4.1% and 2.3%. Requests fell 50.0%, tokens 39.5%, latency 42.5%, and cost 32.8% against the same-model baseline, matching the primary family within a point on requests, tokens, and cost (latency 42.5% against 51.7%). In all, 810 held-out GitHub episodes ran under a compiled artifact on the calibrated model in six studies: 270 on the excerpt-bearing contracts (the 90 primary, these 120, and the 60 post-cutoff records of Appendix G), with this cohort's record 2737 the one compiled-only miss (one-sided bound 1.7%), and 540 on the outcome-only cross-repository contract (120 core, 180 balanced, and 240 gate-frontier episodes over 354 distinct records; Appendices F and H), with none.

E Additional Experimental Detail

Primary workflow families. The three-family result is the main paper's strongest live evidence because it combines distinct tool vocabularies, exact task graders, balanced held-out cohorts, and provider-backed execution. The compiler limits the issue-type family to a shallower two-read prefix; the newer pull-request and backlog families admit deeper pre-model compaction. This difference explains the lower resource savings on the first family and is part of the evidence boundary, not noise to average away.

Cross-repository extension. The cross-repository study is narrower than the primary workflow families. Its task uses two read-only tools and a simpler exact outcome contract, which is precisely

why a fixed template can remain tied or even more efficient than the learned artifact. We keep the result in the main text because it addresses a central scope question — whether guarded specialization survives beyond one repository — while being explicit about the reduced task complexity.

The position invariant, empirically. An early pilot dispatched a compiled *suffix* region at the entry boundary, violating Eq. (2). Because the runtime resolves artifacts before the agent has made the calls the region assumed had already happened, the pilot reordered and duplicated tool calls. The failure is a semantic mismatch between compilation-time position and resolution-time position, not a tuning error, which is why Eq. (2) is a hard constraint in the deployed compiler rather than a scoring penalty.

Omitted from the main claim. The repository retains additional studies that are useful scientifically but less suitable for the main narrative:

- a natural-order issue study that exposed a factual continuation miss for a more aggressive three-read artifact,
- exploratory guarded composite synthesis and a follow-up comparison to a fair pre-model manual program,
- a bounded GEPA prompt-optimization study in which GEPA retained its seed, so the combined arm is a replication rather than evidence of interaction, and
- a portfolio pilot and controlled demonstration suite for runtime boundary coverage, including workloads where removing turns *increases* cost through prefix-cache fragmentation.

These studies remain important to the broader artifact, but they either use a weaker risk level, address a narrower engineering question, or are explicitly exploratory.

F Evidence Breadth and Boundary Audits

The main text covers the registered live GitHub studies, the cross-repository extension, the BIRD live study, and the NESTFUL and AppWorld substrates. This appendix makes the retained breadth visible without pooling incomparable cohorts or promoting screened and exploratory work to deployment evidence.

F.1 Repository-level outcomes on newer records

On the two-read cross-repository PR-outcome task (Section 4), GAC completes 120/120 exact contracts on paired held-out records across four of five repositories and retires `pytorch/pytorch` at admission, where no calibration group is accepted at any threshold. The compiled program ran on 80/120 records: every open-class record abstained on the induced `pr.state` hull (discovery pools held 0–3 open records) and ran the unchanged agent, so requests fall 44.4% (66.7% on dispatched records) and tokens 52.4%. A balanced three-repository rerun with all three classes in discovery, `pytorch/pytorch` included, dispatches on 180/180 and reaches 180/180 at 66.7% fewer requests and 78.4% fewer tokens (Table 17), consistent with a composition-driven refusal. The hand-written program is more efficient in the core protocol and tied in the rerun.

The core `pytorch/pytorch` run mines one candidate from 116 accepted windows and retires at calibration. A provider-free recompilation (`paper/results/iclr_revision/pytorch_core_recompile.json`) recovers $n_\eta = 0$ at every threshold, identifying the empty-sample branch of Eq. (7). Discovery contains 84 closed-unmerged, 30 open, and 2 merged records; the held-out split is balanced at 10 per class. The balanced rerun admits the same task. These observations are consistent with sensitivity to cohort composition, but neither the degenerate bound nor the rerun establishes the cause of the original refusal.

The core result succeeds on four repositories (120/120 held-out pairs) and retains the fifth as a compile-time retirement after 116 exact discovery traces. The balanced rerun has a distinct three-repository cohort and reaches 180/180 held-out pairs. These rows show breadth across repositories drawn from shared pools — the balanced rerun shares 13, 25, and 8 of the 30-record core held-out sets — not independent evidence for an average effect: all runs share the same provider family and the same narrow two-read outcome task.

Table 16: Held-out dispatch in the cross-repository studies. "Dispatched" counts records on which the compiled program ran; the remainder abstained on the named verifier clause and ran the unchanged agent, which is why the request reduction is 44.4% under the core protocol and 66.7% under the balanced one for the same two-read artifact.

Protocol	Repository	Held-out	Dispatched	Fallback clause	Req. ↓
Core	`huggingface/datasets`	30	20/30	`range:pr.state` (10)	44.4%
Core	`pandas-dev/pandas`	30	20/30	`range:pr.state` (10)	44.4%
Core	`psf/requests`	30	20/30	`range:pr.state` (10)	44.4%
Core	`pytorch/pytorch`	—	RETIRE	—	—
Core	`streamlit/streamlit`	30	20/30	`range:pr.state` (10)	44.4%
Balanced	`pandas-dev/pandas`	60	60/60	—	66.7%
Balanced	`psf/requests`	60	60/60	—	66.7%
Balanced	`pytorch/pytorch`	60	60/60	—	66.7%

Table 17: Disaggregated cross-repository outcomes behind Section 5.4. The core protocol uses four completed repositories and one compile-time retirement; the balanced rerun changes the selected repositories and class mix, so it is reported separately rather than pooled with the core protocol. Reductions compare the compiled condition with the paired unchanged baseline.

Protocol	Repository	Held-out	Exact	Req. ↓	Tokens ↓
Core	huggingface/datasets	30	30/30	44.4%	52.2%
Core	pandas-dev/pandas	30	30/30	44.4%	52.5%
Core	psf/requests	30	30/30	44.4%	52.3%
Core	streamlit/streamlit	30	30/30	44.4%	52.7%
Core	pytorch/pytorch	—	RETIRE	—	—
Balanced	pandas-dev/pandas	60	60/60	66.7%	78.6%
Balanced	psf/requests	60	60/60	66.7%	78.7%
Balanced	pytorch/pytorch	60	60/60	66.7%	78.0%

Table 18: All four trace benchmarks used as compiler substrates. NESTFUL and AppWorld are discussed in Section 5.3; API-Bank and executed BFCL v4 gold plans retire on support exactly like NESTFUL. Columns as in Table 3.

Substrate	Traces	Slots (1)	Windows (2)	At floor (2)	Synth. (4)	Replay p/a/**w** (5)	$n_{\max}$ (6)	Adm.
NESTFUL	1,415	96.3%	1,207	32	12	24 / 12 / **0**	26	**0**
API-Bank	212	63.5%	37	8	2	0 / 2 / **0**	8	**0**
BFCL v4	200	68.0%	86	9	4	3 / 3 / **0**	15	**0**
AppWorld	136	63.2%	136	33	1	34 / 0 / **0**	**136**	**1**

F.2 Continuation audit

The earlier natural-order three-read study is excluded from the registered $\alpha = 0.05$ headline because it used $\alpha = 0.10$ and exposed a downstream continuation error. Its retained counterfactual replay nevertheless tests a boundary the exact tool contract cannot see: 17/18 candidate continuations pass unchanged, while issue 6602 has a comment-evidence mismatch. Checked rendering repairs that one case, yielding 18/18 final contracts. This audit makes no provider calls and does not estimate live agent quality; it establishes only that the continuation check caught a concrete failure the compacted tool trace alone would miss.

F.3 External evidence map

Table 20 records every named external benchmark in the retained evidence register and its admissible interpretation.

Obtaining observed results from BFCL. BFCL's reference plans carry no tool results, which is why the plan-format row and the compiler row are listed separately. Its multi-turn categories, how-

ever, ship an executable stateful backend: each task publishes an `initial_config` snapshot and the classes it involves, so executing the *official gold plan* through the upstream helper yields observed results, an entry snapshot, and an in-process replay oracle, with no model in the loop. Five preconditions fail the run closed rather than publishing: an undeclared method, any upstream execution error, a read-like declaration observed mutating state, a compilable declaration observed advancing the scenario RNG, and any inexact re-execution. All held: 200 tasks and 1,142 calls executed cleanly and a second independent pass reproduced every result byte for byte, executing turn-at-a-time where the first pass executed call-at-a-time.

Why the catalog is signed rather than inferred. The same run audits the declarations empirically by snapshotting every involved instance around every call. It observed 1,060 state-mutating calls over 44 of the 81 methods and 94 calls that advanced the seeded scenario RNG, with zero cases of a read-like declaration mutating state or a compilable declaration advancing the RNG. One case is worth naming: a method whose name and docstring present it as a cost getter mutated its instance's internal lookup on all 36 of its calls and is therefore declared a write. A name- or docstring-derived effect label would have admitted a write into a compiled region. Two further methods are read-like but carry no capabilities, because one derives an age from the host date and the other draws from — and advances — the seeded RNG; they are barriers by declaration and account for 35 suppressed spans. Write barriers suppress 2,802 spans in total, against 45 for entry-schema admissibility, 31 for slot ambiguity, and 5 for partial runs.

Obtaining observed results from AppWorld, and why it is the reachable case. AppWorld's shipped `api_calls.json` records method, url, and request data but no responses, so it fails closed for the same reason BFCL's plans do. Executing the official `compiled_solution_code` against the pinned `0.1.3.post1` backend supplies the results; each task's own `specs.json` supplies the entry snapshot; a second execution supplies the replay oracle. Calls are captured at the single dispatch point every `apis.<app>.<api>(...)` call passes through, so the recorded identity is a semantic app/API pair with typed keyword arguments and a decoded JSON result. Of the 147 public train+dev gold solutions, 136 execute cleanly; 11 dev solutions raise against the shipped data version and are reported as upstream compatibility outcomes rather than dropped. The second pass reproduced 7,070/7,070 results byte for byte, minted access tokens included, because AppWorld freezes each task's clock.

The empirical audit attributes every `INSERT`/`UPDATE`/`DELETE`/`CREATE`/`DROP` statement to the enclosing call. It observed 1,320 mutating calls, zero read-like declarations mutating, and — unlike BFCL — zero APIs that mutate on some calls but not others. Every `GET` was clean on every call and every `DELETE` and `PATCH` mutated on every call, so the effect boundary is crisp; the five `*.login` endpoints are the only non-`GET` APIs that never mutate, which is why they are the single contestable declaration and why the run is pre-registered in two arms. Groups are tasks, not scenarios: AppWorld's three variants of a scenario share a supervisor and a database, so the bound is conditional on task-level independence, which this corpus does not establish — the same conditional the live studies carry, neither weakened nor repaired here; at scenario or supervisor level (46 and 79 units) the zero-violation bound is 0.097 and 0.058, both above α. Arm A calibrated one candidate and is compiler-wide; arm B calibrated six (four retired, one dominated, one emitted), so its certificate is per-candidate, with a corrected bound of 0.068. The calibration share is fixed at exactly the 92 groups the bound needs, since a larger set would make the bound easier and a smaller one would make it unreachable. Two held-out replay figures appear for AppWorld and they count different partitions, which Table 3 flags but does not have room to explain. The 34/0/0 in that table is the test side of the *family* partition used to challenge a mined candidate — 102 train and 34 test groups, the same accounting every substrate row reports. The separately reported 11/11 admission-test passes are from the sealed test split of the 11/92/11 dev/calibration/test partition Alg. 1 consumes for admission. Both are genuinely held out; they are held out from different fits, and neither is a subset of the other's training side.

Admissible is not dispatchable. Table 19 separates the two. The admitted region is eligible on essentially every full-code trajectory and essentially no ReAct or plan-and-execute trajectory. Documentation lookups explain some position failures: 3,752 of 8,190 trajectories start with an API-description call, and 61 contain the program away from position 0. They do not explain all missing dispatches: the prologue relaxation recovers only two ReAct runs (Table 29); the admitted call sequence often never occurs. Full-code agents are eligible on 2,339/2,340 trajectories and iterative parallel function-calling agents on 762/1,170. Two secondary observations: the rate is flat across task

Table 19: Structural dispatch eligibility of the admitted AppWorld artifact on the 28 released official baseline runs. Eligibility is the admitted program's exact call sequence at normalized position 0; it is an upper bound on φ, because the manifest check and the calibrated gate are not evaluated and the released logs retain API calls rather than model boundaries. Trajectories repeat tasks across runs and are descriptive rather than independent samples. A miss abstains to the unchanged agent.

Released baseline architecture	Runs	Trajectories	Eligible at pos. 0	Rate
Full code + reflection	8	2,340	2,339	1.000
Iterative parallel function calling	4	1,170	762	0.651
Plan and execute	8	2,340	1	0.000
ReAct	8	2,340	0	0.000
Pooled	28	8,190	3,102	0.379

difficulty (0.377 on `test_normal` against 0.380 on `test_challenge`), consistent with the strong association with agent architecture, without excluding workload effects; and the maximal compilable depth is 2 on the eligible families, so the admitted two-step program sits at the ceiling the effect catalog permits rather than short of it. On the ineligible families the measured depth is a floor, not an estimate, because the API-documentation call is undeclared in the signed catalog and unknown effects are not reads.

F.4 Prefix census over released agent trajectories

The introduction asks how often deployed agents open with a compilable read-only evidence path. No deployment traces are public, so this appendix measures the nearest proxy: released corpora of *executed* multi-step tool-using trajectories from three scaffolds the paper does not otherwise touch, re-measured beside the two substrates it already ingests, all under one set of definitions and with no model in the loop (`paper/scripts/prefix_census.py`; sources, catalogs and aggregates under `paper/results/prefix_census/`). It is descriptive: nothing here is synthesized, calibrated or dispatched, and prefix eligibility is not admission.

Corpora. (i) *OpenHands / SWE-rebench*: 67,074 OpenHands v0.54.0 (Wang et al., 2024b) CodeAct trajectories with native function calling (Qwen3-Coder-480B) over 6,306 real issues in 1,823 repositories (Trofimova et al., 2025). (ii) *SWE-agent*: 80,036 legacy SWE-agent (Yang et al., 2024) text-action ReAct trajectories (fine-tuned Llama-3 action generators, three sizes) over 3,591 real issues (Nebius, 2024). (iii) τ-bench: the 1,980 released historical runs of gpt-4o and claude-3.5-sonnet over the 165 airline and retail tasks, a function-calling conversational agent in a non-coding domain (Yao et al., 2024). (iv) NESTFUL gold sequences and (v) AppWorld's 28 released baseline runs (8,190 trajectories, four architectures), the substrates of Section 5.3 and Table 19. Each corpus is pinned by revision and SHA-256 in `paper/results/datasets/*/source_manifest.json`; raw corpora are parsed in isolation, never executed, and not committed. Candidates rejected and why (SWE-smith, the OpenHands SWE-Gym releases, τ^2-bench, AgentBench and WebArena logs, ToolBench's archive) are recorded in the directory's README.

Definitions, identical for every corpus. The first model boundary is the first agent turn after the task prompt; normalized position 0 is the first tool call. The entry state is the structured task record the scaffold's prompt is filled from (instance, repository, problem statement, workspace path; for τ-bench the user's opening utterance), seeded as pseudo-producer 0 as in Alg. 3. A *read-only opening* is the run of consecutive calls from position 0 whose signed catalog entry is compilable; a new user utterance, an unknown effect, a write, an approval-gated tool or a missing capability ends it. A *groundable opening* is the longest canonical window of the window-mining stage of Alg. 2 that starts at the first boundary, at the registered parameters (depth 2, $\kappa = 3$, $w \in [2, 12]$, $b_{\min} = 2$, entry schema `inputs`): every slot a literal invariant across traces, or grounded to the entry state or a prior result by a bounded transform, exactly Eq. (2). Families are canonical hashes of opening windows with support counted in distinct tasks; $n_{\max}$ is compared with the 92-group floor of Eq. (7). Only the first thirteen calls are ingested ($w_{\max} + 1$), which cannot change a window that starts at the first boundary. One signed catalog per corpus under `benchmarks/contracts/effects/` carries one justification per tool; editor sub-commands are distinct entries (`str_replace_editor.view` is a read, `create`

Table 20: Disposition of every retained external benchmark family. Only the four compiler-measurement rows and the BIRD live study support compiler claims; the remaining rows document why broader benchmark names are not silently converted into GAC results. BFCL appears twice because its reference plans and its executed gold plans are different evidence.

Benchmark	Evidence tier	Retained result / claim boundary
NESTFUL	Compiler measurement	1,415 traces; 24 / 12 / 0 replay; all families retire.
API-Bank	Compiler measurement	212 traces; 0 / 2 / 0 replay; all candidates retire.
BFCL v4 (plans)	Gold-plan checker	200/200 valid plans; plan format only.
BFCL v4 (executed)	Compiler measurement	200 traces, 1,142/1,142 exact re-execution; 3 / 3 / 0 replay; all families retire.
AppWorld (executed)	Compiler measurement	136/147 traces, 7,070/7,070 exact re-execution; $n_{\max} = 136 \geq 92$; one artifact admitted at $U = 0.0498$; under arm B two candidates retire at the margin ($U = 0.051$) and two far from it ($U = 1$).
BIRD (dev and train)	Live-provider compiler study	31 databases, two agent designs, three models: the standard design retires everywhere; the schema-first opening is admitted on 26 of 31 databases (`gpt-5.6-luna`) and 5 of 5 (`gpt-6-luna`); scheduled cohorts are 150 dev questions, those same 150 on a second model, and 630 train questions; 924 complete-case rows are not 924 distinct questions, and noninferiority is unproven (Appendix F.5).
ToolSandbox (Lu et al., 2025)	Live-provider subset	One simulated scenario; compiler not run.
τ^2/τ^3 (Barres et al., 2025)	Live-provider subset	0/4 pass; compiler not run.
ToolBench (Qin et al., 2024)	Adapter smoke	Ten fixtures; full data/backend unavailable.
AgentBench (Liu et al., 2023)	Adapter/preflight	Services and Freebase data gated.
GAIA (Mialon et al., 2023)	Access preflight	Authorization denied; no metric.
SWE-bench Verified (Jimenez et al., 2024)	Task adapter	Official run host-gated.
BrowseComp (Wei et al., 2025)	Live-web subset	1/3 correct, 28 searches; bypassed by the compiler.

a write), the model-side `think` tool is a barrier, and every shell invocation is `UNKNOWN`. Each released corpus has exactly one labelled sensitivity arm, never pooled with the primary row: admitting single-utility shell invocations (`ls`, `cat`, `grep`, `find` without executing flags, read `git` subcommands) as reads for the two coding corpora, and, for τ-bench, measuring from the first tool-issuing turn with the user utterances before it folded into the entry state, the analogue of the AppWorld prologue relaxation of Table 29.

Findings. Table 21 gives the census. Read-only openings of two or more calls are a minority everywhere but one architecture: 16.3% of OpenHands trajectories, 32.5% of SWE-agent trajectories (47.3% with shell reads admitted), 3.9% of τ-bench runs under the strict first-boundary rule (51.7% once the identity-collecting user turns are folded into entry state), and 41.0% of AppWorld's released trajectories, where the split is by architecture (full-code 99.96%, iterative function calling 71.4%, Re-

Table 21: Prefix census. RO≥2: share of trajectories whose read-only opening has at least two calls; Ground. ≥2: share whose longest groundable opening window has at least two calls; Med. len: median length of those windows; Fam. / floor: opening families and those with ≥ 3 supporting tasks (the stage-2 minimum of Table 3); $n_{\max}$: largest family support, † at or above the 92-group floor. Indented rows are the labelled sensitivity arms. AppWorld's released logs retain no responses, so only its read-only column is measurable (its gold-solution substrate reaches $n_{\max} = 136$, Table 3). Trajectories repeat tasks and are descriptive counts, not independent samples.

Corpus	Agent	Traces	Groups	RO≥2	Ground. ≥2	Med. len	Fam. / floor	$n_{\max}$
OpenHands / SWE-rebench	CodeAct, FC	67,074	6,306	16.3%	12.8%	3	27 / 22	752†
+ shell reads admitted		67,074	6,306	16.3%	12.9%	3	153 / 43	752†
SWE-agent (legacy)	ReAct, text	80,036	3,591	32.5%	1.9%	2	154 / 36	187†
+ shell reads admitted		80,036	3,591	47.3%	2.0%	2	184 / 40	187†
τ-bench runs	ReAct, FC, conv.	1,980	165	3.9%	3.0%	2	2 / 2	7
+ pre-tool user turns folded		1,980	165	51.7%	16.6%	3	27 / 9	17
NESTFUL (gold)	no agent	1,415	1,415	100.0%	86.6%	4	1,560 / 241	76
AppWorld released	4 archs	8,190	585	41.0%	–	–	–	–

Act 5.7%, plan-and-execute 2.1%), consistent with Table 19. The first barrier explains most of the remainder and is a scaffold habit rather than a workload property: 87.5% of OpenHands trajectories open with the model-side `think` tool, 47.0% of SWE-agent trajectories open by creating a reproduction script (a write), 99.7% of τ-bench openings are cut by the next user turn, and 45.9% of AppWorld trajectories open with the undeclared API-documentation call.

Groundability under Eq. (2) separates the scaffolds further. On OpenHands, 79% of read-only openings are groundable (12.8% of all trajectories, median three calls): the opening is a run of directory and file views whose paths derive from the workspace path in the entry state, and the largest family spans 752 tasks, above the 92-group floor, with eleven families at or above it. On SWE-agent only 6% of read-only openings are groundable (1.9% of trajectories): the scaffold's first `open`, `search_dir` and `find_file` arguments are model-chosen paths and search terms that no depth-2 transform of the issue text reproduces (24,399 openings blocked on an ungrounded slot, led by `open.path` and `search_dir.search_term`); the two families at the floor (187 tasks each) are two-call search openings. On τ-bench the strict measure grounds 3.0% of runs, all from the claude-3.5-sonnet runs that query a reservation named in the opening utterance; the folded arm grounds 16.6%, and its largest family (17 tasks) cannot reach the floor because the benchmark has 165 tasks. NESTFUL's openings are read-only by construction and 86.6% groundable (the rest have ambiguous slots); its largest opening family spans 76 of 1,415 sequences, against 26 for whole sequences in Table 3. In short, the one-in-six to one-in-three prevalence of read-only openings, the lower prevalence of groundable ones, and the architecture dependence all match what Section 5.3 found on AppWorld alone.

Limitations. These are benchmark runs, not deployments; prevalence in production agents remains unmeasured in the literal sense, and these corpora over-represent software engineering. The catalogs are ours, are not backed by a mutation audit as the BFCL and AppWorld catalogs are, and are conservative by design: every shell command is unknown, so the primary rows are lower bounds on read-only openings and the shell arm is an upper bound under a plausible declaration. The SWE-agent scaffold's editor window is implicit state that `goto`, `scroll` and the no-argument `search_file` read without an argument slot, which Eq. (2) cannot see. Observations longer than 32,000 characters are cut, which can only lose producers. Support is counted in tasks, and trajectories of one task across models or trials are not independent. Finally, an opening family above the floor is a candidate for Alg. 2, nothing more: synthesis, the replay contract and the calibrated gate were not run on any of these corpora.

F.5 A live agent on a public benchmark: BIRD

Design. The study was registered before any provider call (`paper/supplementary/bird-sql-agent-protocol.md`). A tool-using SQL agent on `gpt-5.6-luna`, with the paper's pinned settings, answers BIRD dev questions (Li et al., 2023a) (release `dev_20240627`, question plus evidence) over read-only SQLite files with three tools: `list_tables`, `get_schema(table_names)` with three sample rows, and `run_query`. The contract is BIRD execution accuracy: the final query's result set

Table 22: BIRD: compile-or-retire per database and design, and the schema-first held-out evaluation (30 sealed questions per database). "Repeat" is a second independent run of the unchanged agent. Reductions compare the compiled agent with that warm-cache repeat run (the condition order was not rotated as registered; see text). Standard-design retirements are at synthesis (`ungroundable_slot`) or on calibration support.

		Standard	Schema-first: correct of 30				Compiled vs. warm repeat		
Database	Tables	GAC	Base	Repeat	Compiled	Manual	Requests	Tokens	Cost
`card_games`	6	retire: support 34/92	18	16	**19**	18	44.1%	7.8%	−2.6%
`codebase_community`	8	retire: synthesis	22	23	**22**	21	48.8%	15.2%	6.3%
`formula_1`	13	retire: synthesis	18	19	**20**	19	43.7%	4.2%	1.8%
`student_club`	8	retire: synthesis	26	25	**24**	24	49.6%	19.0%	18.7%
`thrombosis_prediction`	3	retire: synthesis	10	11	**12**	12	39.0%	6.0%	12.1%
Pooled		5 retire	94	94	**97**	94	44.9%	9.8%	6.3%

equals the gold query's. Every dev database with at least 146 questions (16 train, 8 dev, 92 calibration, 30 held out) is a family, which selects five. Two agent designs were registered: *standard*, adapted from LangChain's SQL-agent prompt (LangChain, 2026) ("query the schema of the most relevant tables"), and *schema-first*, whose workflow step reads every table before writing a query, the analogue of the paper's prescribed design. QUALIFY does not filter on answer correctness, the compiler runs with the registered settings, and a strict audit vetoes any admission whose calibration groups show the agent passing different opening arguments than the program derives.

Admission. The standard agent passed 7 to 39 distinct table lists per database and every table in no discovery run on four of them, so `get_schema.table_names` has no consistent expression and four families retire at synthesis (Table 22). On `card_games` the candidate "read only `cards`" (the agent's most common choice, 61 of 132 runs) matched 34 calibration groups and retired on support. In 7 of those groups the agent had read a different table; the compiler labeled them unproductive, which stays in the count, rather than violations. That is the leniency the strict audit was registered to catch, and with more support the compiler alone could have admitted a wrong-shaped program. The schema-first agent passed the full listing in every discovery run, and every family admitted `list_tables` → `get_schema`(all tables) with one candidate at calibration (compiler-wide), 92 groups, zero violations, and a clean audit. The table list is bound as an invariant literal, not as an expression over the listing as the protocol predicted, because the implemented rule tries literals first.

Held-out evaluation. The compiled program dispatched on 150 of 150 sealed questions. Execution accuracy was 97/150 compiled, 94/150 for the unchanged agent, 94/150 for its repeat, and 94/150 for a hand-written schema prefetch; compiled against the unchanged agent there are 5 compiled-only and 2 baseline-only correct answers (McNemar $p = 0.45$; difference +2.0 points, 95% interval $[-1.3, +5.3]$), which meets the registered criterion for no detected accuracy loss. The interval also permits a loss; neither equivalence nor noninferiority was established. The registered rotation of condition order was not implemented: the four conditions ran in a fixed order, so the first baseline met a colder provider prompt cache (cached-input share 0.504 against 0.56–0.59) and its repeat was 11.9% cheaper with the same tokens. Against the warm repeat, the compiled agent made 44.9% fewer model requests $[42.1, 47.4]$, used 9.8% fewer tokens $[4.7, 14.3]$, ran 29.4% faster $[23.6, 35.1]$, and cost 6.3% less $[1.5, 10.6]$; against the first run the cost reduction is 17.5%, and pricing every input token at the uncached rate gives 11–12% either way. The hand-written prefetch reaches 46.3%, 12.4%, 31.5%, and 7.9%, so the discovered program matches the manual ceiling. Savings in tokens and cost are modest because the two removed calls carry the shortest contexts. Total spend was $2.69.

Extensions: order, models, providers, and scale. Four extensions were registered before any of their provider calls (`paper/supplementary/bird-sql-agent-extensions-protocol.md`; Table 23). A rerun with the condition order rotated as originally registered reproduces the result on the same 150 questions (100 against 94 correct; McNemar $p = 0.07$, no superiority claimed) with 46.0% fewer requests and 16.2% lower cost against the first baseline. On a second model family, `gpt-6-luna`, the standard design retires on all five databases and the schema-first opening is admitted on all five, with 49.9% fewer requests and 93 against 91 correct on 149 complete-case questions (150 were scheduled). On the BIRD training split (26 databases with at least 146 questions,

Table 23: BIRD runs. "Std." and "S.-first" count databases admitted per design (the rotated rerun reuses the primary artifacts). Correct is on the held-out questions completed by all four conditions; bold marks the compiled arm, not a significant improvement. Reductions compare the compiled agent with the unchanged agent; †every primary-row reduction is against the warm repeat run because that run's order was not rotated. Cost* prices every input token at the uncached rate.

Run	Model	DBs	Std.	S.-first	n	Correct (base / GAC)	Requests	Tokens	Latency	Cost	Cost*
Primary (fixed order)†	`gpt-5.6-luna`	5	0	5	150	94 / **97**	44.9%	9.8%	29.4%	6.3%	11.2%
E1 rotated rerun	`gpt-5.6-luna`	5	—	—	150	94 / **100**	46.0%	12.3%	33.9%	16.2%	13.9%
E2 second model	`gpt-6-luna`	5	0	5	149	91 / **93**	49.9%	16.8%	38.4%	6.1%	17.5%
E3 second provider	`claude-sonnet-5`	5	0	0	—	—	—	—	—	—	—
E4 train split	`gpt-5.6-luna`	26	0	21	625	401 / **400**	45.9%	12.6%	29.3%	18.1%	14.0%

Table 24: Retrospective BIRD quality over all scheduled questions. Missing runs count as incorrect in their own arm; "Complete" is the historical four-arm intersection. Rotated repeats the primary questions.

Run	Scheduled	Complete	Base	Repeat	GAC	Manual
Primary	150	150	94	94	97	94
Rotated	150	150	94	93	100	96
Second model	150	149	91	95	93	90
Train split	630	625	403	412	402	404

after a pre-execution rule excluded one whose gold SQL references tables missing from the shipped database), the standard design retires on all 26 and the schema-first opening is admitted on 21; the five refusals are databases (11 to 65 tables) where the agent occasionally dropped or reordered a table, four on calibration support (90 or 91 of 92 groups) and one at synthesis. On the 625 four-arm-complete questions across the 21 admitted databases the compiled agent dispatched every time and answered 400 correctly against 401 ($p = 1.0$), with 45.9% fewer requests, 12.6% fewer tokens, 29.3% lower latency, and 18.1% lower cost. Pricing every input token at the uncached rate, which removes provider-cache effects, gives cost reductions of 11–18% in every run. On a second provider, Anthropic `claude-sonnet-5`, the agent followed neither opening as a fixed step: under the schema-first prompt it listed the tables first in 429 of 655 runs and passed every table in 26, and GAC retired all ten (design, database) pairs. GAC compiles what an agent does, not what its prompt asks for. Extension spend was $43.93.

Attrition and all-scheduled denominators. The historical summaries above retain only questions completed by all four arms. This excludes one second-model question because the repeat baseline timed out, and five training-split questions: four manual-arm timeouts and one compiled-arm turn-limit failure. A retrospective audit of the sealed run records counts every scheduled question, treating each missing run as incorrect only in its own arm (Table 24). Training-split accuracy is 402/630 compiled versus 403/630 unchanged, with 17 compiled-only and 18 baseline-only correct ($p = 1$); second-model accuracy is 93/150 versus 91/150 ($p = 0.73$). Neither result establishes noninferiority. Resource comparisons require completed runs: using all baseline/compiled pairs gives request/cost reductions of 45.8/18.0% on 629 training pairs and 49.6/5.8% on 150 second-model pairs. These estimates omit resources consumed by failed runs; aggregate billed cost and time-to-success are not established by them. The script `paper/scripts/audit_bird_denominators.py` regenerates these counts without provider calls. The pinned BIRD release is `dev_20240627`; these selected-family results are not a leaderboard score on the newer cleaned development set.

How much quality loss remains compatible with the data? The all-scheduled audit supports a retrospective uncertainty analysis (Table 25), without selecting a noninferiority margin after seeing the results. Let p_+ denote the probability of a compiled-only correct answer and p_- a baseline-only correct answer. The paired accuracy difference is $\Delta = p_+ - p_-$. Construct a two-sided 97.5% Clopper–Pearson interval for each probability (Clopper & Pearson, 1934); if their endpoints are $[L_+, U_+]$ and $[L_-, U_-]$, then $[L_+ - U_-, U_+ - L_-]$ covers Δ with at least 95% probability under i.i.d. question pairs, by a union bound. Dependence between the gain and loss indicators within a pair is allowed. These conservative, pointwise intervals are wider than the primary study's paired bootstrap interval; neither

Table 25: Retrospective BIRD quality sensitivity on all scheduled questions. All differences are compiled minus unchanged, in percentage points. Intervals assume i.i.d. question pairs. "Delete-one" omits each entire database in turn; it is a descriptive range, not a confidence interval. The last column counts databases with positive, zero, or negative differences.

Cohort	Δ (pp)	Conservative 95% interval	Delete-one range	DBs +/0/−
Primary	+2.00	[−4.40, +8.18]	[+0.83, +4.17]	3/1/1
Rotated	+4.00	[−2.53, +10.11]	[+1.67, +5.83]	3/1/1
Second model	+1.33	[−5.44, +7.97]	[−0.83, +3.33]	3/0/2
Train split	−0.16	[−3.27, +2.96]	[−0.50, +0.33]	5/10/6

Table 26: Every admitted artifact's certificate. m counts candidates that reached calibration; a parenthesized bound uses the candidate-wide budget $\gamma = \delta/(m|\Lambda|)$ or, in the last row, the primary grid. All bounds assume i.i.d. groups; the replay event has no observed occurrence on the GitHub cohorts as labeled (Section 3).

Artifact (where)	Event	m	n_η/k_η	$U_{\hat{\eta}}$	Level at $\alpha = 0.05$
Issue-type routing (Table 1)	replay	1	92/0	0.0498	compiler-wide
PR-outcome audit; backlog-attention routing	replay	2	92/0	0.0498 (0.0569)	per-candidate; compiler-wide at $\alpha \geq 0.057$
Cross-repository PR-outcome, 7 admitted (Appendix F)	replay	1	92/0	0.0498	compiler-wide, frozen search
AppWorld two-call prefix, arm A / B (Appendix F.3)	replay	1 / 6	92/0	0.0498 (0.068)	compiler-wide / per-candidate
BIRD schema-first opening, 26 databases (Appendix F.5)	replay	1	92/0	0.0498	compiler-wide
Post-cutoff PR-outcome, fresh groups (Appendix G)	end-to-end task contract	1	132/0	0.0173 (0.035)	single pre-registered rule

construction proves the sampling assumption. They are not simultaneous across the four rows, and overlapping primary, rotated, and second-model questions are not pooled as independent evidence.

The training interval allows a loss of 3.27 points or a gain of 2.96 points. Deleting one database keeps its observed difference between −0.50 and +0.33 points; 5 databases favor the compiled agent, 10 tie, and 6 favor the unchanged agent. The second-model difference changes sign under a database deletion. This tests concentration of the observed result, without treating the selected databases as a random sample or establishing generalization to new databases. The retained script `paper/scripts/bird_quality_sensitivity.py` regenerates the intervals, database deletions, and source hashes.

Sensitivity to missing execution cost. The same training records retain \$0.891759 of estimated cost for 630 completed baseline runs and \$0.728806 for 629 completed compiled runs. Let $u \geq 0$ be the unrecorded cost of the single failed compiled execution, priced with the same frozen rate card. Net evaluation savings are then \$0.162952 minus u, so they are positive only if $u <$ \$0.162952. This is a break-even condition, not evidence that the unknown cost satisfies it. It includes the failure's quality penalty above but cannot recover its missing usage; discovery, recompilation, and maintenance remain excluded.

F.6 What is certified, and what the statistical stage adds

Each primary artifact was admitted at 92/92 zero-violation calibration groups ($U_{\hat{\eta}} = 0.0498 \leq 0.05$, conditional on i.i.d. groups), a support-sufficiency certificate (Section 3) that is compiler-wide for issue-type routing (one candidate at calibration) and per-candidate for PR-outcome and backlog

routing (two; corrected $U = 0.0569$, so compiler-wide only at $\alpha \geq 0.057$; Corollary 2, Table 5); Table 26 lists every admitted artifact. The one certificate whose event includes the continuation comes from a post-cutoff PR-outcome cohort (pull requests created after the snapshot's last day; Appendix G): the re-derived artifact dispatched on 132 fresh calibration groups and the graded task contract missed 0, a one-sided bound of 0.0173 under the single rule pre-registered for that study and 0.035 under the primary grid. Those groups are 84 open and 48 merged pull requests and no closed-unmerged ones, so the certified population excludes that class; the 60 fresh test records (20 per class) pass under every condition, and hypothetical bounds using days and authors as independent units are 0.0228 and 0.0267; independence of those clusters is unverified.

Denominator and loss conventions. In Eq. (4) the denominator counts groups with at least one attempt admitted by the hard guard and score; a clean interpreter or verifier failure after admission has replay loss zero and remains in that count, and pre-dispatch abstention enters neither the denominator nor the savings. Every certificate except the post-cutoff one instantiates $L(A, x) = 1$ as a replay-contract violation of the compiled region (Appendix C), so the continuation is measured, not certified. The end-to-end certificate bounds absolute error, not excess error over B or noninferiority, and does not attribute errors causally to compilation.

Certifiable selectivity at the floor. The 92-group floor of Eq. (7) also bounds what selectivity is *certifiable*: coverage $c = n_\eta/|\mathcal{K}|$ needs $|\mathcal{K}| \geq \lceil 92/c \rceil$, so on the $|\mathcal{K}| = 92$ primary cohorts, the smallest satisfiable pool, the only outcomes are retirement or full empirical calibration coverage, since abstaining on one group leaves $U_\eta = 0.0503 > \alpha$; intermediate coverage can be *observed* but not certified (Appendix C, Fig. 3). The GitHub calibration labels replay stable call patterns over a pinned snapshot and contain no observed violation. They offer little test of risk discrimination; zero observed failures does not prove that failure is impossible. These certificates function as support-sufficiency statements at full coverage ($n_\eta \geq 92$; $\delta = 0.1$ is the budget at which that floor is satisfiable, $\delta = 0.05$ needs 106) that cannot tell whether q ranks dispatch risk.

What the statistical stage adds at runtime. The addition is measurable, and null on these candidates: full statistical admission, support-only admission, and deterministic guards alone give identical routing and evidence on all 2,724 record–model cases of the AWM comparison, and the score q is constant within each family because the entry state is a single record number (Appendix K). The deterministic barriers do the admission work; the gate contributes its sample floor, which is what refuses the thin benchmarks of Section 5.3 and, on a second provider, backlog routing (Appendix G).

F.7 The registered gate is a support threshold

This audit strengthens the evidentiary record precisely by preserving the negative result: the registered gate behaves as, and at $|\mathcal{K}| = 92$ can only behave as, a support threshold rather than a demonstrated risk–coverage frontier (Table 27); the question of q's ranking quality is open, not answered negatively. The exploratory partial frontiers therefore remain diagnostic rather than evidence for the main claim.

G Extended Limitations

Every protocol in this paper is in-repo and author-held; no external registry was used, and each is dated in its own file.

The certified event is not end-to-end quality. Proposition 1 bounds the replay-contract event of Appendix C. On these cohorts that event has no empirical support: the argument pattern of the compiled steps is invariant across all discovery traces per GitHub family, and the labels replay those calls over recorded results of a pinned snapshot, which is why no retained pool records a violation, every gate is step-like, and admission reduces to the 92-group floor (the second-provider backlog refusal turns on 90 and 91 eligible groups). The misses that do occur on these families are continuation misses (an excerpt that departs from its source), and they are outside the certificate: the one-sided 95% upper bound on compiled-only discordance in Appendix D is an empirical quality statement at a different confidence level, not an admission certificate. An end-to-end certificate needs a pre-registered endpoint that runs the continuation and grades the task contract on fresh calibration groups,

Table 27: Gate-behavior inventory from the retained threshold analysis. A partial frontier would show graded coverage as the threshold changes; none appears among the seven artifacts in the retained threshold analysis.

Evidence set	Artifacts	Partial frontiers	What it licenses
Registered GitHub $\alpha = 0.05$ (threshold analysis)	7	0	Exact per-candidate certificates; all are step-like.
Exploratory or looser-α	3	2	Diagnostic evidence only; not the registered claim.
NESTFUL refusal	12 synthesized	0	$n_{\max} = 26 < 92$; no admission.
API-Bank refusal	2 synthesized	0	$n_{\max} = 8 < 92$; no admission.
BFCL v4 refusal	4 synthesized	0	$n_{\max} = 15 < 92$; no admission, pre-registered.
AppWorld admission	1 admitted / 4 retired	0	$n_{\max} = 136 \geq 92$; the one reachable external bound. Admitted gate is step-like; two retirements are marginal ($U = 0.051$ at 89 and 90 eligible groups), two are not ($U = 1$).

Table 28: Distinct grouping units in the 92-record calibration cohorts, joined to the pinned snapshot: distinct creation days, months, and authors, and the zero-violation bound the registered budget would give if each cluster were one unit (admits only when the count is at least 92).

Cohort	n	Days	Months	Authors	U at days	U at authors
Issue-type routing	92	90	50	82	0.051	0.056
PR-outcome audit	92	86	45	60	0.053	0.075
Backlog-attention routing	92	87	46	73	0.053	0.062
Core: `huggingface/datasets`	92	80	37	46	0.057	0.097
Core: `pandas-dev/pandas`	92	92	66	53	0.050	0.085
Core: `psf/requests`	92	90	66	79	0.051	0.058
Core: `streamlit/streamlit`	92	87	33	25	0.053	0.171

with the sample floor of Eq. (7) applied to that event; one was obtained after the main study for a re-derived PR-outcome artifact on 132 fresh groups (0 misses, bound 0.0173 at 90% confidence; below), and is reported separately from the primary replay-contract certificates.

Dependence sensitivity. The exact binomial bound requires i.i.d. units. One valid cluster-level formulation uses i.i.d. clusters and the maximum violation within each cluster, targeting risk for a new cluster rather than a new record. Distinct days or authors do not establish that assumption; Table 28 is a hypothetical sensitivity calculation. Temporal spacing alone does not ensure independence, and block bootstrap or dependence-aware conformal methods require their own assumptions and guarantees; they do not inherit Eq. (7). For clean-checkout reproduction, the retained `paper/results/iclr_revision/calibration_cluster_rows.json` records the calibration row identifiers, dates, and hashed author equality keys derived from the digest-checked source snapshots.

Table 29: Structural dispatch eligibility before and after admitting an exact read-only prologue (the API-documentation call these architectures open with), per released AppWorld baseline architecture; never pooled. The runtime rule and the measurement rule are the same function.

Released baseline architecture	Runs	Trajectories	Before (pos. 0)	Rate	After (pos. 0 or exact prologue)	Rate
Full code + reflection	8	2,340	2,339	1.000	2,339	1.000
Iterative parallel function calling	4	1,170	762	0.651	762	0.651
Plan and execute	8	2,340	1	0.000	1	0.000
ReAct	8	2,340	0	0.000	2	0.001

A read-only prologue does not recover ReAct dispatch. Section 5.3 shows Eq. (2) refuses essentially every ReAct and plan-and-execute trajectory, which open with an API-documentation read. We built the narrowest relaxation (`paper/supplementary/read-only-prologue-protocol.md`): an artifact may declare an exact, allowlisted, read-only prologue whose calls must equal the committed observations in tool, order, and arguments; the prologue's results become live-ins of the region, its calls are not replayed, and any mismatch, extra call, order change, manifest mismatch, unknown effect, or slot left ungrounded by the extended entry contract falls back. The runtime passes the protocol's acceptance tests, including the archived suffix-dispatch pilot as a regression test. Measured on the 8,190 released trajectories with the documentation call as the prologue (Table 29), it changes little: ReAct eligibility rises from 0 to 2 of 2,340 and plan-and-execute stays at 1. The looser diagnostic — any documentation-only prefix followed by the admitted program — recovers 2 and 3, and the program occurs anywhere in only 16 and 39 of those trajectories. Position was the proximate reason those architectures were refused; the larger limitation is that they seldom execute the admitted region at all. A prologue rule cannot recover those absent executions. The prologue tool is also undeclared in the signed catalog, and the runtime refuses an undeclared prologue; the measurement assumes the declaration a reviewer would have to sign.

Scope of automated workflow-reuse comparison. The cited reuse systems differ from GAC in what they reuse, where the reused unit executes, and what admits it, which is why none could be dropped in as a matched comparator without re-implementation. Agent Workflow Memory induces workflow text from past trajectories and places it in the agent's context, so every step remains a model decision; on the API tasks of Appendix J it issued the baseline's four requests per episode (Wang et al., 2025). Agentic plan caching stores plan templates from completed runs and, at test time, retrieves and adapts one with a lightweight model, so a model stays in the loop and admission is by retrieval match rather than by grounded arguments (Zhang et al., 2025). LLMCompiler lets a planner call emit a DAG of tool calls that execute in parallel within one task and carries nothing across tasks (Kim et al., 2024). EvoC2F compiles a typed plan representation carrying dependency, effect, resource, idempotency, and retry annotations and admits trajectory-derived macro-skills through tests, contracts, and regression checks; Agent JIT Compilation generates and validates code plans for the current web task, selects low-cost candidates, and schedules execution under tool pre- and post-state invariants (Wei et al., 2026; Winston et al., 2026). Those two are the closest compiler comparators, since both carry effect or state annotations and both verify before reuse, but their admission is by test and contract passage rather than by a finite-sample bound over held-out groups, and neither is evaluated on an admit-or-retire decision per candidate region. We therefore execute the comparator whose official implementation we pinned and adapted to shared API tasks, an offline AWM adaptation, in Appendix J. This adds a whole-system quality–cost comparison, but does not reproduce AWM's native browser evaluation or provide an equivalent-placement comparison against another workflow compiler; the compatibility audit retained with that study records why AWO, EvoC2F, Agent JIT, and plan caching were not treated as verified drop-in comparators. That stronger comparator remains missing; internal barrier ablations do not substitute for it.

Table 31 prices the barriers: for each retained hazard it names the guard that caught it and what deterministic replay of the mined region would have done with provenance, effect, position, contract, and risk barriers disabled. On PR-outcome and backlog routing the mined three-read sequence *is* the admitted program, so an unguarded replay executes the same calls and differs only in guards that never fired on in-distribution held-out records; on issue-type routing the compiler refused the three-read candidate, so its price was measured live under the pre-registered protocol (`paper/supplementary/recurrence-only-ablation-protocol.md`). A barrier-free replay of the refused three-read region — no provenance check, guard, verifier, gate, or position check

— handed to one provider request on the same sealed 30 records passed 30/30 exact contracts with one request per record (Table 30); every excerpt was located in its source comment. By the decision rule fixed in advance, on this cohort the provenance refusal cost one request per record without a measured quality benefit, and the guard's value here is the refusal of an unwitnessed argument plus the retained issue-6602 counterexample, not a held-out failure. The measurement concerns one family, one cohort, and in-distribution records; it says nothing about behavior under shift, which the drift protocol below is written to test.

Provider-free drift ablation: the pre-declared null. The drift-robustness protocol (`paper/supplementary/drift-robustness-ablation-protocol.md`; amended before execution and run on 2026-10-06, provider-free) asks whether the induced verifier converts wrong answers into abstentions under the nine declared perturbation families. On the deterministic demonstration substrate it runs two arms over an identical compiled program and identical sealed-test windows, differing only in the verifier (induced versus permissive), with the hard runtime boundary retained in both; the hand-written arm is not run because the demonstration comparators are policy-level macros rather than IR programs. Six admitted artifacts over three demonstrations give 144 paired groups (one window per group). Both arms produce zero wrong outcomes (one-sided 95% upper bound 0.0206 per arm; no discordant pair), so the suite does not separate the arms on this substrate: the simulated tools remain total, and the unverified program's live-outs under nulled fields, duplicated records, padded lists and schema drift equal its unperturbed live-outs. What the verifier does here is abstain, including on invariant perturbations where the answer was unchanged (0.2477 pooled; 0.3333 on every `permissioned_rag` artifact, above the protocol's 0.25 guardrail and below its 0.50 bar), which is a cost without a measured benefit on this substrate. This is a null, not equivalence; it says nothing about the GitHub families or real workloads. Two further observations are reported as such: `incident_triage` admits no artifact under the current compiler although the retained 2026-08 run admitted two, and `mcp_ops` retires in both.

Recorded-replay drift ablation on the primary records: the null, and what it reveals about the oracle. The same question on real records (`paper/supplementary/drift-recorded-replay-protocol.md`; pre-registered and run 2026-10-06, provider-free): the three retained artifacts, recompiled identically from their sealed checkpoints, run with their induced verifier, with a permissive verifier, and beside the family's hand-written pre-model program with a permissive verifier, over the 90 primary held-out records reconstructed on the pinned snapshot (89 paired records; the baseline-only miss record 5189 yields no window of its family). All nine transforms change at least one recorded call per family. All three arms produce zero wrong outcomes (upper bound 0.0331 per arm; no discordant pair; Holm-adjusted exact McNemar $p = 1$ for both comparisons). The reason is structural, not total tools: the suite's oracle is the sequence of derived call arguments, and every admitted GitHub program's only derived argument is the record identifier, bound at entry and re-read unchanged from the first record tool's echo; none of the nine transforms alters that identifier, so no tool-result perturbation in this suite can change a decision and *wrong* is unreachable for these programs. What the suite does measure is the contract: the induced verifier abstains on nulled fields and on duplicated or padded discussion (or label) lists (0.1049 of invariant-family outcomes pooled; 0.0444, 0.1222, 0.1494 per family; padded lists are expected-either and sit outside that denominator), the interpreter abstains on schema drift in both compiled arms, and the hand-written program answers through schema drift unchanged because it echoes what it is given. Those abstentions guard the continuation against malformed evidence, which this endpoint does not grade; a live continuation extension would. Not equivalence; and a program whose later arguments derive from earlier results is where a verifier benefit could appear under this suite.

Post-cutoff cohort and an end-to-end certificate (PR-outcome). After the main study we acquired every `huggingface/datasets` issue and pull request created after the pinned snapshot's last day (2025-06-13; 1,095 records, 1,086 of them passing a preflight that drops any record number a retained result mentions) and pre-registered a study (`paper/supplementary/time-forward-end-to-end-protocol.md`) on the one family whose fresh pool supports it: PR-outcome (issue-type and backlog routing are NO-GO on class availability, as before). From 811 fresh pull requests, three disjoint splits were frozen before any call: 60 test records (20 per class), 132 discovery records and 132 calibration groups. The calibration groups, taken next in rank order after the balanced test draw and the round-robin discovery draw had consumed 64 of the pool's 72 closed-unmerged records,

are 84 open and 48 merged pull requests and no closed-unmerged ones; the certified population is therefore fresh open or merged pull requests, and the 20 closed-unmerged test records below are uncertified. The retained artifact's verifier pins the old snapshot revision, so on fresh records it abstains everywhere and the unchanged agent runs; that is the designed behavior and is reported as such. The compiler, with one frozen candidate, re-derived a byte-identical three-read program, guard, and gate model from the fresh discovery traces (131/132 exact), differing from the retained artifact only in its snapshot pin: a determinism result, not a transfer. The re-derived artifact was then dispatched on all 132 calibration groups and the unchanged continuation run once and graded on the exact task contract: 132 dispatched, 0 misses. Under a single pre-registered rule ($\gamma = \delta = 0.10$, $m = 1$) the one-sided bound on the *end-to-end* violation rate is 0.0173, below $\alpha = 0.05$: the first certificate here whose event includes the continuation. The single rule was adopted prospectively after Table 6 showed the grid buys only its Bonferroni factor; under the primary grid the same count gives 0.035 (0.040 at $m = 2$), and treating the 100 distinct creation days or 85 distinct authors as units gives 0.0228 and 0.0267 under the single rule, so admission does not depend on the rule change. On the 60 fresh test records all four conditions reach 60/60 (20/20 in each class); the re-derived artifact dispatches on every record and cuts requests 75.0%, tokens 82.7%, latency 80.0% and cost 76.6% against the unchanged agent, with no compiled-only miss; the hand-written program ties it. The claim is per-candidate, for one family, conditional on i.i.d. groups from one repository's post-cutoff records; spend was under one dollar.

Continuation-graded drift ablation: an adverse result. The third drift study (`paper/supplementary/drift-continuation-graded-protocol.md`; pre-registered and run 2026-10-07, 1,080 live cells, under one dollar) grades what the two provider-free studies could not: the model's continuation when the evidence it receives is corrupted. On the 60 primary PR-outcome and backlog records, each of the nine perturbations was applied to the tool layer, two arms ran on identical corrupted tools (the retained artifact with its induced verifier, and the same artifact with a permissive verifier; abstention falls back to the unchanged agent on the same corrupted tools), and the final answer was graded against the unperturbed record. The pre-declared adverse branch obtained: the guarded arm produced a silent wrong answer on 39 of 60 records as graded, all under emptied lists, and the permissive arm produced exactly the same 39 (no discordant record); a 40th record (PR 6694) answered "none" against a real comment but passed the grader's substring test because that comment contains "nonetheless", a grader defect since fixed in the family graders (an audit of every retained graded cohort found no other held-out row affected; the 18-record natural-order study keeps its registered grader, under which one discovery row, issue 1741, is the only row the two rules grade differently). An empty discussion or assignee list lies inside the hull the verifier learned, so it accepts, the artifact dispatches, and the model answers faithfully from the emptied evidence ("no comment evidence"; backlog route flipped), which is wrong against the real record. Where the corruption left the hull, dispatch fell back in both arms before any continuation ran: on renamed keys the interpreter's binding failed in both arms; on nulled fields and error payloads the induced verifier's non-null clauses refused in the guarded arm and the composite's typed projection refused in the permissive arm, and the fallback agent, fed the same corrupted tools, then answered wrong on 30/30 in both arms (fallback misses, outside the 39). Where lists were duplicated or padded the verifier abstained on 11–14 of 30 records whose permissive counterparts answered correctly, and one such fallback (backlog record 5971) itself missed comment grounding (guarded fallback rate on invariant families 0.1389; padded lists are expected-either and outside that denominator). The induced contract therefore adds nothing the typed projection does not already catch on out-of-hull corruption, and catches nothing that lands on a legitimate value; on these records the verifier prevented no silent wrong answer and bought abstention, and one fallback miss, only. This is a statement about this contract on these corruptions with the calibrated model, not about production safety; it is reported ahead of every other reading because the protocol said it would be.

A second model family on the same cohorts. Every headline number comes from one model, so a pre-registered transfer (`paper/supplementary/second-model-replication-protocol.md`) reran the three primary families on `gpt-6-luna`, a different model generation at the same efficiency tier, keeping the sealed discovery traces, splits, and 90 held-out records and recompiling each artifact provider-free under a manifest that pins the new model; the recompiled artifacts are identical to the retained ones (same ids, programs, and 92/0 gates). The decision rule fixed in advance puts one finding first: on issue-type routing the compiled arm missed one record the baseline passed (#6532, an excerpt that pluralized one source word), so its preservation hypothesis

Table 30: Recurrence-only replay on issue-type routing: per-record means over the 30 sealed held-out records. The first three rows are the retained arms; the last replays the three reads the compiler refused (`limit` is 100 in 130 of 132 traces and 1 in two, and has no witness) with every barrier disabled and one provider request. All four arms are 30/30 exact (McNemar $p = 1$ against each comparator); the replay is the cheapest arm, which is the price of the refusal on this cohort.

Arm	Exact	Requests	Tokens	Latency (s)	Cost (¢)
Unchanged agent	30/30	4.00	4,259	6.16	0.087
Compiled two-read prefix (GAC)	30/30	2.00	2,576	2.97	0.059
Hand-written bundle	30/30	2.00	1,781	3.25	0.055
Recurrence-only three-read replay	30/30	1.00	1,167	2.67	0.038

Table 31: Mechanism-removal table: every retained hazard, the guard that caught it, and what a recurrence-only replay of the mined region would have done. Each row is backed by a retained result file in the artifact.

Hazard (where observed)	Guard that caught it	Recurrence-only replay	Retained source (`paper/results/`)
Ungrounded argument `issue_get_comments.limit=100`, in 130 of 132 discovery traces (1 in two) (issue-type, held-out #4420 family)	provenance + synthesis (stage 1/4): literal not invariant, no expression in $\mathcal{L}$, `ungroundable_slot`	replays the majority literal; emits the three-read program	`github_natural_replication/results.json compiler.candidates[0]`
Clean tool replay, wrong downstream answer (Markdown URL dropped, issue 6602) (natural-order study, 17/18 → 18/18 after checked rendering)	continuation contract (post-model)	accepts the answer	`github_natural_live/continuation_replay.json`
Getter-named method that mutates state on 36/36 calls (BFCL executed gold plans)	signed effect declaration (stage 2)	compiles a write into the region	`external_benchmarks/bfcl_compiler_execution.json`
API-documentation prologue before the hot prefix (AppWorld released runs, 4,679/4,680 ReAct and plan-and-execute)	position invariant, Eq. (2), at dispatch	dispatches out of position	`external_benchmarks/appworld_dispatch_preflight.json`
Suffix region dispatched at entry: reordered and duplicated calls (archived pilot, 16.7% pass)	position invariant (compile-time `prefix_only`)	ships the fault	`github_live/pilot_2026-08-03/`
Result outside the induced hull (`pr.state=open`; 40/120 core records) (cross-repository core protocol)	induced verifier V; clean fallback	answers from an unseen class unchecked	`github_multirepo_pr_outcome_core/results.json dispatch.reasons`
Projection type mismatch (`merged_at`, record 7199; 13 events) (PR-outcome pilot)	composite projection verifier; fallback	emits a malformed evidence object	`pr_outcome/pilot_v1/failed_attempt_projection_2026-08-05.json`
Recurrent but under-supported families ($n_{\max}$ 26, 8, 15 < 92) (NESTFUL, API-Bank, BFCL)	stage-6 exact gate	deploys on recurrence alone	`nestful/results.json;api_bank_execution.json;bfcl_compiler_execution.json`
Skewed calibration pool (84/30/2 classes) (`pytorch/pytorch`, core protocol)	stage-6 exact gate ($U = 1.0$)	deploys	`github_multirepo_pr_outcome_core/results.json repository_failures[0]`

fails there; a second record (#3859) failed in both arms through line-ending normalization, and backlog record 2648 failed in the baseline only. Every miss is an excerpt-fidelity error by the model on identical evidence, and the retained `gpt-5.6-luna` arms pass all three records. Structure transferred as predicted: requests per record were unchanged in every arm, the artifact dispatched on 30/30 records in each family, and pooled reductions were 66.6% requests, 63.0% tokens, 59.5% latency, and 60.1% cost against 66.6, 63.1, 64.2, and 58.7 on the calibrated model (Table 33). We read the miss as the case the manifest pin exists for: the certificate is conditional on the pinned model, these artifacts were never calibrated on `gpt-6-luna`, and one compiled-only miss in 90 is what an uncertified transfer looks like. Re-discovery on the second model (design A, `paper/supplementary/second-model-rediscovery-protocol.md`) then ran the whole pipeline on `gpt-6-luna`'s own traces over the same sealed records. Issue-type routing reached 128/132 exact discovery traces and PR-outcome 132/132; the compiler admitted the identical artifacts (same ids and programs, 92/0 gates, one and two candidates at calibration as in the primary runs), which dispatched on 30/30 records and passed 30/30 in every arm, with reductions of 50.0/38.9/32.8 and 75.0/80.7/76.0 percent in requests, tokens, and cost; the transfer miss on record 6532 did not recur. Backlog-attention routing refused at compile time: 113 of its 132 discovery traces were exact,

Table 32: Second provider, design A: Anthropic `claude-sonnet-5` on the sealed cohorts through the unchanged pipeline. Reductions are against the same-provider baseline; the compiled-only misses are the finding the protocol reports first, and the backlog refusal is the 92-group floor binding.

	Exact held-out contract			Reduction (%)		
Family	Base	Compiled	Manual	Requests	Tokens	Cost
Issue-type routing	29/30	**28/30**	28/30	48.3	40.4	39.8
PR-outcome audit	27/30	**27/30**	29/30	75.0	84.9	83.4
Backlog-attention routing	RETIRE at calibration (91 and 90 of 92 groups, $U > \alpha$; 116/132 exact traces)					

below the 116 the 16/8/92 split needs, because on 19 records the model skipped the discussion read, took a two-read route, and left its excerpt ungrounded; no held-out arm was run and no artifact is claimed. The exact-trace filter removed exactly those route-deviating groups. Their inclusion need not produce violations: replay can instead abstain, as the BIRD audit demonstrates. That counterfactual was not executed. The certificate inherits the exact-trace selection even on the primary cohorts, where 130, 131, and 132 of 132 traces are exact; it does not certify the unfiltered population. The paper therefore claims one calibrated model family, reports a second on which the pipeline re-derived two of three artifacts and refused the third, and makes no cross-model certificate claim.

A second provider on the same cohorts. The same design ran on Anthropic's `claude-sonnet-5` through the official SDK, with the unchanged Agents-SDK harness behind a provider adapter (`paper/supplementary/second-provider-replication-protocol.md`): adaptive thinking at effort `low`, parallel tool calls disabled, and the same prompts, tools, structured answers, grader, and compiler settings. Discovery reached 126, 128, and 116 exact traces of 132 (12 of backlog's 16 non-exact traces skipped the discussion read, as on `gpt-6-luna`). Issue-type and PR-outcome admitted the identical artifacts (same ids and programs, 92/0 gates, one and two candidates at calibration) and dispatched on 30/30 records; backlog met the split exactly and both candidates retired at calibration because only 91 and 90 groups were accepted at any threshold, leaving the zero-violation bound at 0.0503 and 0.0509, above α: the floor of Fig. 3 binding on a second provider. The decision rule reports the compiled-only misses first: two on issue-type (records 4248 and 6829) and two on PR-outcome (5401 and 6988). Every miss on this provider, in every arm, is an excerpt that drops Markdown link markup or normalizes whitespace, the issue-6602 mechanism, and the baseline itself missed one and three records. Exact contracts were 29/28/28 of 30 on issue-type (one hand-written episode timed out twice and counts as a failure; the issue-type contract is factuality-only by registration, and two compiled episodes, records 6150 and 5514, also issued one redundant read after the compiled region, which the tool-order check flags and the contract does not count) and 27/27/29 on PR-outcome; requests fell 48.3% and 75.0%, tokens 40.4% and 84.9%, and cost 39.8% and 83.4% (Table 32). The structure transfers across providers; excerpt fidelity is the model's, and the exact contract charges it to whichever arm produced it.

A compression comparator ran, and did not engage. Every headline resource number compares a compiled prefix against an *uncompressed* baseline, because prompt compression (Jiang et al., 2023) and, on a comparable workload, deployed compression middleware (Headroom Labs, 2026) report token reductions of the same order by an entirely different mechanism. A live Headroom v0.5.18 ablation adds `headroom_only/compiled_headroom` arms to the PR-outcome and backlog-attention families on the existing 30-record held-out cohort (`paper/supplementary/headroom-ablation-protocol.md`). Headroom attempted every eligible JSON tool result and pre-model evidence object and applied *none* of them, saving zero tokens in either family, with all four paired quality contrasts holding 30/30 at exact p=1. This is a null engagement at this payload shape, not a measured composition or a measured floor — it does not establish Headroom's behavior on larger contexts — but it does retire the narrower concern that our reduction is compression under another name: on this workload the two systems visibly do not touch the same bytes. It is a compression comparator only; it does not stand in for a workflow-reuse or unguarded-replay baseline. That rerun also re-executed the unchanged agent on the same 30 backlog records and scored 30/30, so the single baseline miss of Table 1 did not recur.

Table 33: Same-cohort transfer to `gpt-6-luna`: the three primary families rerun with the sealed records, discovery traces, and recompiled artifacts behind Table 1. Reductions are against the same-model baseline. The issue-type compiled-only miss (record 6532) is the finding the transfer protocol reports first; under re-discovery the compiler re-derives the identical issue-type and PR-outcome artifacts and refuses backlog routing on the second model's own traces.

	Exact held-out contract			Reduction (%)		
Family	Base	Compiled	Manual	Requests	Tokens	Cost
Transfer (design B): retained artifacts recompiled under the new pin						
Issue-type routing	29/30	**28/30**	30/30	50.0	39.2	33.0
PR-outcome audit	30/30	**30/30**	30/30	75.0	80.8	76.3
Backlog-attention routing	29/30	**30/30**	30/30	74.8	81.0	75.4
Weighted total ($n = 90$)	88/90	**88/90**	90/90	66.6	63.0	60.1
Re-discovery (design A): the pipeline run on the second model's own traces						
Issue-type routing	30/30	**30/30**	30/30	50.0	38.9	32.8
PR-outcome audit	30/30	**30/30**	30/30	75.0	80.7	76.0
Backlog-attention routing	RETIRE at compile time (113/132 exact discovery traces)					

Where "unchanged baseline" is exact. Pre-emission rejection is exact on both runtime paths; only a staging-owning runner can restore byte-identical model input after an attempt has been emitted (Appendix B). Without one, a post-commit verifier failure is an incident rather than a fallback, which is why the compiled region is confined to the pre-commit prefix.

How the calibration cohort must be sampled. The calibration cohort must be sampled before filtering on whether the baseline happened to exhibit the mined region, with violation labels defined for every dispatch-eligible member; post-selecting only recurrent successes would not satisfy Proposition 1. This is why the AppWorld substrate fixes its calibration share at exactly the 92 groups the bound requires and draws them by seeded shuffle over all clean tasks, rather than taking the tasks in which the mined family happens to have the deepest support. The GitHub cohorts' exact-trace filter is such a selection (Section 4); the BIRD study drops it (Appendix F.5).

Provider-free challenge and signature audit. The historical live runs and subsequent audit are distinct.

Two operational caveats of the reported runs have since been closed provider-free: the three primary artifacts were compiled with the perturbation suite disabled and registry signature verification off, and a recompilation from the sealed discovery checkpoints (`paper/scripts/recompile_with_challenge.py`) reproduces the identical programs, splits, artifact identifiers, and 92/0 gates with the suite enabled through a sandbox over the pinned snapshot; all nine perturbation families (reordering, duplicates, large and empty collections, nulls, schema drift, formatting, tool errors, timeouts) complete on the eight development windows of each artifact with zero wrong answers and zero hard rejects, so each record now carries `perturbations_claimed: true`. The same recompilation registers each artifact under an HMAC-SHA256 signing key, reloads it with verification on, and confirms that wrong-key, tampered, and the original unsigned registries are refused (`paper/results/iclr_revision/recompile_with_challenge.json`); no reported number changes.

H Prospective Gate-Frontier Protocol

The retained protocol (`paper/supplementary/prospective-gate-frontier-protocol.md`) fixes candidate selection, splits, thresholds, endpoints, and stopping before acquisition. It targets at least 300 paired cases across at least three repositories, comparing the unchanged agent, learned gate, and support-only gate. A positive result requires at least three nonzero coverage levels and better risk at matched coverage; adverse dispatches take reporting priority. Neither gate showing graded coverage is a predeclared negative outcome. Execution reached four of five repositories, as detailed below; the original protocol remains the source for the complete design.

Table 34: Disaggregated outcomes for the gate-frontier study. Exact contracts are reported per arm under intention-to-treat; † one support-only episode (record 6708) raised the 120 s provider timeout and was not re-attempted, while the record passed in the other two arms. "Dispatched" counts held-out records on which the compiled program ran; the remainder fell back to the unchanged agent on the guard clause `range:pr.state`. "Coverage in sweep" lists every calibration coverage the frozen grid produces for the learned gate. The support-only arm ($\alpha = 1$) reproduces the same n and k at every η but, because $\alpha = 1$ appends $\eta = 1.0$ to the grid, its Bonferroni split is $\delta/12$ ($U = 0.0507$ at $n = 92$) and it deploys threshold 1.0 rather than 0.5.

Repository	Held-out	Exact (B/L/S)	Dispatched	Coverage in sweep	Req. ↓
`huggingface/datasets`	60	60/60/60	40/60	0.1087 ($U = 0.375$, rejected), 1.0	44.4%
`pandas-dev/pandas`	60	60/60/60	40/60	1.0	44.4%
`psf/requests`	60	60/60/59†	40/60	1.0	44.4%
`pytorch/pytorch`	60 sealed	RETIRE	—	none ($U = 1$ at every η)	—
`streamlit/streamlit`	60	60/60/60	40/60	1.0	44.4%

The 90-record pilot comparator is invalid for a compiled-effect claim (`gate-frontier-pilot-protocol.md`, correction dated 2026-10-08). Its named compiled arm used a `compiled` policy manifest, while the retained artifact was registered under `base`; exact registry resolution therefore found no artifact. The pilot retained factual grades but no dispatch telemetry. Its reported Markdown-link contrast is a difference between stochastic model/tool runs and is withdrawn as a compiled-versus-unchanged result. The independently recorded primary-study excerpt error on issue 6602 still motivates the link stratum; its predictive value is unestablished.

An amended end-to-end issue gate study also retires (`paper/supplementary/graded-issue-end-to-end-gate-protocol.md`). A pinned 474-issue cohort reserved 200 development, 184 calibration, and 90 untouched test issues, enriched for link-bearing comments. The first attempt stopped after 288 ordinary fallbacks because the historical artifact's manifest did not match the current runtime; its checkpoint and estimated cost are retained. A compatible candidate was then re-derived from 80 disjoint historical discovery issues before any new-cohort compiled result. In the amended run, the candidate dispatched on all 200 development and 184 calibration issues, with 4 and 2 final factual-task errors respectively. The frozen eleven-point grid admitted no threshold at $\alpha = 0.05$ and $\delta = 0.10$: at $\eta = 0.02$, 64/184 groups had no errors but $U = 0.0708$; at $\eta = 0.50$, 153/184 had two errors and $U = 0.0546$. The learned gate therefore retired before the 90-issue test. The support-only ablation accepts 184/184 with two observed errors, but it disables the risk budget; no held-out quality or savings comparison follows from this study.

The comparator feasibility bar has also been checked (spike memo in the same directory): the retained feasibility memo did not identify a confirmed public AWO implementation (Abuzakuk et al., 2026) at that check. This historical no-go does not establish present unavailability or eliminate the comparison gap.

H.1 Observed results

The sealed cohort targets 300 pooled held-out pairs across the five repositories Section 5.4 already uses, 116 discovery and 60 held-out cases each under frozen-candidate selection. Discovery reaches 580/580 exact traces. Four repositories admit an artifact; `pytorch/pytorch` again retires at compile time. This cohort is drawn from the same frozen snapshots with the same selection seed as the core study and excludes none of its records, so it shares 561/580 discovery, 442/460 calibration, and 130/150 held-out records with the core cohort; at twice the core scale it is a larger sealed re-execution, not an independent replication, and the `pytorch/pytorch` retirement is a near-replay of the core one on a 93–95% identical split. The pooled result is therefore 240 of the pre-registered 300 pairs, short of target for the stated reason.

Accounting. 240 records × 3 arms = 720 attempted episodes, 719 completed. Exact task contracts under intention-to-treat: unchanged agent 240/240, learned gate 240/240, support-only 239/240. The incomplete episode is `psf/requests` record 6708 under the support-only arm, which raised the 120 s provider timeout; the same record completed and passed in the other two arms. The protocol fixed no retry rule, so the episode was not re-attempted and counts as a failure; among the 239

records with three completed arms every arm passes 239/239. Two provenance facts belong here. The `psf/requests` block was executed once, on 2026-08-22T18:59Z, as the single-repository pipeline check under the sealed selection; the five-repository run reused it through the driver's resume path rather than re-executing it. Retained spend for successful attempts sums to $0.41; the two failed `streamlit/streamlit` discovery attempts are not retained.

Dispatch. Calibration coverage is 1.0 in every admitting repository, but held-out dispatch is not: the compiled program ran on 160/240 records (learned gate) and 159/239 (support-only). The other 80 — every open-class pull request, 20 per repository — abstained on the induced verifier's `pr.state` hull, which the discovery pools (0–3 open records each) had fixed at {`closed`}, and ran the unchanged agent. Relative to baseline the learned gate reduces provider requests 44.4%, total tokens 52.3%, observed wall latency 51.0%, and estimated cost 48.3%; the support-only gate — $\alpha = 1$ in an otherwise identical second `compile_grc` pass — 44.4%, 52.4%, 47.9%, and 48.8%. Both arms admit the same artifact; they differ in α, in the grid ($\alpha = 1$ appends $\eta = 1.0$, so the split is $\delta/12$ and $U = 0.0507$ at $n = 92$), and in the deployed threshold (1.0 versus 0.5).

Coverage levels. Table 34 lists every coverage the frozen grid produces. Three admitting repositories accept no calibration group below $\eta = 0.11$ and all 92 at $\eta \geq 0.11$; `huggingface/datasets` additionally accepts 10/92 at $\eta \leq 0.08$, whose exact upper bound 0.375 exceeds α. No repository deploys at an intermediate coverage — and none could: with $|\mathcal{K}| = 92$ the registered bound certifies only $n_\eta = 92$, so the pre-registered criterion of three admissible nonzero coverage levels was unattainable at this cohort size whatever q ranks, and the protocol should have said so. Held-out q is moreover constant within each repository (0.0169; 0.0815), because the entry state is a single record number and every entry-observable feature is a function of it. Per the decision rule fixed above in this appendix, this is the fourth listed outcome: neither gate produces graded coverage. We report it as a null about the design, not as evidence about q: the study audits support-threshold behavior at full coverage under two risk budgets and does not test the ranking quality of q. The protocol that could (`paper/supplementary/graded-frontier-protocol.md`) needs $|\mathcal{K}| \geq \lceil 92/c \rceil$ groups per repository and an entry state with informative features; it is written and not run.

I Safeguard Ablations with Live Recovery

These follow-ups ask whether compiled prefixes save model work on unused records, whether individual safeguards change deterministic outcomes, and whether those differences survive a live agent's recovery. Cohorts and endpoints remain separate from the six studies' 810 compiled episodes in Section 7; none changes that historical denominator or supplies a new end-to-end risk certificate.

Fresh-record comparison. On 120 previously unused issues from the same historical snapshot, four live arms complete all 480 episodes. Baseline, GAC, and support-only each pass the registered four-field factuality endpoint on 120/120 records; recurrence-only passes 118/120. On the supplementary seven-field endpoint the respective counts are 119, 120, 119, and 118. GAC halves model requests (4 to 2 per episode) and reduces estimated model cost by 45.8% against baseline. Its dispatch decisions are identical to support-only: 119 dispatches and one fallback, so separate stochastic continuations cannot establish a gate effect. The two recurrence-only misses are strict formatting/case differences, not demonstrated semantic false assertions. Prefix length and context also differ, preventing an isolated safeguard attribution. The supplementary conservative 95% interval for GAC minus baseline is $[-3.59, 3.59]$ points on the registered endpoint; equivalence and noninferiority are not established. The previously admitted artifact is fixed, without new discovery or calibration; these are not time-forward records (`experiments/fresh_safeguards/analysis/`).

Deterministic mechanism diagnostic. A disjoint cohort supplies 24 discovery, 96 development, 192 calibration, and 96 held-out issue groups. The study specifies a three-comment prefetch, a requested-limit entry hull, and a response-completeness check; production helpers fit argument bindings and the gate, but end-to-end mining of these contracts is not tested. Each held-out issue receives requested limits 1, 3, and 5 under ordinary responses and responses capped at two comments. Five arms (full, baseline, and removal of scope, completeness, or gate) yield 2,880 provider-free evaluations. Under ordinary responses, full is correct on 288/288 cases versus 260/288 without scope admission; under the cap, full is 288/288 versus 252/288 without completeness checking. The baseline reference policy is correct throughout. The verifier compensates for scope removal un-

Table 35: Live completion: correct answers / scheduled episodes, including failures. No scope removes the requested-limit hull; no completeness removes only that check. All four timeouts occur in confirmation: one each in ordinary baseline, full, and no scope, plus one in capped full.

Cohort	Responses	Baseline	Full	No scope	No completeness
Confirmation	Ordinary	204/216	205/216	186/216	210/216
Confirmation	Capped	211/216	205/216	207/216	204/216
Replication	Ordinary	71/72	70/72	67/72	70/72
Replication	Capped	70/72	68/72	69/72	69/72

der the cap, where that ablation remains 288/288. The fitted gate score is constant (0.00247); all eleven thresholds admit the same 192 calibration groups, and removing it changes no held-out decision. These are deterministic outcomes, not live accuracy or evidence that recovery is impossible (`experiments/safeguard_mechanisms/`).

Stopped pilot and separately sealed completion. The initial 24-issue live pilot attempted 153 of 576 scheduled episodes: 152 completed, 151 were correct, one raised an execution error, and 423 remained unattempted. A separate diagnostic returned the correct answer but reproduced an accounting failure: the old validator rejected reported cache-write tokens. The original failure retained no usage, so this does not identify its exact cause. The completion study prices cache writes at the existing frozen rate and preserves partial usage and unknown-cost reserves. It retains the pilot's prompt, model, tools, answer contract, prefetch, guards, source, timeout, and retry policy; its new protocol and schedule were sealed before execution. Pilot and diagnostic outcomes are excluded from the completion comparison.

Live design and statistical unit. The completion uses all 96 offline held-out issues: 72 *confirmation* groups untouched by the live pilot or diagnostic, and 24 *replication* groups selected for the pilot, including issues it never reached. Every issue has prior offline exposure; replication is not independent of the pilot. Three limits × two response conditions × four arms produce 24 episodes per issue, or 2,304 overall. Full, no-scope-guard, and no-completeness-check differ by one intervention; baseline receives no prefetch. Every arm has the same live continuation and recovery tools, including original call arguments and completeness information. Tool reads use the pinned local snapshot, not a remote GitHub service. The constant gate is retained without a redundant fifth live arm.

The model is `gpt-5.6-luna`, with low reasoning and verbosity, `store=false`, no parallel tool calls, at most 2,048 output tokens per request and eight turns, a 120 s request timeout, and no SDK or transport retries. Four workers execute issue blocks; all 24 arm permutations occur three times in confirmation and once in replication within each condition/limit. All scheduled episodes were attempted, with no replacements: 2,300 completed, 2,186 were correct, 114 completed with incorrect answers, and four timed out. Failures count as unsuccessful scheduled attempts (Table 35).

The prespecified paired analysis averages the three limit variants within each issue, then bootstraps issues 10,000 times (seed 20261018), separately for each cohort and response condition. The 95% percentile intervals in Table 36 are descriptive and unadjusted for multiple contrasts. They assume suitable independence between issue groups; author or repository dependence remains possible. The 2,304 episodes are not independent samples. No noninferiority margin or new live risk certificate is asserted.

Scope protection and recoverability. Under ordinary responses, scope admission improves confirmation success by 8.80 points. Full-only correct cases occur in 20 issue groups and ablation-only correct cases in five. Replication gives +4.17 points with interval $[-1.39, 9.72]$: directionally consistent but not decisive. A post-hoc sensitivity on the 214 ordinary pairs where both arms complete gives 23 full-only versus four no-scope-only correct cases; timeout imbalance does not create the net 19-case advantage, but this does not replace the primary analysis.

Incomplete accepted prefetches are defined by source positions and requested scope before inspecting live answers. Without scope admission, only 5/20 such ordinary confirmation cases trigger more reads and only four finish correctly; in replication, 4/8 read further and finish correctly. Without completeness checking under the cap, all 27 confirmation cases read further and 22 finish correctly;

Table 36: Confirmation contrasts: full minus comparator, in percentage points. Discordances count full-only / comparator-only correct cases. Intervals resample 72 issue groups, not 216 independent requests, and are descriptive/unadjusted.

Responses	Comparator	Discordances	Difference	95% interval
Ordinary	No scope	24 / 5	+8.80	[3.70, 13.89]
Ordinary	No completeness	4 / 9	−2.31	[−5.56, 0.93]
Ordinary	Baseline	11 / 10	+0.46	[−3.70, 4.63]
Capped	No scope	6 / 8	−0.93	[−4.17, 2.31]
Capped	No completeness	8 / 7	+0.46	[−3.70, 4.63]
Capped	Baseline	4 / 10	−2.78	[−6.48, 0.46]

Table 37: Complete 24-group replication accounting, baseline → full. Model-cost reductions exclude tool-service and lifecycle costs.

Responses	Model requests	Model cost	Total tools	Correct / 72
Ordinary	2.417 → 1.931	−18.2%	1.417 → 1.597	71 → 70
Capped	2.722 → 2.458	−9.1%	1.722 → 2.125	70 → 68

all nine replication cases read further and eight finish correctly. The agent often repairs capped evidence, but frequently treats a page complete for a three-comment call as sufficient for a larger request. Requests requiring more than the two-comment cap number 54 in confirmation and 18 in replication.

Full versus no completeness under ordinary responses has identical initial model inputs: differing accuracy reflects separate stochastic continuations, not a verifier effect. Under capped responses, completeness can compensate for the removed scope guard. Neither the unresolved completeness contrast nor the earlier deterministic gap establishes that every safeguard is necessary. Correct prefetches also leave semantic errors after full dispatch; accepted dispatch is not an end-to-end answer certificate.

Resource trade-offs and accounting. Every confirmation contrast involving full has an execution failure; therefore its completed-only resource means are descriptive, and overall saving intervals are withheld. Replication has complete accounting (Table 37). Model requests and estimated model cost fall, while total tools increase and slightly fewer answers are correct. There is no demonstrated quality equivalence or overall system-cost win. All discarded prefetch and continuation reads count. Local snapshot reads, concurrency, caching, frozen rates, and omitted engineering/human costs limit transfer to deployment.

The paired model-cost difference intervals are [−0.00010541, −0.00005827] dollars per episode for ordinary responses and [−0.00007654, −0.00001820] for capped responses. New-run recorded cost, including known partial usage, is $0.952696; four failed-attempt reserves and the carried pilot/diagnostic accounting give $7.001762 in estimates plus reserves. These are neither invoices nor bounds on actual billing. All 448,212 cache-write tokens were priced; the prior cache-write rejection did not recur.

Interpretation and provenance. The live evidence supports request-scope admission on this constructed workload; it does not establish a completeness-check accuracy benefit, learned-gate value, or superiority to baseline. It does not test end-to-end contract mining or transfer of this guard benefit across models, natural requests, or time. The broader paper's cross-model and post-cutoff studies concern other questions. Source regrading is consistency checking, not independent human gold attestation. All incorrect answers, failure stages, source expectations, metric totals, and paired intervals are retained in `experiments/live_recovery_completion/analysis.json`; the checkpoint, prospective seal, diagnostic, and protocol remain unchanged. The post-run audit regrades answers, checks tool-output digests, and recomputes costs and outcomes. Table 41 indexes the drivers and preceding studies.

Table 38: Primary comparison on 120 fresh issue groups. Correct is the seven-field endpoint; factual is the historical four-field endpoint. Requests and tools are observed calls per attempted episode; USD/1k is estimated deployment-model cost per 1,000 attempted episodes, excluding one-time induction. This is an adapted AWM API-task comparison.

Arm	Correct	Factual	Requests	Tools	USD/1k
Baseline	114/120	117/120	4.00	3.00	0.892
GAC	116/120	119/120	2.00	3.00	0.624
AWM adapted	114/120	117/120	4.00	3.00	0.902

J Matched Comparison with Adapted Workflow Memory

This prospective study compares an ordinary agent, the real GAC SDK adapter, and an offline AWM workflow-memory adaptation (Wang et al., 2025) on fresh issues from the same historical snapshot. It addresses a narrower gap than a native browser-benchmark reproduction or a comparison between equally placed compilers. AWM adds induced workflow text to the model's context; GAC replaces selected model decisions with compiled SDK tool calls. The comparison measures these whole systems, without isolating a safeguard's causal effect.

Pinned adaptation and shared evidence. The official AWM source is pinned to commit `8c0ff8cd11d648c8fceb99e4e42f37e3b75381b1`. Its offline induction, example formatting, workflow filtering, and workflow-only retrieval function bodies are executed unchanged; the Apache-2.0 files and checksums are retained. We adapt the navigation-domain phrase to read-only API tasks, append the actual tool schemas, and serialize recorded discovery tasks and tool arguments into the upstream action representation. The original one-shot example is retained. Both methods receive access to the same 130 compiler-eligible traces from the historical 132-trace discovery pool. GAC retains its original split and admitted candidate; AWM induces from all 130 eligible traces. No held-out answer, source oracle, or evaluation outcome enters induction. Workflow-only retrieval uses upstream top-$k = 0$ for concrete examples, without online updates.

One induction request uses the shared model through the Responses transport with low reasoning/verbosity and a 1,024-token output limit, rather than the upstream Chat Completions transport with temperature zero. The resulting memory is frozen before evaluation, without manual rewriting or output selection. The AWM browser evaluation loop is not run. GAC's existing reconstruction checks program, admission counts, and split identity; changed runtime contract components remain disclosed in the seal, without a new risk certificate.

Prospective evaluation. A conservative exclusion audit removes issue IDs appearing in earlier retained studies, and hash ranking selects six execution-pilot issues and 120 disjoint primary issues from 937 available records. All six arm orders occur once in the pilot and 20 times in the primary cohort. The pilot's continuation gate checks execution, accounting, output schema, and actual GAC compaction, not favorable accuracy. All arms share the natural extractive task, output schema, snapshot tools, model, low-reasoning settings, 2,048-token per-request output limit, eight-turn limit, 120-second timeout, and no-retry policy. Each starts a fresh conversation; four issue groups run concurrently, with arms sequential within an issue. Provider caching is measured but not controlled.

The primary endpoint requires all seven answer fields to pass the existing source grader. The historical four-field factuality endpoint is secondary; tool-call contract agreement is reported separately. Failed and unattempted scheduled episodes remain in accuracy denominators. Paired issue bootstrap intervals use 10,000 draws and are descriptive; exact McNemar tests receive Holm correction across three accuracy contrasts. Exact marginal binomial intervals are also retained. GAC minus adapted AWM is the primary contrast; baseline contrasts are secondary. The sample is not powered for equivalence or noninferiority.

Results and uncertainty. All 18 pilot episodes completed correctly. All 360 primary episodes were attempted: 359 completed, with one GAC output-schema rejection retained as a failure and never retried. Primary seven-field success is 116/120 for GAC and 114/120 for each other arm; four-field factuality is 119/120 versus 117/120. GAC has two correct-only pairs and no incorrect-only pair

against either comparator. The accuracy difference is +1.67 percentage points, with a descriptive paired-bootstrap interval $[0, 4.17]$, but exact McNemar $p = 0.50$ and Holm-adjusted $p = 1.00$: an accuracy improvement is not established. A post-run source audit finds that issue 377 cannot supply the required 20-character excerpt: its title has six characters, its body is empty and it has no comments. All arms fail that issue (GAC at schema validation, the other two at source matching); it remains in every original denominator. This was the only issue failing this narrow feasibility screen among the 126 selected records. The failed GAC output is not retained, so its exact invalid field is unknown. Each arm also has three strict category/evidence-label errors involving `dataset bug`, where the exact-label contract expects `other/none`. These are not established semantic falsehoods; independent human adjudication remains missing.

The exact marginal 95% intervals are $[91.69, 99.08]$% for GAC and $[89.43, 98.14]$% for each comparator. The positive point estimate and the boundary bootstrap do not establish equality, noninferiority or superiority.

GAC makes two real provider requests per attempted primary episode versus four for both comparators; all use three snapshot tool calls. Observed deployment model cost is 30.8% lower than adapted AWM and 30.1% lower than baseline, including the failed episode's observed usage. All 1,260 evaluation provider requests have usage records, including cache writes; no usage is silently zero-filled. For the 119 completed GAC–AWM pairs, mean cost difference is −$0.0002963 per task, descriptive 95% interval $[-\$0.0003126, -\$0.0002800]$; the latency difference is −2.109 seconds $[-2.268, -1.954]$. Those paired resource intervals exclude the failed pair, whereas the table reports all attempted episodes. Cache state and concurrent scheduling limit latency interpretation.

The single AWM induction costs an estimated $0.0025674 ($0.00002140 per primary task when amortized over 120 tasks) and takes 4.745 seconds. GAC's reconstruction preflight takes 0.646 seconds, including source loading and compatibility checking; these timings do not isolate comparable optimization stages. New known study spend, including pilot and induction, is $0.30665104. The failed episode retains a $1 unknown-cost reserve despite having recorded response usage. Aggregate prior and new estimates/reserves total $8.57705193 within the authorized $400 cap; these estimates are not provider invoices.

Scope and artifacts. These cohorts remain separate from the six-study 810-episode historical summary. The study has one workflow, one model, one historical snapshot, one induction output and one stochastic evaluation per arm/issue. It establishes neither cross-domain superiority, a learned-gate benefit, independent human answer validity, nor a new answer-risk certificate. Historical shared discovery acquisition is a sunk cost; induction is priced separately, while reconstruction preflight time includes source loading and compatibility checks. Total lifetime authoring and optimization effort is not priced.

The protocol, code and dependency hashes, upstream pins, discovery prompt, cohorts, order, memory seal, checkpoint, usage and failure reserves, source regrading, tool-output replay, and complete contrasts are retained under `experiments/matched_reuse_baseline/`. The compatibility audit also records why AWO, EvoC2F, AgentJIT and plan caching were not treated as verified drop-in comparators; it does not claim their implementations are permanently unavailable.

K Fresh Workflow Generalization and Gate Attribution

This separately frozen follow-up extends the adapted AWM comparison in Appendix J to three workflow families and two model strings: `gpt-5.6-luna` and `gpt-6-luna`. All records come from the same pinned repository snapshot. Thus the study tests workflow and model transfer, not cross-domain or cross-provider generalization.

Prospective selection and execution. All record identifiers in prior retained studies are excluded. Backlog tasks are allocated first, followed by issue triage and pull-request outcome tasks, with fixed hash ranking within source-only eligibility. Titles must be representable by the answer schema, and issue tasks must expose a source excerpt meeting the minimum length. No successful dispatch or provider answer is an eligibility criterion. Six pilot records per family precede 144 backlog, 600 issue, and 600 pull-request primary records. Families are disjoint; models share records. The design schedules 8,172 episodes, including 108 pilot episodes, with all six three-arm permutations balanced

within each cell. Models alternate order by record. Eight record groups may execute concurrently; all arms and models within a record execute sequentially.

The primary class mix is 169 bug, 59 enhancement and 372 other issues; 517 merged, 82 closed-unmerged and one open pull request; and 109 discussed unowned plus 35 awaiting-response backlog records. There are no owned backlog records in the fresh cohort. The hash-selected residual population therefore does not supply balanced branch coverage, and the open/owned branches cannot support generalization claims from this follow-up. The source audit finds no structural title/excerpt/owner representability defect, but flags 27 issue records with nonliteral bug labels; exact-label errors can differ from semantic classification errors. This is an author audit, not independent adjudication.

The baseline, GAC, and offline adapted AWM arms share each family's prompt, source tools, output schema and factual grader. GAC uses the existing SDK model wrapper for issue triage and the outer pre-model composite runner for backlog and pull-request tasks; refusals use the unchanged agent. These placements are part of the treatment. Historical discovery is reconstructed under each model's manifest, without model-specific rediscovery. AWM induces one memory per cell from that family's eligible historical action traces, using the pinned adapter in Appendix J; the memory is frozen before pilots, without manual rewriting or online updates. The reused adapter retains its generic "Software,Issue triage,github" induction header for every family; the action examples and tool schemas are family-specific, and every retained memory names that family's three tools. This transport/domain adaptation is not native AWM evaluation. GAC retains its historical split. Induction and reconstruction costs are reported separately from deployment.

Settings are low reasoning/verbosity, no parallel tools, 2,048 output tokens, eight SDK turns, a 120-second timeout and zero SDK/HTTP retries. Pilot continuation requires execution, schema and usage completeness and actual compaction in every cell; favorable accuracy is not a continuation criterion. Failed and unattempted scheduled tasks remain incorrect in accuracy denominators. There are no replacement tasks or repeat attempts.

Accuracy margin and precision. The primary contrast is GAC minus adapted AWM on exact source-grounded answer fields; tool-order compliance is reported separately. The registered non-inferiority margin is five percentage points, a research tolerance rather than an operational safety threshold. For each cell, let L count AWM-correct/GAC-incorrect pairs among n scheduled records. A one-sided Clopper–Pearson upper bound U on L/n at error probability $0.05/6$ yields a conservative lower bound $-U$ on the accuracy difference, since it ignores GAC-only successes. Noninferiority requires all scheduled pairs attempted and $U < 0.05$. Bonferroni controls the six comparisons under the record-sampling assumptions without requiring model independence. Models are never counted as additional independent records in a pooled interval.

At $n = 600$, at most 17 losses pass; power is 93.9% under a 2% loss probability, but 46.7% under a 3% probability. At $n = 144$, only zero or one loss passes, and corresponding power is 21.5% or 6.8%. The backlog cell is therefore a low-precision replication, not an adequately powered test under those alternatives. Tighter one- and two-point margins are sensitivity analyses. Resource intervals use 10,000 paired bootstrap draws on completed pairs; all-attempted resource totals also retain observed failure usage. Concurrent wall-clock measurements include shared runner/checkpoint overhead. Missing usage is never interpreted as zero cost. Task/author/time independence is unproven.

Gate attribution. Before paid execution, the real dispatcher runs each selected record under full statistical admission, support-only admission, and deterministic guards. Support-only requires the retained 92-group calibration floor but disables the score and risk admission; deterministic guards also remove the support-count check. Both ablations preserve the exact candidate program, hard guards, verifier, effect checks and continuation contract. They are uncertified research ablations on already admitted candidates, not alternative certified compilers.

All three paths produce identical decisions, calls and continuation evidence on all 2,724 record/model cases, including pilots. Recorded gate scores are constant within each family on both models: 0.1237 for backlog, 0.0815 for issue triage, and 0.1308 for pull requests. The selected entry features therefore provide no observed risk ranking within these cohorts. Backlog and pull-request candidates dispatch throughout; issue candidates dispatch on 600 of 606 records per model and refuse the same six records under every variant. Consequently, only one live GAC path is run: duplicate stochastic answers would

not constitute independent evidence that identical routing policies differ. This is a null result for incremental runtime gate value on these candidates. It neither establishes universal gate equivalence nor measures whether statistical admission excluded bad candidates during compilation. The earlier learned end-to-end gate retirement remains unchanged.

Completed results. All 8,172 scheduled episodes were attempted: 8,171 completed and one baseline PR episode on `gpt-5.6-luna` failed with a provider server error. It remains incorrect, with no retry. All 2,724 live GAC decisions match their frozen provider-free probes, with zero dispatch mismatches. The pilot has 107/108 correct answers; primary results are separate in Tables 39 and 40.

Family	Model	n	GAC	AWM	Base	$100U$	NI
Backlog	5.6 Luna	144	143	142	144	3.27	yes
Backlog	6 Luna	144	142	143	144	5.86	no
Issue triage	5.6 Luna	600	584	583	590	3.61	yes
Issue triage	6 Luna	600	583	583	582	4.04	yes
PR outcome	5.6 Luna	600	597	598	597	1.43	yes
PR outcome	6 Luna	600	598	600	599	1.43	yes

Table 39: Primary exact-answer successes over scheduled records. AWM denotes the offline API-task adaptation. $100U$ is the simultaneous conservative accuracy-loss bound in percentage points, not an observed accuracy difference. NI uses the registered five-point margin.

Five cells pass the five-point margin. Second-model backlog has two AWM-only successes and one GAC-only success; its 5.86-point bound does not establish noninferiority or inferiority. Only the two PR cells pass a two-point margin, and none passes one point. Thus this study does not establish exact accuracy equality or a uniformly small quality loss across all cells.

Family	Model	GAC req.	AWM req.	GAC USD/1k	AWM USD/1k	Saving
Backlog	5.6 Luna	1.00	4.00	0.204	0.806	74.7%
Backlog	6 Luna	1.00	4.00	0.088	0.380	76.8%
Issue triage	5.6 Luna	2.00	4.00	0.588	0.931	36.9%
Issue triage	6 Luna	2.00	4.00	0.210	0.411	48.9%
PR outcome	5.6 Luna	1.00	4.00	0.164	0.739	77.8%
PR outcome	6 Luna	1.00	4.00	0.074	0.353	79.1%

Table 40: Observed deployment resources per attempted primary episode; USD/1k scales estimated cost to 1,000 episodes, excluding induction. Savings are relative to adapted AWM. Both compared arms have complete usage in every cell. Frozen API prices are estimates, not provider invoices.

Paired cost and latency differences favor GAC in every cell; the descriptive 95% intervals exclude zero, with 144 or 600 complete pairs. Tool-call means are equal between the compared arms. The full resource intervals are retained in `analysis.json`; latency includes shared runner overhead. New observed cost, including six inductions, is $3.864269; induction alone is $0.011211. One baseline request has no returned usage; its unknown charge is not treated as zero. The aggregate estimate plus prior and new failure reserves is $13.441321 against the authorized $400 cap. No baseline-cost superiority claim depends on the missing charge.

The post-registration author audit retains all 113 incorrect or failed episodes, including one pilot error. Overlapping failed checks include 50 issue-excerpt, 45 category, 46 evidence-label, 12 comment-grounding and five title checks; 45 error episodes involve nonliteral bug labels. No labels or outcomes are changed. Author/day cluster bootstraps are descriptive sensitivity, not replacement primary tests: the issue cohort has 386 authors (largest cluster 70 records), and the PR cohort 190 (largest 120). Their intervals do not establish independent sampling or repair the inconclusive backlog cell. Per-class outcomes and every failed answer are retained in `error_audit.json`. No independent human adjudication is claimed.

Artifacts and boundaries. The prospective protocol, code/environment/source hashes, frozen candidates, cohorts, schedule, per-record gate decisions, induction memories, checkpoints and analysis

are retained under `experiments/generalization_gate/`. Model aliases may drift; resolved immutable provider versions are not retained. This evidence remains separate from the historical 810-episode six-study summary. The study does not establish native AWM benchmark reproduction, independent human adjudication, total lifetime optimization cost, a new risk certificate, or cross-domain superiority.

L Claims and Evidence

Table 41 maps the principal empirical claims to retained artifacts. Numerical results and their interpretation boundaries appear in the preceding sections; this index supplies source locations without repeating them. The source manifests and publication manifest bind those locations to specific retained files. The statistical assumptions in Appendices A and C apply throughout.

Table 41: Retained evidence index. Paths beginning with a result directory are relative to `paper/results/`; other paths are repository-relative. Angle brackets and braces denote study-specific expansions.

Claim or study	Retained evidence
Preservation and reductions on the primary families	`github_workflow_families/summary.json` (`overall`, `families[].reductions`); per-record arms in `github_natural_replication/results.json` and `github_workflow_families/pr_outcome/final/results.json`, `github_workflow_families/backlog_attention/final/results.json`.
Discordance	`iclr_revision/paired_statistics.json` (`discordance`); `github_workflow_families/backlog_attention/headroom_ablation/results.json`.
Certificate level: per-candidate versus compiler-wide	`iclr_revision/multiplicity_accounting.json` (`u_at_92`, `n_min`); `iclr_revision/recompiled_candidates_pr_outcome.json`, `iclr_revision/recompiled_candidates_backlog_attention.json`.
The step-like gate is a sample-size floor	`src/guarded_agentic_compaction/grc/calibrate.py` (`clopper_pearson_upper`, $k = 0$); `gate_selectivity_analysis.json` (`summary`); `iclr_revision/gate_frontier_reanalysis.json` (`grid_rows`); `paper/iclr/figures/gate_floor.pdf`.
810 compiled episodes, one compiled-only failure	Per-record `results[]` of the six studies' result files; counted by `validate_live_extensions` in `paper/scripts/validate_artifacts.py`.
Cross-repository core protocol (newer records)	`github_multirepo_pr_outcome_core/results.json`; `iclr_revision/gate_frontier_reanalysis.json` (`core`); `iclr_revision/pytorch_core_recompile.json`.
Balanced three-repository rerun	`github_multirepo_pr_outcome_balanced/results.json`; `github_multirepo_pr_outcome_balanced/repos/*/results.json`.
Prospective gate-frontier study	`github_multirepo_gate_frontier/results.json`; `iclr_revision/gate_frontier_reanalysis.json` (`gate_frontier`).
Amended issue gate: actual dispatch and retirement	`graded_issue_gate/preflight.json`; `graded_issue_gate/abort_v1.json`; `graded_issue_gate/candidate_v2/summary.json`; `graded_issue_gate/results_v2.json`; `graded_issue_gate/audit_v2.json`; `paper/supplementary/graded-issue-end-to-end-gate-protocol.md`.
External substrates retire at the exact floor	`nestful/results.json` (`compiler.exact_gate`); `external_benchmarks/api_bank_execution.json`; `external_benchmarks/bfcl_compiler_execution.json`, `external_benchmarks/bfcl_gold_execution.json`.
AppWorld admits one artifact	`external_benchmarks/appworld_compiler_execution.json` (`admission`, `exact_replay`, `effect_audit`); `paper/supplementary/appworld-compiler-protocol.md`.

Claim or study (continued)	Retained evidence
Admissible is not dispatchable	external_benchmarks/appworld_dispatch_preflight.json (pooled, by_agent_family, first_call_in_released_trajectories).
Recurrence-only replay	recurrence_only_ablation/results.json (aggregate, per_field_exact); paper/supplementary/recurrence-only-ablation-protocol.md.
gpt-6-luna transfer (design B)	second_model_replication/summary.json; second_model_replication/issue_type/results.json; github_workflow_families/pr_outcome/gpt6_luna/results.json, github_workflow_families/backlog_attention/gpt6_luna/results.json.
gpt-6-luna re-discovery (design A)	second_model_replication/issue_type_rediscovery/results.json; github_workflow_families/pr_outcome/gpt6_luna_rediscovery/results.json; github_workflow_families/backlog_attention/gpt6_luna_rediscovery/failure.json; paper/supplementary/second-model-rediscovery-protocol.md.
Second provider: Anthropic claude-sonnet-5 (design A)	second_provider_replication/issue_type/results.json; github_workflow_families/pr_outcome/anthropic_sonnet5_rediscovery/results.json; github_workflow_families/backlog_attention/anthropic_sonnet5_rediscovery/failure.json; paper/supplementary/second-provider-replication-protocol.md.
Multiplicity repair: NO-GO	multiplicity_repair/preflight.json; paper/supplementary/multiplicity-repair-protocol.md.
Group independence	iclr_revision/effective_units.json (primary_ranges).
Continuation check (introduction example)	github_natural_live/continuation_replay.json (summary).
Extended issue-type held-out	issue_type_extended_heldout/results.json (compiled_only_bound, dispatch); paper/supplementary/issue-type-extended-heldout-protocol.md.
Read-only prologue	external_benchmarks/appworld_dispatch_prologue_preflight.json (by_agent_family); tests/unit/test_dispatch_prologue.py; paper/supplementary/read-only-prologue-protocol.md.
Perturbation challenge and signed registries	iclr_revision/recompile_with_challenge.json; paper/scripts/recompile_with_challenge.py.
Drift ablation (provider-free, demonstration substrate)	drift_ablation/results.json (pooled, decision); paper/scripts/drift_ablation_study.py; paper/supplementary/drift-robustness-ablation-protocol.md.
Recorded-replay drift ablation (provider-free, 90 primary records)	drift_recorded_replay/results.json (pooled, families.*.applicability); paper/scripts/drift_recorded_replay_study.py; paper/supplementary/drift-recorded-replay-protocol.md.
Post-cutoff cohort; end-to-end certificate (PR-outcome)	time_forward/pr_outcome/certificate.json, time_forward/pr_outcome/evaluation.json, time_forward/pr_outcome/selection.json; paper/scripts/time_forward_pr_outcome_study.py; paper/supplementary/time-forward-end-to-end-protocol.md.
Continuation-graded drift ablation (live, 60 primary records)	drift_continuation_graded/results.json (per_cell, decision); paper/scripts/drift_continuation_graded_study.py; paper/supplementary/drift-continuation-graded-protocol.md.
BIRD: a live SQL agent on a public benchmark (five databases, two designs)	bird/summary.json, bird/<design>/<database>/compile.json, bird/<design>/<database>/evaluation.json; paper/scripts/bird_sql_agent_study.py; paper/supplementary/bird-sql-agent-protocol.md.

Claim or study (continued)	Retained evidence
BIRD extensions: rotated rerun, second model, second provider, train split	`bird/extensions_summary.json`, `bird/{rotated_rerun,replications,train}/`; `paper/scripts/bird_sql_agent_study.py`; `paper/supplementary/bird-sql-agent-extensions-protocol.md`. All-scheduled denominators: `bird/denominator_audit.json`; `paper/scripts/audit_bird_denominators.py`. Paired uncertainty and database/cost sensitivity: `bird/quality_sensitivity.json`; `paper/scripts/bird_quality_sensitivity.py`.
Fresh-record safeguard comparison (480 live episodes)	`experiments/fresh_safeguards/analysis/analysis.json`, `experiments/fresh_safeguards/analysis/results.md`, `experiments/fresh_safeguards/analysis/error_audit.md`; `paper/scripts/fresh_safeguard_comparison.py`.
Deterministic scope/completeness diagnostic	`experiments/safeguard_mechanisms/artifacts/run/summary.json`, `experiments/safeguard_mechanisms/artifacts/run/outcomes.jsonl`, `experiments/safeguard_mechanisms/artifacts/seal.json`; `paper/scripts/safeguard_mechanism_study.py`.
Stopped live pilot and diagnostic	`experiments/live_safeguard_recovery/OBSERVED.md`; `experiments/live_recovery_completion/diagnostic.json`, `experiments/live_recovery_completion/diagnostic_driver.py.txt`. Pilot outcomes are excluded from the completion comparison.
Completed live recovery (2,304 attempts)	`experiments/live_recovery_completion/README.md`, `experiments/live_recovery_completion/seal.json`, `experiments/live_recovery_completion/checkpoint.json`, `experiments/live_recovery_completion/analysis.json`, `experiments/live_recovery_completion/FINDINGS.md`; `paper/scripts/complete_live_recovery.py`, `paper/scripts/recovery_observation.py`.
Matched offline AWM adaptation (fresh API tasks)	`experiments/matched_reuse_baseline/analysis.json`, `experiments/matched_reuse_baseline/seal.json`, `experiments/matched_reuse_baseline/memory_seal.json`, `experiments/matched_reuse_baseline/checkpoint.json`; `paper/scripts/matched_reuse_baseline.py`.
Fresh three-workflow, two-model comparison and gate attribution	`experiments/generalization_gate/README.md`, `experiments/generalization_gate/seal.json`, `experiments/generalization_gate/checkpoint.json`, `experiments/generalization_gate/analysis.json`, `experiments/generalization_gate/error_audit.json`, `experiments/generalization_gate/source_audit.json`; `paper/scripts/generalization_gate_study.py`.